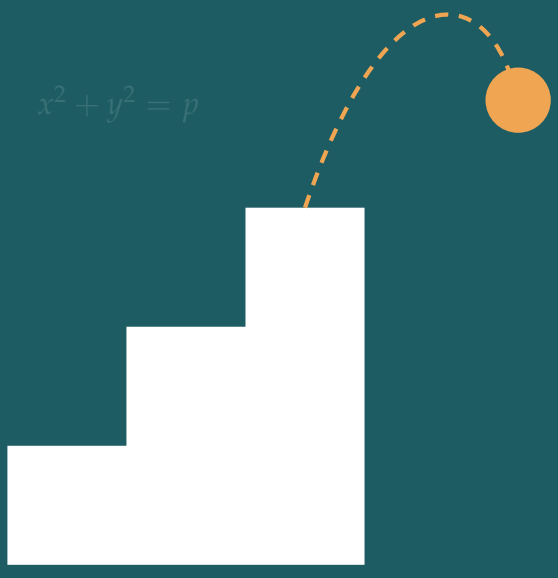
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2000 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2000 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2000 Worked Solutions

Worked solutions to all 42 questions of the 2000 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2000, Question 1

Pure • Logarithms and Leading Digits

STEP I 2000, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q1

The key idea

The leading digit of a large number N is decided entirely by the *fractional part* of $\log_{10} N$. If $\log_{10} N = k + f$ with k an integer and $0 \leq f < 1$, then $N = 10^f \times 10^k$, and the first digit of N is d exactly when $\log_{10} d \leq f < \log_{10}(d+1)$. So all we need are the logs of the digits 1 to 9, which come from $\log_{10} 2$ and $\log_{10} 3$, and a careful eye on how much rounding error we can afford.

Solution

Throughout, log means $\log_{10}$. The interval $[0, 1)$ splits into nine pieces, one for each possible leading digit:



A fractional part f landing in the piece labelled d gives leading digit d.

Part (i)

Since $5 = 10/2$ and $6 = 2 \times 3$,

$$\log 5 = 1 - \log 2 = 1 - 0.301029996 = 0.698970004 \approx 0.699,$$
$$\log 6 = \log 2 + \log 3 = 0.301029996 + 0.477121255 = 0.778151251 \approx 0.778,$$

each to three decimal places.

Now

$$\log 3^{100} = 100 \log 3 = 47.7121255.$$

The given value of $\log 3$ is correct to within 5×10^{-10}, so this is correct to within

5×10^{-8}; certainly $47.712 < \log 3^{100} < 47.713$. Comparing with

$$\log(5 \times 10^{47}) = 47 + \log 5 \approx 47.699 \quad \text{and} \quad \log(6 \times 10^{47}) = 47 + \log 6 \approx 47.778,$$

we have

$$\log(5 \times 10^{47}) < \log 3^{100} < \log(6 \times 10^{47}).$$

The margins (about 0.013 and 0.066) dwarf any rounding error, and log is strictly increasing, so

$$5 \times 10^{47} < 3^{100} < 6 \times 10^{47},$$

as required. Hence 3^{100} is $5.\ldots \times 10^{47}$: its first digit is $\boxed{5}$ (and it has 48 digits).

Part (ii)

We use the same idea with $\log 2 = 0.301029996$, which is in error by at most 5×10^{-10}. Multiplying by 10^n multiplies that error by 10^n, so we must check that the fractional part stays safely inside one piece of the number line.

2^{1000}. $\log 2^{1000} = 301.029996$, with error at most 5×10^{-7}. The fractional part, about 0.030, lies between $\log 1 = 0$ and $\log 2 \approx 0.301$, so $2^{1000} = 1.\ldots \times 10^{301}$ and its first digit is $\boxed{1}$.

$2^{10\,000}$. $\log 2^{10\,000} = 3010.29996$, with error at most 5×10^{-6}. The fractional part is between 0.299955 and 0.299965.

Ask yourself The fractional part 0.29996 is very close to $\log 2 = 0.30103$. Are we sure the first digit isn't 2?

Yes: the gap is $0.30103 - 0.29996 \approx 0.0011$, which is more than two hundred times the largest possible error of 5×10^{-6}. So the fractional part really is less than $\log 2$, and $10^{3010} < 2^{10\,000} < 2 \times 10^{3010}$. (Equivalently, $2^{10\,000} < 2 \times 10^{3010}$ says $2^{9999} < 10^{3010}$, i.e. $\log 2 < 3010/9999 = 0.3010301\ldots$, which is true.) The first digit is $\boxed{1}$.

$2^{100\,000}$. $\log 2^{100\,000} = 30102.9996$, with error at most 5×10^{-5}, so the fractional part lies between 0.99955 and 0.99965. We need to know which piece of the number line this is in, and the top piece starts at

$$\log 9 = 2 \log 3 = 0.954242510.$$

Since $0.954 < 0.99955$ and $0.99965 < 1$, we get $9 \times 10^{30102} < 2^{100\,000} < 10^{30103}$, and the first digit is $\boxed{9}$.

Ask yourself Couldn't the error push the fractional part past 1, making the number $1.\ldots \times 10^{30103}$?

Only if the true value of $100\,000 \log 2$ were at least 30103, which would need an error of 0.0004 in it, eight times more than the maximum 5×10^{-5}. So the first digit is 9 and $2^{100\,000}$ has 30 103 digits.

Checks

Exact integer arithmetic by computer confirms everything: 3^{100} has 48 digits and begins $515\ldots$; 2^{1000} begins $107\ldots$ (302 digits); $2^{10\,000}$ begins $199\ldots$ (3011 digits); and $2^{100\,000}$ begins $999\ldots$ (30 103 digits). A quick sanity check on the method: $\log 2^{10} = 3.0103$ has fractional part $0.0103 < \log 2$, matching $2^{10} = 1024$.

Common slips

- Reading the first digit from the integer part of the logarithm. The integer part only gives the number of digits (one more than it); the leading digit lives in the fractional part.
- Forgetting that multiplying by 100 000 also multiplies the rounding error in $\log 2$. Nine decimal places are just enough here, and it's worth saying why.
- Seeing $f = 0.9996 \approx 1$ and concluding the digit is 1. A fractional part just below 1 means the number is just *below* a power of ten, so it starts with 9.

Beyond STEP

The leading digits of 2^n are governed by the fractional parts of $n \log 2$. Because $\log 2$ is irrational, these fractional parts are *equidistributed* in $[0, 1)$ (Weyl's equidistribution theorem), so the proportion of powers of 2 starting with digit d tends to $\log(1 + 1/d)$: this is Benford's law, and about 30% of powers of 2 begin with a 1. For the irrationality and approximation side of the story, Davenport's *The Higher Arithmetic* is a lovely read.

WORKED SOLUTION

STEP I 2000, Question 2

Pure • Binomial Expansion Coefficients

STEP I 2000, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q2

The key idea

Expanding two binomials with fractional terms and then hunting for matching powers is messy. It is much cleaner to pull out the negative powers of x first, so that each bracket becomes a polynomial in a single power of x (x^6 or x^2). Then the second product turns out to be the first one in disguise, thanks to the factorisation $x^6 - 1 = (x^2 - 1)(x^4 + x^2 + 1)$, so the "hence" is almost free.

Solution

The first expansion

Taking out the lowest power of x from each bracket,

$$\left(x^4 - \frac{1}{x^2}\right)^5 = \left(\frac{x^6 - 1}{x^2}\right)^5 = x^{-10}\,(x^6 - 1)^5,$$

$$\left(x - \frac{1}{x}\right)^6 = \left(\frac{x^2 - 1}{x}\right)^6 = x^{-6}\,(x^2 - 1)^6.$$

So the whole expression is $x^{-16}P(x)$, where

$$P(x) = (x^6 - 1)^5\,(x^2 - 1)^6.$$

The coefficient of x^n in the original expansion is therefore the coefficient of x^{n+16} in the polynomial $P(x)$.

By the binomial theorem,

$$(x^6 - 1)^5 = \sum_{j=0}^{5} \binom{5}{j} (-1)^{5-j} x^{6j}, \qquad (x^2 - 1)^6 = \sum_{k=0}^{6} \binom{6}{k} (-1)^{6-k} x^{2k},$$

so the coefficient of x^m in $P(x)$ is the sum of

$$\binom{5}{j} (-1)^{5-j} \binom{6}{k} (-1)^{6-k}$$

over all pairs with $6j + 2k = m$, $0 \le j \le 5$, $0 \le k \le 6$.

Coefficient of x^{-12}. We need $m = -12 + 16 = 4$, that is $6j + 2k = 4$. Since $6j \le 4$ forces $j = 0$, the only pair is $(j, k) = (0, 2)$, giving

$$\binom{5}{0}(-1)^5 \cdot \binom{6}{2}(-1)^4 = (-1)(15) = -15,$$

as required.

Coefficient of x^2. Now $m = 2 + 16 = 18$, so $6j + 2k = 18$, i.e. $3j + k = 9$ with $k \le 6$. That allows $j = 1, 2, 3$:

(j,k)	from $(x^6-1)^5$	from $(x^2-1)^6$	product
$(3,0)$	$\binom{5}{3}(-1)^2 = 10$	$\binom{6}{0}(-1)^6 = 1$	10
$(2,3)$	$\binom{5}{2}(-1)^3 = -10$	$\binom{6}{3}(-1)^3 = -20$	200
$(1,6)$	$\binom{5}{1}(-1)^4 = 5$	$\binom{6}{6}(-1)^0 = 1$	5

So the coefficient of x^2 is $10 + 200 + 5 = 215$.

Ask yourself Why can't $j = 0$ or $j = 4$ contribute to x^{18}?

With $j = 0$ we would need $k = 9$, and with $j = 4$ we would need $k = -3$; neither lies in $0 \le k \le 6$. It is easy to miss a pair (or invent one) here, so it pays to list the possibilities systematically.

The second expansion

Ask yourself What has $x^4 + x^2 + 1$ got to do with the first part?

It is the other factor of $x^6 - 1$ when we treat it as a difference of cubes in x^2:

$$x^6 - 1 = (x^2)^3 - 1 = (x^2 - 1)(x^4 + x^2 + 1).$$

Splitting $(x^2 - 1)^{11}$ as $(x^2 - 1)^5 (x^2 - 1)^6$,

$$(x^2-1)^{11}(x^4+x^2+1)^5 = \left[(x^2-1)(x^4+x^2+1)\right]^5 (x^2-1)^6 = (x^6-1)^5(x^2-1)^6 = P(x).$$

So the second expansion is exactly $P(x)$, which is x^{16} times the first expansion.

Coefficient of x^4. This is the coefficient of x^4 in $P(x)$, which we found above (it corresponded to x^{-12} in the first expansion). It is -15.

Coefficient of x^{38}. This corresponds to $6j + 2k = 38$, i.e. $3j + k = 19$ with $j \le 5$, $k \le 6$. Since $k \le 6$ we need $3j \ge 13$, so $j = 5$ and $k = 4$, giving

$$\binom{5}{5}(-1)^0 \cdot \binom{6}{4}(-1)^2 = 15.$$

So the coefficient of x^{38} is 15.

There is a neat way to see this without counting. P has degree $30 + 12 = 42$, and replacing x by $1/x$ gives

$$x^{42}P(1/x) = (1 - x^6)^5(1 - x^2)^6 = (-1)^{11}P(x) = -P(x).$$

Comparing coefficients, the coefficient of x^{42-m} in P is minus the coefficient of x^m. With $m = 4$, the coefficient of x^{38} is $-(-15) = 15$, which agrees.

Checks

- Put $x = 1$ in $P(x)$: it vanishes, so all the coefficients of P add up to 0. That's a quick sanity check if you expand fully.
- We multiplied out both products exactly by computer (as lists of coefficients). We found -15 for x^{-12} and 215 for x^2 in the first, and -15 for x^4 and 15 for x^{38} in the second, and we confirmed that the second product has degree 42 and equals x^{16} times the first.
- Evaluating the original expression at random rational values of x matches the computed expansion exactly.

Common slips

- Sign errors: with x^{6j} and x^{2k} as the chosen powers, the sign of a term is $(-1)^{5-j}(-1)^{6-k} = -(-1)^{j+k}$, not $(-1)^{j+k}$. Using the wrong one flips every answer.
- Forgetting the $(1, 6)$ or $(3, 0)$ terms in the x^2 coefficient, and getting 200 or 205.
- Expanding $(x^2 - 1)^{11}(x^4 + x^2 + 1)^5$ directly without spotting the factorisation: possible, but long and error-prone.

Beyond STEP

Writing a product of sums as a single sum and picking out one power of x is the idea behind *generating functions*. A coefficient of a product counts the ways to make a total from the parts, exactly as we counted pairs (j, k) above. The symmetry $x^{42}P(1/x) = -P(x)$ says that P is an (anti)*palindromic* polynomial; such polynomials turn up in the study of cyclotomic polynomials and roots of unity, of which $x^6 - 1$ is a first example. For more problems in this spirit, Siklos's *Advanced Problems in Mathematics* is a good next step, and Allenby's *Rings, Fields and Groups* covers polynomial factorisation properly.

WORKED SOLUTION

STEP I 2000, Question 3

Pure • Integrals of Step Functions

STEP I 2000, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q3

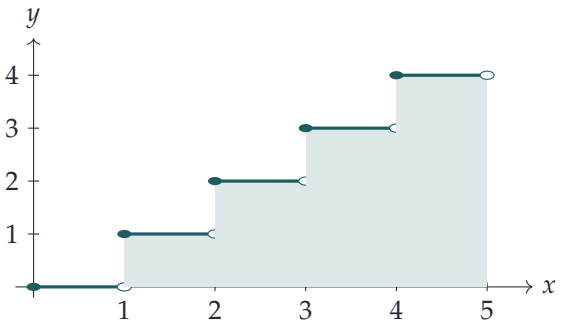
The key idea

The integer part of anything is a step function: it stays constant on intervals and then jumps by one. So the integral is just a sum of rectangle areas, and the only real work is finding where each step starts and stops. For $[\mathrm{e}^x]$ the jumps happen where e^x passes an integer, that is, at $x = \ln k$, and the resulting sum of logarithms telescopes neatly.

Solution

The graph of $y = [x]$

On each interval $k \leq x < k+1$ (with k an integer) we have $[x] = k$. So the graph is a staircase of horizontal segments, each of length 1, with a filled dot at the left end (the value is attained) and an open dot at the right end (where the function jumps up by 1).



The integral is the shaded area, made of rectangles of width 1 and heights $0, 1, 2, 3, 4$ (the value at the single points $x = 1, 2, 3, 4$ makes no difference to the area). Hence

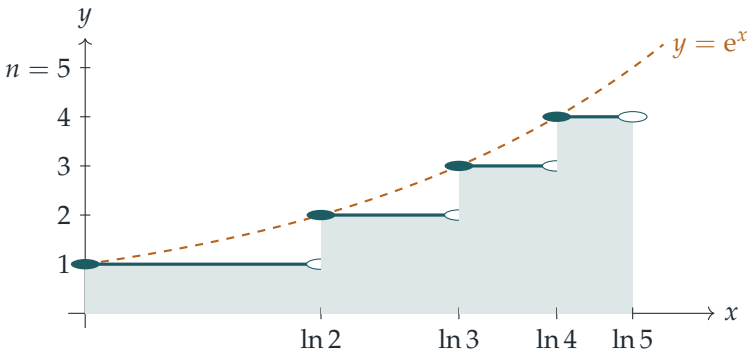
$$\int_0^5 [x]\,\mathrm{d}x = 0 + 1 + 2 + 3 + 4 = 10.$$

The graph of $y = [\mathrm{e}^x]$

For $0 \le x < \ln n$ we have $1 \le \mathrm{e}^x < n$. Since e^x is increasing,

$$[\mathrm{e}^x] = k \iff k \le \mathrm{e}^x < k+1 \iff \ln k \le x < \ln(k+1),$$

for $k = 1, 2, \ldots, n-1$. So the graph is again a staircase with steps of height 1, but the steps now get shorter as x increases: the step at height k has width $\ln(k+1) - \ln k = \ln\left(1 + \frac{1}{k}\right)$. The sketch shows $n = 5$, with the curve $y = \mathrm{e}^x$ dashed; the staircase touches the curve at the left end of each step.



The integral

Adding the rectangle areas, height k times width $\ln(k+1) - \ln k$,

$$\int_0^{\ln n} [\mathrm{e}^x]\,\mathrm{d}x = \sum_{k=1}^{n-1} k\big(\ln(k+1) - \ln k\big) = \sum_{k=1}^{n-1} k\ln(k+1) - \sum_{k=1}^{n-1} k\ln k.$$

? **Ask yourself** How do we make the two sums cancel?

Each $\ln j$ appears in both sums, with different coefficients. Substituting $j = k+1$ in the first sum lines them up:

$$\begin{aligned}\sum_{k=1}^{n-1} k\ln(k+1) - \sum_{k=1}^{n-1} k\ln k &= \sum_{j=2}^{n}(j-1)\ln j - \sum_{j=1}^{n-1} j\ln j \\ &= (n-1)\ln n + \sum_{j=2}^{n-1}\big((j-1) - j\big)\ln j - 1\cdot\ln 1 \\ &= (n-1)\ln n - \sum_{j=2}^{n-1}\ln j.\end{aligned}$$

Here we took the $j = n$ term out of the first sum, and the $j = 1$ term out of the second (it is zero, since $\ln 1 = 0$). Now $\sum_{j=2}^{n-1}\ln j = \ln\big((n-1)!\big)$, and $(n-1)\ln n = n\ln n - \ln n$, so

$$\int_0^{\ln n} [\mathrm{e}^x]\,\mathrm{d}x = n\ln n - \big(\ln n + \ln((n-1)!)\big) = n\ln n - \ln(n!),$$

as required. (For $n = 1$ the integral is over an interval of length zero, and both sides are 0.)

A picture-proof (optional)

The rectangle of width $\ln n$ and height n has area $n \ln n$. The part of it lying *above* the staircase is made of horizontal strips of height 1: between heights $m-1$ and m, for $m = 1, \ldots, n$, the strip runs from $x = 0$ to $x = \ln m$, because $[\mathrm{e}^x] < m$ exactly when $x < \ln m$. Its area is $\ln m$, so the region above the steps has area $\ln 1 + \ln 2 + \cdots + \ln n = \ln(n!)$. Subtracting from $n \ln n$ gives the result again. This is the discrete version of integrating with respect to y instead of x.

Checks

For $n = 2$ the staircase is a single rectangle of height 1 and width $\ln 2$, and the formula gives $2\ln 2 - \ln 2 = \ln 2$. For $n = 4$ the formula gives $4\ln 4 - \ln 24 \approx 2.3671$; a midpoint-rule evaluation of $\int_0^{\ln 4} [\mathrm{e}^x]\,\mathrm{d}x$ agrees to five decimal places, and so it does for every n up to 8. The answer must also be less than $\int_0^{\ln n} \mathrm{e}^x\,\mathrm{d}x = n - 1$, since the steps sit under the curve: for $n = 4$, $2.37 < 3$.

Common slips

- Taking the step at height k to have width 1 (copying the first part), rather than $\ln(k+1) - \ln k$.
- Starting the staircase at height 0: on $0 \le x < \ln n$ we have $\mathrm{e}^x \ge 1$, so the lowest step is at height 1.
- Mis-shifting the index and losing the $(n-1)\ln n$ term, or not noticing it must combine with $\ln((n-1)!)$ to give $\ln(n!)$.

Beyond STEP

Since the steps lie under $y = \mathrm{e}^x$, the integral is less than $n - 1$, which rearranges to $\ln(n!) > n\ln n - n + 1$. Bounding the steps from above in the same way leads towards Stirling's formula, $\ln(n!) \approx n\ln n - n$, a key estimate in analysis, probability and combinatorics. The swap from a sum of rectangles to a sum of strips is also an example of "summation by parts", the discrete cousin of integration by parts. Abbott's *Understanding Analysis* is a good next step for putting integrals of step functions on a rigorous footing.

WORKED SOLUTION

STEP I 2000, Question 4

Pure • Bounding Integrals

STEP I 2000, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q4

The key idea

We can't integrate $1/(x^2+1)^4$ easily by elementary means, so the question splits it into two pieces: an exact derivative, which integrates for free, and a leftover term that is small. Part (ii) builds the split and part (i) supplies the bound on the leftover. To bound an integral, we usually bound the integrand and multiply by the length of the interval.

Solution

Part (i)

Write $f(x) = \dfrac{x^6}{(x^2+1)^4}$. Since f depends only on x^2, put $t = x^2$, so that t runs over $[0,1]$ as x does and $f = g(t)$ with

$$g(t) = \frac{t^3}{(1+t)^4}.$$

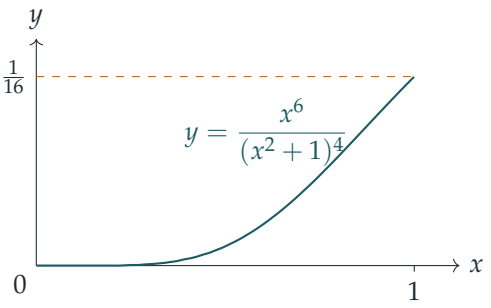
By the quotient rule,

$$g'(t) = \frac{3t^2(1+t)^4 - 4t^3(1+t)^3}{(1+t)^8} = \frac{t^2\big(3(1+t) - 4t\big)}{(1+t)^5} = \frac{t^2(3-t)}{(1+t)^5}.$$

For $0 < t \leq 1$ every factor is positive (in particular $3 - t \geq 2$), so $g'(t) > 0$ and g is strictly increasing on $[0,1]$. Its largest value is therefore at $t = 1$, that is $x = 1$:

$$f(1) = \frac{1}{2^4} = \frac{1}{16},$$

as required. We also have $f(x) \geq 0$ throughout, with $f(0) = 0$ the smallest value.



Part (ii)

Differentiate the bracket with the quotient rule. With $N = Ax^5 + Bx^3 + Cx$ and denominator $(x^2+1)^3$, whose derivative is $6x(x^2+1)^2$, one factor of $(x^2+1)^2$ cancels:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{N}{(x^2+1)^3}\right) = \frac{(5Ax^4+3Bx^2+C)(x^2+1) - 6x(Ax^5+Bx^3+Cx)}{(x^2+1)^4}.$$

Expanding the numerator and collecting powers of x:

$$\begin{aligned} &5Ax^6 + 5Ax^4 + 3Bx^4 + 3Bx^2 + Cx^2 + C - 6Ax^6 - 6Bx^4 - 6Cx^2 \\ &\quad = -Ax^6 + (5A-3B)x^4 + (3B-5C)x^2 + C. \end{aligned}$$

Everything now has denominator $(x^2+1)^4$, so the identity holds for all x exactly when the numerators agree as polynomials:

$$(D-A)x^6 + (5A-3B)x^4 + (3B-5C)x^2 + C = 1.$$

Comparing coefficients from the bottom up: $C = 1$; then $3B = 5C = 5$; then $5A = 3B = 5$; then $D = A$. So

$$A = 1, \qquad B = \tfrac{5}{3}, \qquad C = 1, \qquad D = 1.$$

Part (iii)

Integrate the identity from part (ii) over $[0,1]$. The first term on the right is an exact derivative, so

$$\begin{aligned} \int_0^1 \frac{\mathrm{d}x}{(x^2+1)^4} &= \left[\frac{x^5 + \frac{5}{3}x^3 + x}{(x^2+1)^3}\right]_0^1 + \int_0^1 \frac{x^6}{(x^2+1)^4}\,\mathrm{d}x \\ &= \frac{1+\frac{5}{3}+1}{8} - 0 + J = \frac{11}{24} + J, \end{aligned}$$

where $J = \displaystyle\int_0^1 \frac{x^6}{(x^2+1)^4}\,\mathrm{d}x$.

Ask yourself How do we bound J without working it out?

By part (i), the integrand of J satisfies $0 \le f(x) \le \frac{1}{16}$ for every x in $[0,1]$. Integrating these inequalities over an interval of length 1 gives

$$0 \le J \le \frac{1}{16} \times 1 = \frac{1}{16}.$$

Adding $\frac{11}{24}$ throughout,

$$\frac{11}{24} \le \int_0^1 \frac{\mathrm{d}x}{(x^2+1)^4} \le \frac{11}{24} + \frac{1}{16},$$

which is the required result. (Both inequalities are in fact strict, since f is continuous and equals neither bound on the whole interval, but the question only asks for $\le$.)

Checks

- Put $x = 0$ in the identity of part (ii): the derivative term is $C = 1$ and the D term vanishes, giving $1 = 1$. At $x = 1$ the derivative's numerator is $-1 + 0 + 0 + 1 = 0$, so the identity reads $\frac{1}{16} = \frac{D}{16}$, consistent with $D = 1$.
- Numerically, Simpson's rule gives $\displaystyle\int_0^1 \frac{\mathrm{d}x}{(x^2+1)^4} \approx 0.47460$, comfortably between $\frac{11}{24} \approx 0.45833$ and $\frac{11}{24} + \frac{1}{16} \approx 0.52083$. The leftover $J \approx 0.01627$ is much less than $\frac{1}{16}$, since f is tiny except near $x = 1$.

Common slips

- Forgetting to cancel the common factor $(x^2+1)^2$ in the quotient rule, which leaves a denominator of $(x^2+1)^6$ and makes the coefficient comparison a mess.
- In part (i), only checking that $x = 1$ is a stationary point or an endpoint: you need to show f is increasing (or compare all candidates) to know the endpoint value is the largest.

Beyond STEP

The substitution $x = \tan\theta$ turns the integral into $\int_0^{\pi/4} \cos^6\theta \, \mathrm{d}\theta$, which can be done exactly with a reduction formula: the value is $\frac{11}{48} + \frac{5\pi}{64}$, matching our numerical estimate. Splitting an integrand into "exact derivative plus something small" is the idea behind reduction formulae, and bounding integrals by bounding integrands is the basic estimate of real analysis. Bartle and Sherbert, *Introduction to Real Analysis*, develops these estimates carefully.

WORKED SOLUTION

STEP I 2000, Question 5

Pure • Vectors in Three Dimensions

STEP I 2000, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q5

The key idea

Everything happens in the inclined plane, so set up two perpendicular unit vectors *in that plane*: one pointing straight up the line of greatest slope, the other horizontal. Arthur's and Bertha's positions are then simple combinations of these, and the sum-to-product formulae tidy up the difference at once. For the angle with the vertical, we only need to know how each of the two in-plane directions sits relative to the vertical, and that is where θ enters.

Solution

Setting up

Let $\mathbf{e}$ be the unit vector pointing up the steepest line through O, and let $\mathbf{f}$ be the unit vector in the plane, perpendicular to $\mathbf{e}$, on the side where both walkers are. Since $\mathbf{f}$ is perpendicular to the line of greatest slope, it is horizontal. Let $\mathbf{k}$ be the unit vector vertically upwards. Then

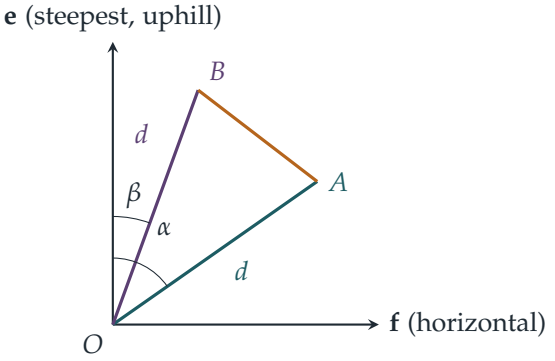
$$\mathbf{e}\cdot\mathbf{k} = \sin\theta, \qquad \mathbf{f}\cdot\mathbf{k} = 0, \qquad \mathbf{e}\cdot\mathbf{f} = 0.$$

(The first holds because $\mathbf{e}$ makes angle θ with the horizontal, hence angle $\frac{\pi}{2} - \theta$ with $\mathbf{k}$.)

Both walk at the same speed, so when Arthur has walked a distance d, so has Bertha. Their positions relative to O are

$$\mathbf{a} = d(\cos\alpha\,\mathbf{e} + \sin\alpha\,\mathbf{f}), \qquad \mathbf{b} = d(\cos\beta\,\mathbf{e} + \sin\beta\,\mathbf{f}).$$

Walking uphill on the same side of the steepest line means $0 \le \alpha, \beta < \frac{\pi}{2}$.



The figure is drawn in the inclined plane itself, looking straight at it.

The distance between them

Using the sum-to-product formulae,

$$\begin{aligned}\mathbf{a}-\mathbf{b} &= d\big[(\cos\alpha-\cos\beta)\,\mathbf{e}+(\sin\alpha-\sin\beta)\,\mathbf{f}\big]\\ &= d\big[-2\sin\tfrac{1}{2}(\alpha+\beta)\sin\tfrac{1}{2}(\alpha-\beta)\,\mathbf{e}+2\cos\tfrac{1}{2}(\alpha+\beta)\sin\tfrac{1}{2}(\alpha-\beta)\,\mathbf{f}\big]\\ &= 2d\sin\tfrac{1}{2}(\alpha-\beta)\,\big[-\sin\tfrac{1}{2}(\alpha+\beta)\,\mathbf{e}+\cos\tfrac{1}{2}(\alpha+\beta)\,\mathbf{f}\big].\end{aligned}$$

Write $s=\frac{1}{2}(\alpha+\beta)$. The vector $\mathbf{u}=-\sin s\,\mathbf{e}+\cos s\,\mathbf{f}$ is a unit vector, because $\mathbf{e}$ and $\mathbf{f}$ are perpendicular unit vectors and $\sin^2 s+\cos^2 s=1$. Hence

$$AB=|\mathbf{a}-\mathbf{b}|=2d\,\big|\sin\tfrac{1}{2}(\alpha-\beta)\big|,$$

as required. (Geometrically: OAB is isosceles with equal sides d and apex angle $|\alpha-\beta|$; dropping the perpendicular from O gives the same answer.)

The angle with the vertical

When $\alpha\neq\beta$ we have $\sin\frac{1}{2}(\alpha-\beta)\neq 0$, since $0<|\alpha-\beta|<\frac{\pi}{2}$, so $A\neq B$ and the line AB is well defined. The calculation above shows that it is parallel to the unit vector $\mathbf{u}$, or equivalently to $\mathbf{v}=-\mathbf{u}=\sin s\,\mathbf{e}-\cos s\,\mathbf{f}$.

Ask yourself A line has two directions. Which one should we compare with the vertical?

The angle between a line and the vertical is the acute (or right) angle, so we use whichever direction along the line points upwards, or at least not downwards. Now

$$\mathbf{v}\cdot\mathbf{k}=\sin s\,(\mathbf{e}\cdot\mathbf{k})-\cos s\,(\mathbf{f}\cdot\mathbf{k})=\sin\theta\sin s.$$

Since $0\leq s<\frac{\pi}{2}$ and $0<\theta<\frac{\pi}{2}$, this is non-negative, so $\mathbf{v}$ is the upward direction. As $\mathbf{v}$ and $\mathbf{k}$ are both unit vectors, the angle ϕ between them satisfies

$$\cos\phi=\mathbf{v}\cdot\mathbf{k}=\sin\theta\sin\tfrac{1}{2}(\alpha+\beta),$$

which is the required result. Notice that it does not depend on d: the line AB moves parallel to itself as they walk.

Checks

- If Arthur walks horizontally ($\alpha = \frac{\pi}{2}$, the boundary case) and Bertha straight up ($\beta = 0$), then $AB = 2d \sin\frac{\pi}{4} = \sqrt{2}\,d$, which is Pythagoras for two perpendicular paths of length d. With $\theta = 30^\circ$ and $d = 1$, coordinates give $A = (1, 0, 0)$, $B = (0, \frac{\sqrt{3}}{2}, \frac{1}{2})$, so the vertical drop from B to A is $\frac{1}{2}$ over a distance $\sqrt{2}$: $\cos\phi = \frac{1}{2\sqrt{2}} \approx 0.354$, matching $\sin 30^\circ \sin 45^\circ$, so $\phi \approx 69.3^\circ$.
- If the plane is horizontal ($\theta = 0$), $\cos\phi = 0$: the line is horizontal, as it must be.
- A computer check with 10^5 random choices of θ, α, β, d in genuine 3D coordinates agreed with both formulae to within 10^{-12}.

Common slips

- Measuring the angle in the plane between AB and the steepest line instead of the angle in space with the vertical; θ must appear in the answer.
- Forgetting the modulus: $\sin\frac{1}{2}(\alpha - \beta)$ is negative when $\beta > \alpha$, but a distance is not.
- Taking the downward direction along AB and getting $\cos\phi = -\sin\theta \sin s$; the angle of a line with the vertical is never obtuse.

Beyond STEP

The method here is to choose an orthonormal basis adapted to the problem (two vectors in the plane, then relate them to the ambient vertical) and to read off lengths and angles from dot products. In a first-year linear algebra course this becomes the idea of changing basis and of orthogonal projection: $\mathbf{v} \cdot \mathbf{k}$ is the length of the projection of $\mathbf{v}$ onto the vertical. Strang's *Introduction to Linear Algebra* develops these ideas carefully, with plenty of geometry.

WORKED SOLUTION

STEP I 2000, Question 6

Pure • Regions Bounded by Lines

STEP I 2000, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q6

The key idea

A quadratic expression in x and y that factorises into two linear factors describes a pair of straight lines. A product of two factors is positive exactly when the factors have the same sign, and negative when they have opposite signs, so each inequality picks out a pair of opposite "wedges" between two crossing lines. Once both regions are drawn on the same axes, finding a common point is a matter of looking.

Solution

The first factorisation and region

Expanding the right-hand side,

$$\begin{aligned}(x-y+2)(x+y-1) &= x^2+xy-x-xy-y^2+y+2x+2y-2\\ &= x^2-y^2+x+3y-2,\end{aligned}$$

as required. So the first inequality, after subtracting 2 from both sides, is

$$(x-y+2)(x+y-1)>0.$$

The boundary consists of the two lines

$$L_1:\ y=x+2, \qquad L_2:\ y=1-x,$$

which have gradients 1 and -1 and meet where $x+2=1-x$, that is at $\left(-\frac{1}{2},\frac{3}{2}\right)$.

The product is positive when both factors have the same sign:

- both positive: $y<x+2$ and $y>1-x$, i.e. below L_1 and above L_2. This is the wedge to the *right* of the crossing point (it contains $(5,0)$, say);
- both negative: $y>x+2$ and $y<1-x$, i.e. above L_1 and below L_2. This is the wedge to the *left* of the crossing point (it contains $(-5,0)$).

So the region is the pair of opposite wedges, left and right, between the lines $y = x + 2$ and $y = 1 - x$, with the lines themselves excluded (on them the product is 0, not > 0). A quick test: at the origin the expression is $-2 < 0$, and the origin lies in the lower wedge, which is correctly excluded.

The second region

Ask yourself How do we know this one factorises too, and what should the factors look like?

The quadratic terms $x^2 - 4y^2 = (x + 2y)(x - 2y)$ suggest trying

$$x^2 - 4y^2 + 3x - 2y + 2 = (x + 2y + a)(x - 2y + b).$$

The right-hand side expands to $x^2 - 4y^2 + (a + b)x + (2b - 2a)y + ab$. Matching coefficients gives $a + b = 3$ and $b - a = -1$, so $a = 2$, $b = 1$, and then the constant term $ab = 2$ also matches. Hence, adding 2 to both sides of the second inequality,

$$x^2 - 4y^2 + 3x - 2y < -2 \iff (x + 2y + 2)(x - 2y + 1) < 0.$$

The boundary lines are

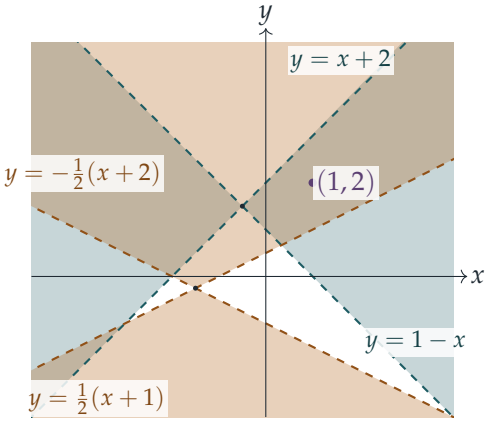
$$L_3 : \ y = -\tfrac{1}{2}(x + 2), \qquad L_4 : \ y = \tfrac{1}{2}(x + 1),$$

with gradients $-\frac{1}{2}$ and $\frac{1}{2}$, meeting where $-x - 2 = x + 1$, that is at $\left(-\frac{3}{2}, -\frac{1}{4}\right)$.

Now we need the factors to have *opposite* signs:

- $x + 2y + 2 > 0$ and $x - 2y + 1 < 0$: $y > -\frac{1}{2}(x + 2)$ and $y > \frac{1}{2}(x + 1)$, i.e. above both lines. This is the *upper* wedge (it contains $(0, 5)$);
- $x + 2y + 2 < 0$ and $x - 2y + 1 > 0$: below both lines. This is the *lower* wedge (it contains $(0, -5)$).

So the second region is the pair of opposite wedges, above and below, between the lines L_3 and L_4, again with the boundary excluded.



In the sketch the first region is shaded teal (left and right wedges), the second amber (upper and lower wedges), so the darker overlap is where both hold; the dashed lines are not included.

A point in both regions

The two pairs of wedges certainly overlap: the right-hand teal wedge opens out to the right between gradients ± 1, while the upper amber wedge contains everything above both of the shallower lines of gradient $\pm\frac{1}{2}$. For $x > -\frac{1}{2}$ the top edge $y = x + 2$ of the teal wedge rises faster than the line $y = \frac{1}{2}(x+1)$, so there is room between them.

Take, for instance, $(x, y) = (1, 2)$. Then

$$x^2 - y^2 + x + 3y = 1 - 4 + 1 + 6 = 4 > 2$$

and

$$x^2 - 4y^2 + 3x - 2y = 1 - 16 + 3 - 4 = -16 < -2,$$

so $(1, 2)$ satisfies both inequalities. (Checking against the factors: $(x - y + 2)(x + y - 1) = 1 \cdot 2 > 0$ and $(x + 2y + 2)(x - 2y + 1) = 7 \cdot (-2) < 0$.)

Checks

- Each factorisation can be checked at a random point: at $(3, 1)$, $x^2 - y^2 + x + 3y - 2 = 12$ and $(x - y + 2)(x + y - 1) = 4 \cdot 3 = 12$; at the same point $x^2 - 4y^2 + 3x - 2y + 2 = 14$ and $(x + 2y + 2)(x - 2y + 1) = 7 \cdot 2 = 14$.
- Test one point in each wedge, e.g. $(0, -5)$ gives $x^2 - 4y^2 + 3x - 2y = -90 < -2$, confirming the lower amber wedge belongs to the second region.
- A numerical check over 200 000 random points in $[-10, 10]^2$ found no disagreement between the inequalities and the wedge descriptions above.

Common slips

- Shading the wrong pair of wedges. Always test a point (the origin is the easiest) rather than relying on "above" and "below".
- Forgetting the -2 and $+2$ when moving the constants across: the first inequality is $(x - y + 2)(x + y - 1) > 0$, the second $(x + 2y + 2)(x - 2y + 1) < 0$.
- Drawing the boundary lines as solid: the inequalities are strict, so the lines are excluded.

Beyond STEP

A quadratic in two variables, $ax^2 + bxy + cy^2 + dx + ey + f$, defines a conic; when it factorises into two linear factors the conic is *degenerate*, a pair of lines, and this happens

exactly when a certain 3×3 determinant built from the coefficients vanishes. The sign pattern of a quadratic form (here $x^2 - y^2$ and $x^2 - 4y^2$, both indefinite) is studied in first-year linear algebra through symmetric matrices and their eigenvalues. Strang's *Introduction to Linear Algebra* is a good place to see this.

WORKED SOLUTION

STEP I 2000, Question 7

Pure • Bounding a Derivative

STEP I 2000, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q7

The key idea

Differentiate, and the condition $f'(x) \geq 0$ becomes "a is at least as big as a certain function of x". So everything hinges on the *largest* value of that function. It depends only on x^2, and after the substitution $u = 1 + x^2$ it becomes a quadratic in $1/u$. Completing the square then gives the maximum, and shows exactly where it occurs.

Solution

The derivative

By the quotient rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x^3}{1+x^2}\right) = \frac{3x^2(1+x^2) - x^3 \cdot 2x}{(1+x^2)^2} = \frac{x^4 + 3x^2}{(1+x^2)^2},$$

so

$$f'(x) = a - g(x), \qquad \text{where } g(x) = \frac{x^2(x^2+3)}{(1+x^2)^2}.$$

Hence $f'(x) \geq 0$ for all x exactly when $a \geq g(x)$ for all x. We need the largest value of g.

The largest value of g

Put $u = 1 + x^2$, so that $u \geq 1$ and $x^2 = u - 1$. Then

$$g = \frac{(u-1)(u+2)}{u^2} = \frac{u^2 + u - 2}{u^2} = 1 + \frac{1}{u} - \frac{2}{u^2}.$$

? **Ask yourself** Why is it a good idea to write this in terms of $1/u$ rather than u?

In terms of $s = 1/u$, $g = 1 + s - 2s^2$, which is just a quadratic, and the maximum of a

quadratic comes from completing the square, with no calculus needed:

$$1 + s - 2s^2 = \frac{9}{8} - 2\left(s - \frac{1}{4}\right)^2.$$

Since a square is never negative, $g \leq \frac{9}{8}$ for every x, with equality only when $s = \frac{1}{4}$, that is $u = 4$, $x = \pm\sqrt{3}$.

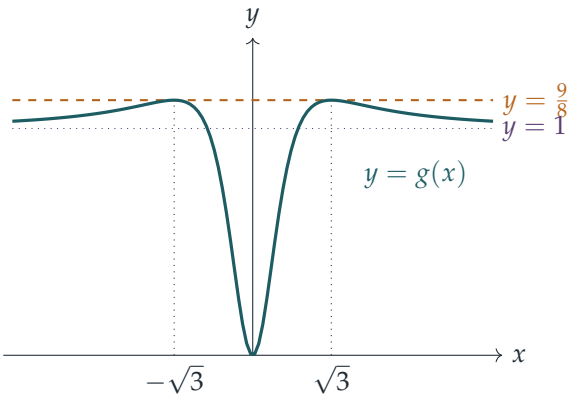
Putting this back in terms of x (using $s - \frac{1}{4} = \frac{3-x^2}{4(1+x^2)}$) gives a tidy identity for the derivative:

$$\mathrm{f}'(x) = \left(a - \frac{9}{8}\right) + \frac{(3-x^2)^2}{8(1+x^2)^2}.$$

Conclusion

If $a \geq \frac{9}{8}$, the first bracket is non-negative and the second term is a square divided by something positive, so it is non-negative too. Therefore $\mathrm{f}'(x) \geq 0$ for all x, as required.

The constant $\frac{9}{8}$ is also the best possible: if $a < \frac{9}{8}$ then $\mathrm{f}'(\sqrt{3}) = a - \frac{9}{8} < 0$. When $a = \frac{9}{8}$, f' is zero only at $x = \pm\sqrt{3}$, so f is still strictly increasing, with two stationary points of inflection.



The graph of g climbs from 0, overshoots its limit 1, peaks at $\frac{9}{8}$ when $x = \pm\sqrt{3}$, and then falls back towards 1. The derivative $\mathrm{f}'(x)$ is the height of the line $y = a$ above this curve.

Checks

- At $x = \sqrt{3}$: $g = \dfrac{3 \cdot 6}{4^2} = \dfrac{18}{16} = \dfrac{9}{8}$. ✓
- As $x \to \infty$, $g \to 1 < \frac{9}{8}$. This makes sense, since $\frac{x^3}{1+x^2}$ behaves like x for large x, so its slope tends to 1.

- Numerically, with $a = \frac{9}{8}$ the smallest value of $\mathrm{f}'(x)$ on a grid of step 0.001 over $[-100, 100]$ is about 2×10^{-10} (essentially 0, at $x \approx \pm\sqrt{3}$). With $a = 1.1$, $\mathrm{f}'(\sqrt{3}) = -0.025 < 0$.

Common slips

- Treating g's limit at infinity, 1, as its largest value. The curve overshoots, which is why the answer is $\frac{9}{8}$ and not 1.
- Differentiating g directly and getting lost in the algebra. The substitution $s = 1/(1+x^2)$ turns it into a one-line completion of the square.

Beyond STEP

Here a "for all x" statement was reduced to finding a *supremum*: $\mathrm{f}' \geq 0$ everywhere exactly when $a \geq \sup g$. The fact that g has a limit at infinity but is largest at a finite point is a good warning that a limit need not be a bound. In analysis you'll prove that a function with non-negative derivative is increasing (via the Mean Value Theorem), which is why this question is really about whether f is monotonic. Abbott's *Understanding Analysis* develops these ideas carefully.

WORKED SOLUTION

STEP I 2000, Question 8

Pure • Integrals of Moduli

STEP I 2000, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q8

The key idea

A modulus sign inside an integral is dealt with by finding where the expression inside changes sign. On each interval between sign changes, $|f|$ is either f or $-f$, so we split the range there and integrate ordinary functions. All the real work is in locating the zeros: by inspection for xe^x, by factorising for the cubic, and by writing $\sin x + \cos x$ as a single sine wave.

Solution

The first integral

Since $e^x > 0$ for every x, the sign of xe^x is the sign of x. So on $[-1,0]$ we have $xe^x \le 0$ and $|xe^x| = -xe^x$, while on $[0,1]$ we have $xe^x \ge 0$ and $|xe^x| = xe^x$. Splitting the range at 0,

$$\int_{-1}^{1} |xe^x|\,dx = \int_{-1}^{0} (-xe^x)\,dx + \int_{0}^{1} xe^x\,dx = -\int_{-1}^{0} xe^x\,dx + \int_{0}^{1} xe^x\,dx,$$

which is the required identity.

To evaluate it, integrate by parts with $u = x$ and $dv/dx = e^x$:

$$\int xe^x\,dx = xe^x - \int e^x\,dx = (x-1)e^x + c.$$

Hence

$$\int_{-1}^{0} xe^x\,dx = \left[(x-1)e^x\right]_{-1}^{0} = -1 - (-2e^{-1}) = \frac{2}{e} - 1,$$
$$\int_{0}^{1} xe^x\,dx = \left[(x-1)e^x\right]_{0}^{1} = 0 - (-1) = 1.$$

The first of these is negative (as it must be, since the integrand is negative there), and

$$\int_{-1}^{1} |xe^x|\,dx = \left(1 - \frac{2}{e}\right) + 1 = 2 - \frac{2}{e}.$$

Part (i)

Let $p(x) = x^3 - 2x^2 - x + 2$. Grouping terms,

$$p(x) = x^2(x-2) - (x-2) = (x-2)(x^2-1) = (x+1)(x-1)(x-2).$$

The only zeros in $[0,4]$ are $x = 1$ and $x = 2$. A positive-leading cubic with simple roots $-1, 1, 2$ is positive between -1 and 1, negative between 1 and 2, and positive beyond 2. So

$$|p(x)| = \begin{cases} p(x), & 0 \le x \le 1, \\ -p(x), & 1 \le x \le 2, \\ p(x), & 2 \le x \le 4. \end{cases}$$

Take the antiderivative

$$F(x) = \tfrac{1}{4}x^4 - \tfrac{2}{3}x^3 - \tfrac{1}{2}x^2 + 2x,$$

for which

$$F(0) = 0, \qquad F(1) = \tfrac{1}{4} - \tfrac{2}{3} - \tfrac{1}{2} + 2 = \tfrac{13}{12},$$
$$F(2) = 4 - \tfrac{16}{3} - 2 + 4 = \tfrac{2}{3}, \qquad F(4) = 64 - \tfrac{128}{3} - 8 + 8 = \tfrac{64}{3}.$$

Ask yourself Why can't we just compute $F(4) - F(0)$ and take its modulus?

Because the modulus has to be applied to the integrand, not to the answer. On $[1,2]$ the cubic is negative, and $F(4) - F(0)$ would subtract that area instead of adding it. So we treat the three pieces separately:

$$\begin{aligned} \int_0^4 |p(x)|\,\mathrm{d}x &= \big(F(1) - F(0)\big) - \big(F(2) - F(1)\big) + \big(F(4) - F(2)\big) \\ &= \tfrac{13}{12} - \big(\tfrac{2}{3} - \tfrac{13}{12}\big) + \big(\tfrac{64}{3} - \tfrac{2}{3}\big) \\ &= \tfrac{13}{12} + \tfrac{5}{12} + \tfrac{62}{3} = \tfrac{3}{2} + \tfrac{62}{3} = \frac{133}{6}. \end{aligned}$$

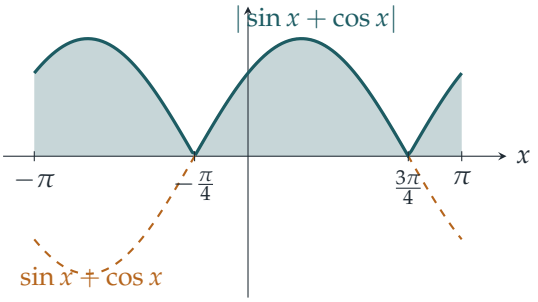
Each bracket is positive, as it should be: $\frac{13}{12}$, $\frac{5}{12}$ and $\frac{62}{3}$ are the three separate areas. (For comparison, $F(4) - F(0) = \frac{64}{3}$, which is smaller by $2 \times \frac{5}{12}$.)

Part (ii)

Using $\sin(x + \frac{\pi}{4}) = \frac{1}{\sqrt{2}}(\sin x + \cos x)$,

$$\sin x + \cos x = \sqrt{2}\,\sin\big(x + \tfrac{\pi}{4}\big).$$

This vanishes at $x = -\frac{\pi}{4}$ and $x = \frac{3\pi}{4}$ in $[-\pi, \pi]$. It is negative on $[-\pi, -\frac{\pi}{4}]$ and on $[\frac{3\pi}{4}, \pi]$, and positive in between.



We could split into three pieces as before, but there is a neater route. Substitute $u = x + \frac{\pi}{4}$:

$$\int_{-\pi}^{\pi} |\sin x + \cos x| \, \mathrm{d}x = \sqrt{2} \int_{-3\pi/4}^{5\pi/4} |\sin u| \, \mathrm{d}u.$$

The function $|\sin u|$ has period π, and the new range has length 2π, so it covers exactly two periods. The integral of a periodic function over a whole number of periods doesn't depend on where we start, so

$$\sqrt{2} \int_{-3\pi/4}^{5\pi/4} |\sin u| \, \mathrm{d}u = \sqrt{2} \times 2 \int_{0}^{\pi} \sin u \, \mathrm{d}u = \sqrt{2} \times 2 \times 2 = 4\sqrt{2}.$$

If you prefer the direct split, $G(x) = \sin x - \cos x$ is an antiderivative of $\sin x + \cos x$, with $G(-\pi) = 1$, $G(-\frac{\pi}{4}) = -\sqrt{2}$, $G(\frac{3\pi}{4}) = \sqrt{2}$ and $G(\pi) = 1$. The three areas are $1 + \sqrt{2}$, $2\sqrt{2}$ and $\sqrt{2} - 1$, which add to $4\sqrt{2}$.

Checks

- Simpson's rule on each piece (where the integrand is smooth) gives 1.264241… for the first integral, matching $2 - 2/\mathrm{e} \approx 1.264241$.
- For (i), numerical integration gives 22.16667, matching $\frac{133}{6}$. It is also slightly more than $F(4) - F(0) = \frac{64}{3} \approx 21.33$, as it must be, since the modulus turns a small negative area into a positive one.
- For (ii), numerical integration gives 5.656854, matching $4\sqrt{2}$. A sanity check: $|\sin x + \cos x|$ has the same shape as $\sqrt{2}|\sin x|$, which averages $\frac{2\sqrt{2}}{\pi}$, and $2\pi \times \frac{2\sqrt{2}}{\pi} = 4\sqrt{2}$.

Common slips

- Integrating straight across a sign change and taking the modulus at the end. This undercounts every negative region.
- Missing the root at $x = 1$ in (i). Always factorise fully, or at least check small integer roots with the factor theorem.
- Sign errors in $\left[(x-1)\mathrm{e}^x\right]_{-1}^{0}$: at $x = -1$ the value is $-2\mathrm{e}^{-1}$, so subtracting it adds $2/\mathrm{e}$.

Beyond STEP

The integral $\int_a^b |f(x)|\,\mathrm{d}x$ is the total area between a curve and the axis. At university it reappears as the L^1 norm of f, one way of measuring the size of a function and the distance between two functions. Proving that $|f|$ is integrable whenever f is, and that $\left|\int f\right| \leq \int |f|$, is a standard early exercise in real analysis. Abbott's *Understanding Analysis* develops the Riemann integral carefully enough to prove both.

WORKED SOLUTION

STEP I 2000, Question 9

Mechanics • Projectiles in Accelerating Frames

STEP I 2000, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q9

The key idea

The range is measured *in the carriage*, so work in the carriage's frame. Since the carriage accelerates north at a, the shell's acceleration relative to the floor is g downwards plus a southwards, which is the direction the cannon fires in. The motion is still a projectile with constant acceleration, just with an extra horizontal term. Once the range is written in terms of 2θ, maximising it is a one-line differentiation.

Solution

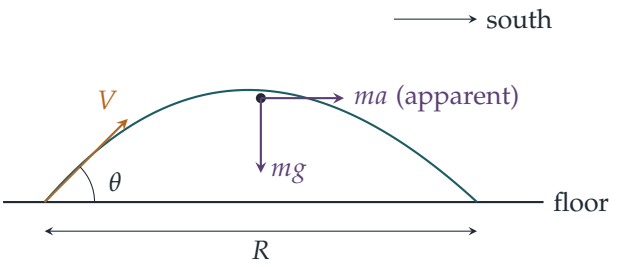
The range in the carriage

Model the shell as a particle and ignore air resistance. Assume it is fired from floor level and lands on the floor, and that the carriage is long enough for it to land inside. Take x horizontal and pointing south (along the firing direction) and y vertically up, both measured relative to the carriage.

In the ground frame the shell has acceleration g downwards and no horizontal acceleration. The carriage has acceleration a northwards. So the shell's acceleration *relative to the carriage* is

$$\ddot{x} = 0 - (-a) = a, \qquad \ddot{y} = -g.$$

Equivalently, in the carriage's frame there is an apparent force ma southwards on a shell of mass m, as well as its weight.



The initial velocity relative to the carriage is $V\cos\theta$ south and $V\sin\theta$ up, so

$$x = Vt\cos\theta + \tfrac{1}{2}at^2, \qquad y = Vt\sin\theta - \tfrac{1}{2}gt^2.$$

The vertical motion is unchanged by a: the shell lands when $y = 0$ with $t > 0$, that is at $T = \dfrac{2V\sin\theta}{g}$. Its range in the carriage is therefore

$$\begin{aligned} R = x(T) &= \frac{2V^2\sin\theta\cos\theta}{g} + \frac{a}{2}\cdot\frac{4V^2\sin^2\theta}{g^2} \\ &= \frac{V^2}{g}\left(\sin 2\theta + \frac{a}{g}(1-\cos 2\theta)\right), \end{aligned}$$

using $2\sin\theta\cos\theta = \sin 2\theta$ and $2\sin^2\theta = 1 - \cos 2\theta$.

Maximising the range

Differentiating with respect to θ,

$$\frac{\mathrm{d}R}{\mathrm{d}\theta} = \frac{2V^2}{g}\left(\cos 2\theta + \frac{a}{g}\sin 2\theta\right),$$

which is zero when

$$\boxed{\tan 2\theta = -\frac{g}{a}.}$$

Ask yourself A negative tangent? Does that mean something has gone wrong?

No. For $0 < \theta < \frac{\pi}{2}$ we have $0 < 2\theta < \pi$, and $\tan 2\theta < 0$ simply puts 2θ in $(\frac{\pi}{2}, \pi)$, so $\theta > \frac{\pi}{4}$. That makes sense: the extra push a acts for as long as the shell is in the air, so it pays to fire a little higher than 45° and buy more flight time. With $a = 0$ we would have $\cos 2\theta = 0$, giving the familiar $\theta = \frac{\pi}{4}$.

To see that this is a maximum, write $\tan\varphi = \dfrac{a}{g}$ with $0 < \varphi < \frac{\pi}{2}$. Then

$$\sin 2\theta - \frac{a}{g}\cos 2\theta = \frac{\sin(2\theta - \varphi)}{\cos\varphi}, \qquad \text{so} \qquad R = \frac{V^2}{g}\left(\frac{\sin(2\theta-\varphi)}{\cos\varphi} + \frac{a}{g}\right).$$

This is largest when $2\theta - \varphi = \frac{\pi}{2}$, that is when $2\theta = \frac{\pi}{2} + \varphi$, which is the root of $\tan 2\theta = -g/a$ in $(\frac{\pi}{2}, \pi)$. So the optimal angle is exactly

$$\theta = \frac{\pi}{4} + \frac{1}{2}\arctan\frac{a}{g},$$

and, since $\cos\varphi = g/\sqrt{g^2+a^2}$, the maximum range is exactly

$$R_{\max} = \frac{V^2}{g}\left(\frac{\sqrt{g^2+a^2}}{g} + \frac{a}{g}\right) = \frac{V^2}{g^2}\left(a + \sqrt{g^2+a^2}\right).$$

Small a

For small u, $\arctan u = u + O(u^3)$. With $u = a/g$,

$$\theta = \frac{\pi}{4} + \frac{1}{2}\left(\frac{a}{g} + O\left(\frac{a^3}{g^3}\right)\right) \approx \frac{\pi}{4} + \frac{a}{2g},$$

as required. For the range, $\sqrt{g^2+a^2} = g\sqrt{1+a^2/g^2} = g + O(a^2/g)$, so

$$R_{\max} = \frac{V^2}{g^2}\left(a + g + O(a^2/g)\right) \approx \frac{V^2}{g} + \frac{V^2 a}{g^2},$$

which is the second required result. Both approximations are correct to first order in a/g; the neglected terms are of order $(a/g)^2$ or smaller.

If you prefer to stay with $\tan 2\theta$, write $2\theta = \frac{\pi}{2} + \varepsilon$. Then $\tan 2\theta = -\cot\varepsilon \approx -1/\varepsilon$, so $\varepsilon \approx a/g$. Substituting $\sin 2\theta = \cos\varepsilon \approx 1$ and $\cos 2\theta = -\sin\varepsilon \approx -\varepsilon$ into R gives $R \approx \frac{V^2}{g}(1 + \frac{a}{g})$ to first order.

Checks

- $a = 0$: the formulae reduce to $\theta = \frac{\pi}{4}$ and $R_{\max} = V^2/g$, the standard results on level ground.
- Dimensions: a/g is dimensionless and V^2/g is a length, as it should be.
- Numerically, with $V = 7$, $g = 9.8$ and $a = 1.3$, stepping the motion in the ground frame and subtracting the carriage's position reproduces the formula for R (e.g. $R \approx 5.478$ at $\theta = 0.7$). A brute-force search over θ for several random a found the maximum where $\tan 2\theta = -g/a$, with $R_{\max}$ agreeing with $\frac{V^2}{g^2}(a + \sqrt{g^2+a^2})$ to 10 significant figures. For $a = 0.05$ the exact optimal angle $0.787949\ldots$ matches $\frac{\pi}{4} + \frac{a}{2g}$ to seven decimal places.

Common slips

- Getting the sign of a wrong. The carriage accelerates *away* from the shell's direction of fire, so relative to the floor the shell gains speed southwards: the term $\frac{1}{2}aT^2$ is added, not subtracted.

- Measuring the range on the ground rather than in the carriage. The question asks for the distance along the carriage floor.
- Taking $2\theta = \arctan(-g/a)$ from a calculator, which gives a negative angle. The correct root lies in $(\frac{\pi}{2}, \pi)$.

Beyond STEP

Working in an accelerating frame and adding an apparent force $-m\mathbf{a}$ is the simplest example of a *fictitious force*. Rotating frames bring the centrifugal and Coriolis forces in the same way. Here the apparent force simply tilts "effective gravity" to magnitude $\sqrt{g^2 + a^2}$, which is one way to see why the answer contains $\sqrt{g^2 + a^2}$. Non-inertial frames are treated clearly in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 2000, Question 10

Mechanics • Collisions and Restitution

STEP I 2000, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q10

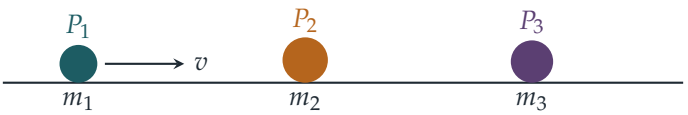
The key idea

Each collision gives us two equations: conservation of momentum and Newton's law of restitution. Here the outcomes are partly known (P_1 stops; P_3 leaves with speed v), so instead of solving for the velocities we solve for the unknown coefficients. The feasibility conditions then come from the one physical constraint on any coefficient of restitution, $0 \leq e \leq 1$, and the final energy becomes a function of the single free mass m_2, which we optimise over the interval those conditions allow.

Solution

Model

We treat the three bodies as particles moving along one line on a smooth table, so no horizontal external forces act and momentum is conserved in each collision. Take the direction of P_1's initial motion as positive.



The first collision

Let P_2 move off with velocity u. Since P_1 is brought to rest,

$$\text{momentum:} \qquad m_1 v = m_2 u \qquad \Longrightarrow \qquad u = \frac{m_1 v}{m_2},$$

$$\text{restitution:} \qquad u - 0 = e(v - 0) \qquad \Longrightarrow \qquad e = \frac{u}{v} = \frac{m_1}{m_2}.$$

The second collision

P_2 now hits the stationary P_3 with speed u. Let P_2's velocity afterwards be w; we are told P_3's is v. Momentum gives

$$m_2u = m_2w + m_3v \quad \Longrightarrow \quad m_1v = m_2w + m_3v \quad \Longrightarrow \quad w = \frac{(m_1 - m_3)v}{m_2},$$

using $m_2u = m_1v$ from the first collision. Restitution (speed of separation $= e' \times$ speed of approach) gives

$$v - w = e'u.$$

Substituting for w and u and multiplying through by m_2/v (which is non-zero):

$$m_2 - (m_1 - m_3) = e'm_1 \quad \Longrightarrow \quad e' = \frac{m_2 + m_3 - m_1}{m_1},$$

which is the required result.

When are such collisions possible?

Any coefficient of restitution lies between 0 and 1. Applying $0 \le e' \le 1$ and multiplying by $m_1 > 0$:

$$0 \le m_2 + m_3 - m_1 \le m_1 \quad \Longleftrightarrow \quad m_1 \le m_2 + m_3 \le 2m_1,$$

which is the inequality asked for. So it is necessary for the collisions to be possible.

? **Ask yourself** Have we used every constraint? What about e?

We also need $0 \le e \le 1$. Since $e = m_1/m_2$ this holds exactly when $m_2 \ge m_1$. That condition is not part of the result we were asked to show (and in fact it makes the left-hand inequality $m_2 + m_3 \ge m_1$ automatic), but it matters in the final part. Putting everything together, the collisions are possible exactly when

$$m_1 \le m_2 \le 2m_1 - m_3.$$

For this interval to be non-empty we need $m_3 \le m_1$. Then $w = (m_1 - m_3)v/m_2 \ge 0$, so after the second collision P_2 keeps moving forwards (or stops) and never comes back to hit P_1, which is at rest. Also $v - w = e'u \ge 0$, so P_2 does not catch P_3 again. These are therefore the only two collisions.

The final energy

Finally P_1 is at rest, P_2 has velocity w and P_3 has velocity v, so the final kinetic energy is

$$E = \tfrac{1}{2}m_2w^2 + \tfrac{1}{2}m_3v^2 = \frac{(m_1 - m_3)^2v^2}{2m_2} + \tfrac{1}{2}m_3v^2.$$

With m_1, m_3 and v fixed, only m_2 can vary, over $m_1 \le m_2 \le 2m_1 - m_3$. The first term is a non-negative constant divided by m_2, so E is a decreasing function of m_2 (constant if $m_1 = m_3$, when the interval is the single point $m_2 = m_1$ anyway).

The largest energy occurs at $m_2 = m_1$ (the first collision is perfectly elastic, $e = 1$):

$$E_{\max} = \frac{(m_1 - m_3)^2 v^2}{2m_1} + \frac{m_3 v^2}{2} = \frac{(m_1^2 - m_1 m_3 + m_3^2)\, v^2}{2m_1}.$$

The smallest energy occurs at $m_2 = 2m_1 - m_3$ (the second collision is perfectly elastic, $e' = 1$):

$$E_{\min} = \frac{(m_1 - m_3)^2 v^2}{2(2m_1 - m_3)} + \frac{m_3 v^2}{2} = \frac{(m_1^2 - 2m_1 m_3 + m_3^2 + 2m_1 m_3 - m_3^2) v^2}{2(2m_1 - m_3)} = \frac{m_1^2 v^2}{2(2m_1 - m_3)}.$$

Checks

- Energy can never increase in collisions, so we need $E_{\max} \le \frac{1}{2} m_1 v^2$. This is $m_1^2 - m_1 m_3 + m_3^2 \le m_1^2$, that is $m_3(m_3 - m_1) \le 0$, true since $m_3 \le m_1$.
- One can check that $E_{\max} - E_{\min} = \dfrac{(m_1 - m_3)^3 v^2}{2m_1(2m_1 - m_3)} \ge 0$, and both answers equal $\frac{1}{2} m_1 v^2$ when $m_1 = m_3$ (then $m_2 = m_1$, both collisions are elastic and no energy is lost).
- Numerically, with $m_1 = 2$, $m_3 = 1$, $v = 1$: the initial energy is 1, $E_{\max} = \frac{3}{4}$ and $E_{\min} = \frac{2}{3}$. Stepping m_2 through $[2, 3]$ in exact arithmetic reproduces these extremes, and confirms that $e, e' \in [0, 1]$ throughout and fail outside.

Common slips

- Optimising over $m_1 - m_3 \le m_2 \le 2m_1 - m_3$ (from e' alone). This gives a "maximum" of $\frac{1}{2} m_1 v^2$, the full initial energy, at a value of m_2 for which $e = m_1 / m_2 > 1$, which is impossible.
- Getting the restitution law the wrong way round: it is the speed of *separation* that equals e times the speed of *approach*.

Beyond STEP

The pattern here, where each constraint carves out an allowed interval for a parameter and we then optimise a monotonic function over that interval, is the simplest case of constrained optimisation, which grows into linear programming and Lagrange multipliers. On the physics side, collisions are much cleaner in the centre-of-mass frame, where the coefficient of restitution simply scales every velocity by $-e$ and the energy lost is $\frac{1}{2}\mu(1 - e^2) \times (\text{relative speed})^2$ with μ the reduced mass. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops this view of collisions.

WORKED SOLUTION

STEP I 2000, Question 11

Mechanics • Ladder Friction Equilibrium

STEP I 2000, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q11

The key idea

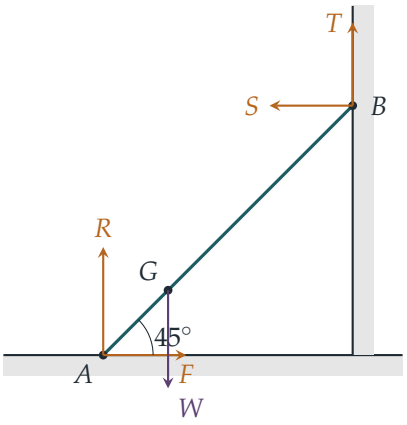
This is a ladder problem with friction at both ends. "Limiting at both A and B" fixes every frictional force in terms of a normal reaction, so the two resolving equations determine all four contact forces. The moment equation is then one condition too many: either it happens to hold, or it fails. For the first part we show that it fails. For the second part we use the position of the extra mass as the unknown that makes it hold.

Solution

The model and the equations

We model the rod as a rigid rod with its weight $W = 5g\,\mathrm{N}$ acting at G. The contacts at A (floor) and B (wall) are both rough. The forces are:

- at A: normal reaction R upwards and friction F horizontally;
- at B: normal reaction S horizontally away from the wall and friction T vertically (taken as upwards, so $T < 0$ would mean downwards);
- the weight W downwards at G.



The only horizontal forces are F and S, and $S \geq 0$ pushes the rod away from the wall. So F must act towards the wall, with

$$F = S. \tag{1}$$

Resolving vertically,

$$R + T = W. \tag{2}$$

For moments about A, put $AB = 0.81$ and $c = \cos 45^\circ = \sin 45^\circ$. Then B is a horizontal distance $0.81c$ and a height $0.81c$ from A, and G is a horizontal distance $0.21c$ from A. Taking clockwise and anticlockwise moments,

$$S\,(0.81c) + T\,(0.81c) = W\,(0.21c),$$

and cancelling c gives

$$0.81\,(S + T) = 0.21\,W. \tag{3}$$

This is where the 45° helps: the factor c cancels from every term.

Friction cannot be limiting at both ends

Suppose friction is limiting at both ends. Then $F = 0.2R$ and $|T| = 1.0\,S = S$.

Ask yourself Which way does the friction at B act when it is limiting?

The natural case is that the rod is about to slip with A moving away from the wall and B sliding down, so that $T = +S$. To be thorough we check both signs.

Case $T = S$. From (1), $S = F = 0.2R$, and (2) gives $R + 0.2R = W$. So $R = \frac{5}{6}W$ and $S = T = \frac{1}{6}W$. Then the left-hand side of (3) is

$$0.81\left(\tfrac{1}{6}W + \tfrac{1}{6}W\right) = 0.27\,W \neq 0.21\,W.$$

So the moment equation fails.

Case $T = -S$. The left-hand side of (3) is 0, but the right-hand side is $0.21W > 0$, which is impossible.

In both cases the rod could not be in equilibrium. So the friction cannot be limiting at both A and B, as required.

(Equilibrium is certainly possible in the non-limiting case: for instance $T = S = \frac{7}{54}W$ satisfies (3) and gives $R = \frac{47}{54}W$, with $F = S \leq 0.2R$.)

Adding the mass at P

Let $AP = x$ m. The extra weight is also $W = 5g$ N, acting at P, so the total weight is $2W$. Equations (1) and (2) become

$$F = S, \qquad R + T = 2W,$$

and the moment equation about A gains a term $W(xc)$. After cancelling c it reads

$$0.81\,(S + T) = 0.21\,W + x\,W. \tag{4}$$

Now friction is limiting at both ends. The case $T = -S$ is again impossible, because then the left-hand side of (4) is 0 but the right-hand side is positive. So $T = S$ and $F = 0.2R$, and therefore

$$S = 0.2R, \qquad R + 0.2R = 2W \quad \Longrightarrow \quad R = \tfrac{5}{3}W, \quad S = T = \tfrac{1}{3}W.$$

Substituting into (4) gives

$$0.81 \times \tfrac{2}{3}\,W = 0.54\,W = 0.21\,W + x\,W,$$

so $x = 0.33$. Hence

$$AP = 0.33 \text{ m}.$$

This lies between 0 and 0.81, so P is on the rod. It is beyond G (and beyond the midpoint), so the extra mass pushes the combined centre of gravity up the rod, and that tends to make a ladder slip.

Checks

- With the mass at P, the forces in units of W are $R = \frac{5}{3}$ and $F = S = T = \frac{1}{3}$. Then $F/R = 0.2$ and $T/S = 1$, so both are limiting. Vertically $\frac{5}{3} + \frac{1}{3} = 2$, and the moments give $0.81 \times \frac{2}{3} = 0.54 = 0.21 + 0.33$.
- The combined centre of gravity is at $\frac{1}{2}(0.21 + 0.33) = 0.27$ m from A. That is exactly the position the first part needed for limiting friction at both ends ($0.27W$ against $0.21W$), which makes sense because the coefficients are the same and only the total weight has doubled.
- The mass and g cancel throughout, so the answer is independent of them. A numerical substitution with $g = 9.8$ and the true 45° moments leaves residuals of order 10^{-15}.

Common slips

- Taking the friction at A in the wrong direction. Since S is the only other horizontal force, F must point towards the wall.
- Forgetting that the friction at B is vertical, so it appears in the vertical resolution and reduces R below the total weight.
- Using the midpoint of the rod instead of G. The rod is not uniform.

Beyond STEP

A rigid body in a plane gives three equilibrium equations, so a problem with four unknown contact forces is *statically indeterminate*. Rigid-body statics alone cannot tell you how the friction is shared between the ends. Imposing "limiting at both ends" adds two more equations, which over-determines the system, and that is exactly why the first part has no solution. In engineering, indeterminacy is resolved by modelling the elasticity of the contacts. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully at first-year level.

WORKED SOLUTION

STEP I 2000, Question 12

Probability and Statistics • Geometric Distribution

STEP I 2000, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q12

The key idea

In each case, ask: *given that every attempt so far has failed, what is the chance that the next one succeeds?* Then multiply the failure probabilities for the first $n-1$ attempts by the success probability on the nth. The three rules give three very different shapes of answer: one where the conditional chance grows, one where it stays fixed from the start, and one where it stays fixed only after the first go.

Throughout, let N be the number of attempts needed, so we want $\mathrm{P}(N = n)$ for $n = 1, 2, 3, \ldots$. Call the key that fits the door the *right* key.

Solution

Part (i)

Here I never repeat a key, so I am trying the keys in a uniformly random order. By symmetry, the right key is equally likely to be in any of the k positions of that order. Hence

$$\mathrm{P}(N = n) = \frac{1}{k} \quad \text{for } n = 1, 2, \ldots, k, \qquad \mathrm{P}(N = n) = 0 \text{ for } n > k.$$

If the symmetry argument feels too slick, multiply conditional probabilities instead. After j failures there are $k - j$ untried keys, one of them right, so for $1 \leq n \leq k$

$$\mathrm{P}(N = n) = \frac{k-1}{k} \cdot \frac{k-2}{k-1} \cdots \frac{k-n+1}{k-n+2} \cdot \frac{1}{k-n+1} = \frac{1}{k},$$

since the product telescopes.

Part (ii)

Now each attempt is a fresh, independent choice from all k keys, so each attempt succeeds with probability $\frac{1}{k}$, whatever has happened before. To need exactly n attempts,

the first $n-1$ must fail and the nth must succeed:

$$\mathrm{P}(N=n)=\left(\frac{k-1}{k}\right)^{n-1}\frac{1}{k}, \qquad n=1,2,3,\ldots$$

This is a geometric distribution with parameter $\frac{1}{k}$, and there is no upper limit on n.

Part (iii)

The first attempt succeeds with probability $\frac{1}{k}$, so $\mathrm{P}(N=1)=\frac{1}{k}$.

Suppose now that attempt m (for some $m \geq 1$) has failed. Then the key tried at attempt m was a wrong key. At attempt $m+1$ I choose uniformly from the other $k-1$ keys.

Ask yourself Does the history before attempt m affect the chance of success at attempt $m+1$?

No. The only thing that matters is which key was excluded, and it is always a *wrong* key. So the $k-1$ keys on offer always include the right one, and attempt $m+1$ succeeds with probability $\frac{1}{k-1}$ and fails with probability $\frac{k-2}{k-1}$, whatever happened earlier.

So for $n \geq 2$: the first attempt fails (probability $\frac{k-1}{k}$), attempts $2,\ldots,n-1$ fail (each with probability $\frac{k-2}{k-1}$), and attempt n succeeds (probability $\frac{1}{k-1}$). Therefore

$$\mathrm{P}(N=n)=\frac{k-1}{k}\left(\frac{k-2}{k-1}\right)^{n-2}\frac{1}{k-1}=\frac{1}{k}\left(\frac{k-2}{k-1}\right)^{n-2}, \qquad n \geq 2,$$

together with $\mathrm{P}(N=1)=\frac{1}{k}$.

Interestingly, $\mathrm{P}(N=2)=\frac{1}{k}$ as well, the same as in part (i). (When $k=2$ the rule forces me to alternate keys, so N is 1 or 2, each with probability $\frac{1}{2}$; the formula gives this if we read 0^0 as 1.)

Checks

- Each distribution sums to 1. In (iii), $\frac{1}{k}+\frac{1}{k}\sum_{j\geq 0}\left(\frac{k-2}{k-1}\right)^j=\frac{1}{k}+\frac{1}{k}(k-1)=1$.
- The means behave sensibly: $\frac{k+1}{2}$ in (i), k in (ii), and $1+\frac{(k-1)^2}{k}$ in (iii). Not repeating the last key helps a little compared with (ii), but much less than never repeating any key.
- We checked all three formulas exactly for $k=2,\ldots,5$ and $n \leq 7$ by enumerating every sequence of choices with fractions. A simulation of 2×10^5 trials of (iii) with $k=5$ gave $\mathrm{P}(N=3)\approx 0.1496$, against the exact $\frac{1}{5}\cdot\frac{3}{4}=0.15$.

Common slips

- In (iii), using $\frac{k-2}{k-1}$ for the first attempt too. The first choice is from all k keys, so the first factor is $\frac{k-1}{k}$.
- In (i), forgetting to say that $\mathrm{P}(N = n) = 0$ when $n > k$.
- In (iii), thinking the excluded key could be the right one. It is only ever a key that has just failed.

Beyond STEP

Part (iii) is a small *Markov chain*. What happens next depends only on the current state (here, which key was tried last), not on the whole history. Spotting a state that makes the past irrelevant is the central trick in the theory of Markov chains, and it turns messy conditional probabilities into simple products. Grimmett and Welsh, *Probability: An Introduction*, introduces Markov chains at exactly this level.

WORKED SOLUTION

STEP I 2000, Question 13

Probability and Statistics • Conditional Probability

STEP I 2000, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q13

The key idea

Everything comes down to tracking one gene from each parent. In (i) each child of two AB parents is AA or BB with the same small probability, and the four children are independent, so we count arrangements and multiply. Part (ii) is a "reverse" conditional probability: we know the outcome (the child) and want the cause (the father). That is Bayes' theorem, and a tree diagram over the father's three possible genotypes organises it.

Solution

Part (i)

Each parent is AB, so each passes on A or B with probability $\frac{1}{2}$, and the two parents act independently. For a single child,

$$\mathrm{P}(AA) = \tfrac{1}{2}\cdot\tfrac{1}{2} = \tfrac{1}{4}, \qquad \mathrm{P}(BB) = \tfrac{1}{4}, \qquad \mathrm{P}(AB) = \tfrac{1}{2},$$

where the last one counts both orders (A from the mother and B from the father, or the other way round).

Different children inherit independently of one another. A particular ordering of the four children with two AA and two BB, such as AA, AA, BB, BB, has probability $\left(\frac{1}{4}\right)^4$. The number of ways to choose which two of the four children are the AA ones is $\binom{4}{2} = 6$. So

$$\mathrm{P}(\text{two } AA \text{ and two } BB) = 6 \times \left(\tfrac{1}{4}\right)^4 = \frac{6}{256} = \frac{3}{128},$$

as required.

Part (ii)

Let F_{AA}, F_{AB}, F_{BB} be the events that my father is AA, AB, BB, and let C be the event that I am AA. We take my father to be a random member of the population, chosen

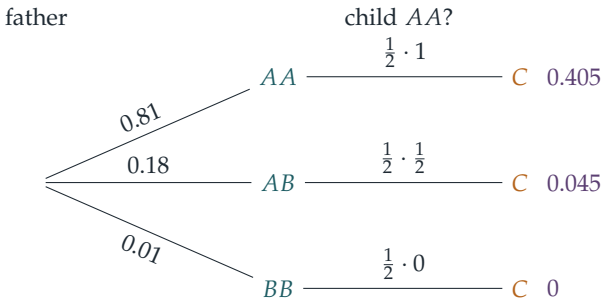
independently of my mother's genotype, so

$$\mathrm{P}(F_{AA}) = 0.81, \qquad \mathrm{P}(F_{AB}) = 0.18, \qquad \mathrm{P}(F_{BB}) = 0.01.$$

Ask yourself Why is 0.18 not already the answer?

Because we have extra information: I am AA. That rules out a BB father completely and makes an AA father more likely, since an AA father always passes on A. So we must condition on C.

For me to be AA I need an A from my mother and an A from my father. My mother is AB, so she gives A with probability $\frac{1}{2}$ whoever my father is. The father gives A with probability 1, $\frac{1}{2}$ or 0 according to his genotype. The tree:



By the law of total probability,

$$\begin{aligned}\mathrm{P}(C) &= 0.81 \cdot \tfrac{1}{2} \cdot 1 + 0.18 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} + 0.01 \cdot \tfrac{1}{2} \cdot 0 \\ &= 0.405 + 0.045 + 0 = 0.45.\end{aligned}$$

Then by Bayes' theorem,

$$\mathrm{P}(F_{AB} \mid C) = \frac{\mathrm{P}(F_{AB})\,\mathrm{P}(C \mid F_{AB})}{\mathrm{P}(C)} = \frac{0.045}{0.45} = \frac{1}{10}.$$

So the probability that my father is AB is $\frac{1}{10}$.

Checks

- In (i), the probabilities for the four children's genotype counts should total 1; the $\binom{4}{2}$ factor is the same one that appears in the multinomial expansion of $\left(\frac{1}{4} + \frac{1}{2} + \frac{1}{4}\right)^4$, of which $3/128 \approx 0.0234$ is one term. An exact enumeration of all 2^8 ways the eight genes can be inherited gives $3/128$.
- In (ii), the answer has dropped from the prior 0.18 to 0.1, as it should once a BB father is excluded and an AA father is favoured. A simulation of 10^6 families gave 0.0998.

Common slips

- Forgetting the factor $\binom{4}{2} = 6$ in (i), and getting $1/256$.
- Answering 0.18 in (ii), or computing $\mathrm{P}(C \mid F_{AB}) = \frac{1}{4}$ instead of $\mathrm{P}(F_{AB} \mid C)$.
- Leaving out the mother's factor $\frac{1}{2}$ in only some branches: it cancels in the end, but only if it is used consistently.

Beyond STEP

The population proportions 81%, 18%, 1% are p^2, $2pq$, q^2 with $p = 0.9$, $q = 0.1$: the Hardy–Weinberg equilibrium, which is preserved from generation to generation under random mating. Inferring a hidden cause from an observed outcome is the heart of Bayesian statistics, and the same calculation underlies genetic counselling and paternity testing. Blitzstein and Hwang, *Introduction to Probability*, has a thorough treatment of conditioning and Bayes' theorem with many examples like this one.

WORKED SOLUTION

STEP I 2000, Question 14

Probability and Statistics • Moments of Uniform Distributions

STEP I 2000, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q14

The key idea

Everything here is a moment of a uniform distribution, and the moments of X on $[-1,1]$ are easy: the odd ones vanish by symmetry and the even ones are $1/(k+1)$. For $Z = Y - X$ we don't need the distribution of Z at all. Expanding $(Y-X)^2$ and $(Y-X)^4$ binomially and using independence turns every expectation into a product of moments we already know.

Solution

Moments of X

X has density $\frac{1}{2}$ on $[-1,1]$, so for a whole number k

$$\mathrm{E}(X^k) = \frac{1}{2}\int_{-1}^{1} x^k \,\mathrm{d}x = \begin{cases} 0, & k \text{ odd (the integrand is odd)}, \\ \dfrac{1}{k+1}, & k \text{ even}. \end{cases}$$

In particular $\mathrm{E}(X) = \mathrm{E}(X^3) = 0$, $\mathrm{E}(X^2) = \frac{1}{3}$ and $\mathrm{E}(X^4) = \frac{1}{5}$. Hence

$$\mathrm{E}(X^2) = \frac{1}{3}, \qquad \mathrm{Var}(X^2) = \mathrm{E}(X^4) - \big(\mathrm{E}(X^2)\big)^2 = \frac{1}{5} - \frac{1}{9} = \frac{4}{45}.$$

Y has the same distribution, so it has the same moments.

The mean of Z^2

Expanding, $Z^2 = Y^2 - 2XY + X^2$. Since X and Y are independent, $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y) = 0$, so

$$\mathrm{E}(Z^2) = \frac{1}{3} - 0 + \frac{1}{3} = \frac{2}{3}.$$

The variance of Z^2

We need $\mathrm{E}(Z^4)$. By the binomial theorem,

$$Z^4 = (Y - X)^4 = Y^4 - 4Y^3X + 6Y^2X^2 - 4YX^3 + X^4.$$

Ask yourself Can we really take the expectation of Y^3X as $\mathrm{E}(Y^3)\,\mathrm{E}(X)$?

Yes: because X and Y are independent, so are any functions of them, such as Y^3 and X, and the expectation of a product of independent random variables is the product of the expectations. Each of the two middle cross terms contains an odd power of X or of Y, so its expectation is zero. Therefore

$$\begin{aligned}\mathrm{E}(Z^4) &= \mathrm{E}(Y^4) + 6\,\mathrm{E}(Y^2)\,\mathrm{E}(X^2) + \mathrm{E}(X^4)\\ &= \frac{1}{5} + 6\cdot\frac{1}{3}\cdot\frac{1}{3} + \frac{1}{5} = \frac{2}{5} + \frac{2}{3} = \frac{16}{15}.\end{aligned}$$

So

$$\mathrm{Var}(Z^2) = \mathrm{E}(Z^4) - \big(\mathrm{E}(Z^2)\big)^2 = \frac{16}{15} - \frac{4}{9} = \frac{48-20}{45} = \frac{28}{45}.$$

Since $\frac{28}{45} = 7 \times \frac{4}{45}$, we have $\mathrm{Var}(Z^2) = 7\,\mathrm{Var}(X^2)$, as required.

Checks

- Z has the triangular density $f(z) = \frac{1}{4}(2 - |z|)$ on $[-2, 2]$. Directly, $\mathrm{E}(Z^4) = 2\int_0^2 \frac{1}{4}z^4(2-z)\,\mathrm{d}z = \frac{1}{2}\left(\frac{64}{5} - \frac{32}{3}\right) = \frac{16}{15}$, agreeing with the expansion.
- $\mathrm{E}(Z^2) = \mathrm{Var}(Y - X) = \mathrm{Var}(Y) + \mathrm{Var}(X) = \frac{1}{3} + \frac{1}{3}$, since $\mathrm{E}(Z) = 0$.
- A simulation of a million pairs (X, Y) gave $\mathrm{Var}(X^2) \approx 0.0889$ and $\mathrm{Var}(Z^2) \approx 0.623$, against $4/45 \approx 0.0889$ and $28/45 \approx 0.622$.

Common slips

- Writing $\mathrm{Var}(Z^2) = \mathrm{Var}(Y^2) + \mathrm{Var}(X^2)$. This ignores the term $-2XY$ inside Z^2, which has mean zero but contributes $4\,\mathrm{Var}(XY) = \frac{4}{9} = \frac{20}{45}$; the slip gives $\frac{8}{45}$ instead of $\frac{28}{45}$.
- Forgetting the binomial coefficient 6 on the Y^2X^2 term.
- Using density 1 instead of $\frac{1}{2}$ on $[-1, 1]$, which doubles every moment.

Beyond STEP

The quantities $\mathrm{E}(X^k)$ are the moments of X, and the neat way to handle sums of independent variables is the moment generating function $M_X(t) = \mathrm{E}(\mathrm{e}^{tX})$, for which $M_{Y-X}(t) = M_Y(t)M_X(-t)$. Expanding this product as a power series in t gives every

moment of Z at once. Moment and characteristic functions are central in a first probability course: they are the main tool in the proof of the central limit theorem. Grimmett and Welsh, *Probability: An Introduction*, develops them clearly.

WORKED SOLUTION

STEP II 2000, Question 1

Pure • Egyptian Fractions

STEP II 2000, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q1

The key idea

The two examples share a pattern: the first denominator is one more than N, and the second is the product of N and that. For the uniqueness parts, clearing denominators and "completing the rectangle" turns $\frac{1}{a}+\frac{1}{b}=\frac{1}{N}$ into a product of two integers equal to N^2. A prime squared has very few factorisations, so there are very few candidates for a and b, and we can simply list them.

Solution

The general result

In $\frac{1}{2}=\frac{1}{3}+\frac{1}{6}$ and $\frac{1}{3}=\frac{1}{4}+\frac{1}{12}$ the denominators are $N+1$ and $N(N+1)$ (since $6=2\times 3$ and $12=3\times 4$). So we guess

$$\frac{1}{N}=\frac{1}{N+1}+\frac{1}{N(N+1)}.$$

To prove it, put the right-hand side over a common denominator:

$$\frac{1}{N+1}+\frac{1}{N(N+1)}=\frac{N+1}{N(N+1)}=\frac{1}{N}.$$

Both denominators are integers greater than 1, so both terms are unit fractions. They are distinct because $N(N+1)>N+1$ when $N>1$. Hence every unit fraction is a sum of two distinct unit fractions.

Uniqueness when N is prime

Multiplying $\frac{1}{N}=\frac{1}{a}+\frac{1}{b}$ by abN gives $ab=Nb+Na$, that is

$$ab-Na-Nb=0 \quad\Longleftrightarrow\quad ab-Na-Nb+N^2=N^2 \quad\Longleftrightarrow\quad (a-N)(b-N)=N^2.$$

Each step is reversible (as a, b, N are non-zero), so the solutions of $(*)$ are exactly the solutions of $(a-N)(b-N)=N^2$.

Ask yourself Could $a - N$ and $b - N$ both be negative?

No. Since $\frac{1}{b} > 0$, we have $\frac{1}{a} < \frac{1}{N}$, so $a > N$; similarly $b > N$. So $a - N$ and $b - N$ are *positive* integers whose product is N^2. (It's worth noticing: without this, the pair $(-1, -N^2)$ would be another factorisation to rule out.)

Now let $N = p$ be prime. The positive divisors of p^2 are only 1, p and p^2, so the pair $(a - p,\ b - p)$ must be one of

$$(1,\ p^2), \qquad (p,\ p), \qquad (p^2,\ 1).$$

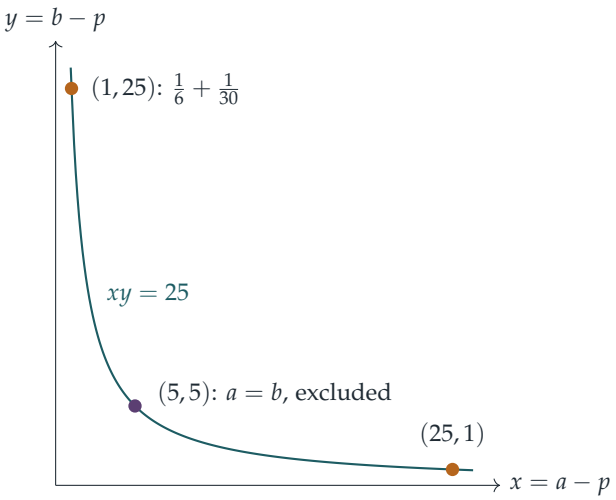
- (p, p) gives $a = b = 2p$, i.e. $\frac{1}{p} = \frac{1}{2p} + \frac{1}{2p}$: the two unit fractions are not distinct, so this is excluded.
- $(1, p^2)$ gives $a = p + 1$, $b = p^2 + p = p(p + 1)$, which is the representation found above; $(p^2, 1)$ is the same pair in the other order.

So, apart from the order of the two terms, the only way is

$$\frac{1}{p} = \frac{1}{p+1} + \frac{1}{p(p+1)},$$

as required.

The picture shows why: the solutions are the lattice points on the hyperbola $xy = p^2$ in the positive quadrant, where $x = a - p$, $y = b - p$. Here $p = 5$.



Fractions of the form $2/N$

Now suppose $\frac{2}{N} = \frac{1}{a} + \frac{1}{b}$ with $N = p$ an odd prime. Multiplying by abp gives $2ab = pa + pb$. To complete the rectangle we first multiply by 2:

$$4ab - 2pa - 2pb = 0 \quad \Longleftrightarrow \quad 4ab - 2pa - 2pb + p^2 = p^2 \quad \Longleftrightarrow \quad (2a - p)(2b - p) = p^2.$$

Ask yourself Why multiply by 2 before factorising?

Factorising $2ab - pa - pb$ directly would need terms like $(a - \frac{p}{2})$, which aren't integers. Multiplying by 2 keeps everything in whole numbers, so the factors $2a - p$ and $2b - p$ are integers and we can use divisibility.

As before, $\frac{1}{a} < \frac{2}{p}$, so $2a > p$ and $2a - p > 0$; likewise $2b - p > 0$. The positive divisors of p^2 are 1, p, p^2, so $(2a - p, 2b - p)$ is one of

$$(1, p^2), \qquad (p, p), \qquad (p^2, 1).$$

- (p, p) gives $a = b = p$, i.e. $\frac{2}{p} = \frac{1}{p} + \frac{1}{p}$: not distinct, so excluded.
- $(1, p^2)$ gives

  $$a = \frac{p+1}{2}, \qquad b = \frac{p^2 + p}{2} = \frac{p(p+1)}{2}.$$

  Since p is odd, $p + 1$ is even, so both are integers. Since $p \geq 3$, $a \geq 2$, so both are greater than 1, and $b = pa > a$, so they are distinct. The case $(p^2, 1)$ is the same pair in the other order.

Checking directly:

$$\frac{2}{p+1} + \frac{2}{p(p+1)} = \frac{2(p+1)}{p(p+1)} = \frac{2}{p}.$$

So $\frac{2}{p}$ can be written as a sum of two distinct unit fractions, and the only way (up to order) is

$$\frac{2}{p} = \frac{1}{\frac{1}{2}(p+1)} + \frac{1}{\frac{1}{2}p(p+1)},$$

which proves the final result.

Checks

- $p = 3$: $\frac{2}{3} = \frac{1}{2} + \frac{1}{6}$; $p = 5$: $\frac{2}{5} = \frac{1}{3} + \frac{1}{15}$; $p = 7$: $\frac{2}{7} = \frac{1}{4} + \frac{1}{28}$.
- A computer search over all $N < 300$ found exactly one representation of $\frac{1}{N}$ for every prime N and exactly one of $\frac{2}{N}$ for every odd prime N, matching the formulas above.
- Primality really matters: $N = 6$ has four, e.g. $\frac{1}{6} = \frac{1}{7} + \frac{1}{42} = \frac{1}{8} + \frac{1}{24} = \frac{1}{9} + \frac{1}{18} = \frac{1}{10} + \frac{1}{15}$, one for each factorisation $36 = 1 \times 36 = 2 \times 18 = 3 \times 12 = 4 \times 9$ (the pair 6×6 gives $a = b$).
- Why $p > 2$: for $p = 2$, $\frac{2}{2} = 1$ is not of the form $\frac{1}{a} + \frac{1}{b}$ with distinct $a, b > 1$, and indeed $(p + 1)/2$ is not an integer.

Common slips

- Forgetting to show $a - N > 0$ and $b - N > 0$, and so ignoring negative factorisations of N^2.

- Not ruling out the factorisation $N \times N$, which gives equal unit fractions.
- In the $2/N$ case, not checking that $\frac{1}{2}(p+1)$ is an integer (this is exactly where $p \neq 2$ is used).

Beyond STEP

Writing fractions as sums of distinct unit fractions goes back to the ancient Egyptians, and "Egyptian fractions" still hold open problems: the Erdős–Straus conjecture says $\frac{4}{N}$ is always a sum of three unit fractions for $N \geq 2$, and nobody has proved it. The trick of turning $xy - Nx - Ny = 0$ into $(x-N)(y-N) = N^2$ is a standard tool for Diophantine equations, and counting solutions becomes counting divisors. Davenport's *The Higher Arithmetic* is a lovely next step into divisors and primes.

WORKED SOLUTION

STEP II 2000, Question 2

Pure • Repeated Roots of Polynomials

STEP II 2000, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q2

The key idea

A repeated factor survives differentiation, but loses one power each time. So a fourth-power factor $(x-a)^4$ forces p and its first three derivatives all to vanish at a. For the sextic, the third derivative is only a cubic, and its roots give a very short list of candidates for a. We test those, then choose k last, since k disappears as soon as we differentiate.

Solution

The first result

Suppose $(x-a)^2$ is a factor, so $\mathrm{p}(x) = (x-a)^2\,\mathrm{q}(x)$ for some polynomial q. By the product rule,

$$\mathrm{p}'(x) = 2(x-a)\,\mathrm{q}(x) + (x-a)^2\,\mathrm{q}'(x) = (x-a)\big[\,2\mathrm{q}(x) + (x-a)\mathrm{q}'(x)\,\big].$$

Putting $x = a$ gives $\mathrm{p}'(a) = 0$, as required.

The corresponding result

The same calculation works for any power. If $\mathrm{f}(x) = (x-a)^m\,\mathrm{g}(x)$ with $m \geq 1$ and g a polynomial, then

$$\mathrm{f}'(x) = (x-a)^{m-1}\big[\,m\,\mathrm{g}(x) + (x-a)\,\mathrm{g}'(x)\,\big],$$

and the bracket is again a polynomial. So *if* $(x-a)^m$ *divides* f, *then* $(x-a)^{m-1}$ *divides* f'.

Now let $(x-a)^4$ be a factor of $\mathrm{p}(x)$. Applying this fact three times,

$$(x-a)^3 \text{ divides } \mathrm{p}'(x), \qquad (x-a)^2 \text{ divides } \mathrm{p}''(x), \qquad (x-a) \text{ divides } \mathrm{p}'''(x).$$

Together with $(x-a)^4$ dividing $\mathrm{p}(x)$ itself, setting $x = a$ gives

$$\mathrm{p}(a) = \mathrm{p}'(a) = \mathrm{p}''(a) = \mathrm{p}'''(a) = 0.$$

Finding k

Let $\mathrm{p}(x) = x^6 + 4x^5 - 5x^4 - 40x^3 - 40x^2 + 32x + k$. Differentiating three times,

$$\begin{aligned}
\mathrm{p}'(x) &= 6x^5 + 20x^4 - 20x^3 - 120x^2 - 80x + 32, \\
\mathrm{p}''(x) &= 30x^4 + 80x^3 - 60x^2 - 240x - 80 = 10\,(3x^4 + 8x^3 - 6x^2 - 24x - 8), \\
\mathrm{p}'''(x) &= 120x^3 + 240x^2 - 120x - 240 = 120\,(x^3 + 2x^2 - x - 2).
\end{aligned}$$

Ask yourself Which of the four conditions should we use first?

Use the one that is easiest to solve. The constant k has disappeared from p''', and p''' is only a cubic, which factorises by grouping:

$$x^3 + 2x^2 - x - 2 = x^2(x+2) - (x+2) = (x+2)(x-1)(x+1).$$

So by the corresponding result, a must be -2, 1 or -1. We test each in p'':

$$\begin{aligned}
\tfrac{1}{10}\mathrm{p}''(1) &= 3 + 8 - 6 - 24 - 8 = -27 \neq 0, \\
\tfrac{1}{10}\mathrm{p}''(-1) &= 3 - 8 - 6 + 24 - 8 = 5 \neq 0, \\
\tfrac{1}{10}\mathrm{p}''(-2) &= 48 - 64 - 24 + 48 - 8 = 0.
\end{aligned}$$

Only $a = -2$ survives. It also passes the p' test:

$$\mathrm{p}'(-2) = -192 + 320 + 160 - 480 + 160 + 32 = 0.$$

Finally, $\mathrm{p}(-2) = 0$ fixes k:

$$\mathrm{p}(-2) = 64 - 128 - 80 + 320 - 160 - 64 + k = k - 48 = 0, \qquad \text{so} \qquad \boxed{k = 48}.$$

Ask yourself We have only used *necessary* conditions. How do we know that $(x+2)^4$ really is a factor when $k = 48$?

The four conditions $\mathrm{p}(a) = \mathrm{p}'(a) = \mathrm{p}''(a) = \mathrm{p}'''(a) = 0$ do in fact imply that $(x-a)^4$ is a factor (by Taylor's theorem for polynomials). The quickest confirmation, though, is to multiply out. Since $(x+2)^4 = x^4 + 8x^3 + 24x^2 + 32x + 16$, the quotient must be a monic quadratic with constant term $48/16 = 3$. Matching the x^5 coefficient gives $x^2 - 4x + 3$, and indeed

$$(x+2)^4(x^2 - 4x + 3) = x^6 + 4x^5 - 5x^4 - 40x^3 - 40x^2 + 32x + 48.$$

So, with $k = 48$,

$$x^6 + 4x^5 - 5x^4 - 40x^3 - 40x^2 + 32x + 48 = (x+2)^4(x-1)(x-3),$$

and the factor has the required form, with $a = -2$.

Checks

- Expand $(x+2)^4(x-1)(x-3)$ in full: every coefficient matches the given polynomial, and the constant term is $16 \times 3 = 48$.
- Test the sum of the coefficients: $\mathrm{p}(1) = 1+4-5-40-40+32+48 = 0$, as it must be since $(x-1)$ is a factor.
- A computer confirmed the general result on random polynomials $(x-a)^4\mathrm{q}(x)$, found the roots $-2, \pm 1$ of p''' with p'' vanishing only at -2, and gave $k = 48$.

Common slips

- Starting with $\mathrm{p}(a) = 0$ or $\mathrm{p}'(a) = 0$. These involve a quintic or k itself, and lead nowhere quickly. Start from the lowest-degree derivative.
- Stopping at a root of p'''. The candidates $a = \pm 1$ are roots of p''', but they are not roots of p''.
- Only stating the corresponding result as "$\mathrm{p}'''(a) = 0$". Make it the full set $\mathrm{p}(a) = \mathrm{p}'(a) = \mathrm{p}''(a) = \mathrm{p}'''(a) = 0$, and prove it.

Beyond STEP

The converse holds too: $(x-a)^m$ divides p exactly when p and its first $m-1$ derivatives vanish at a. This is because a polynomial equals its own Taylor expansion about a. In a first-year algebra course, the same idea defines the *multiplicity* of a root. It also gives a test for repeated roots: compute $\gcd(\mathrm{p}, \mathrm{p}')$ with Euclid's algorithm. Over fields such as the integers modulo a prime, the formal derivative behaves in surprising ways. Allenby's *Rings, Fields and Groups* develops this.

WORKED SOLUTION

STEP II 2000, Question 3

Pure • Error Propagation

STEP II 2000, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q3

The key idea

The cosine rule gives a^2 exactly in terms of b, c and A, so the whole question is about how a responds when each of these is nudged slightly. Expand every ingredient to first order in the small quantities, throw away products of small quantities, and finally take a square root using $\sqrt{1+x} \approx 1 + \frac{1}{2}x$. For the range, notice that η is a linear combination of independent errors, so it is largest when each term is pushed to its extreme with the helpful sign.

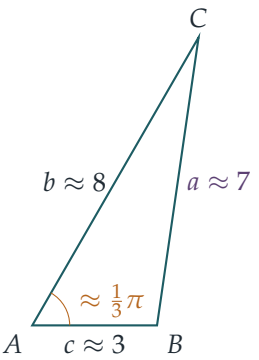
Solution

The approximation for a

With the unperturbed values $b = 8$, $c = 3$, $A = \frac{1}{3}\pi$, the cosine rule gives

$$a^2 = 64 + 9 - 2 \cdot 8 \cdot 3 \cdot \tfrac{1}{2} = 49,$$

so $a = 7$. This explains the leading term.



Now perturb. Throughout, "≈" means we keep terms of first order in $\epsilon_1, \epsilon_2, \epsilon_3$ and

discard anything quadratic or higher, such as ϵ_1^2 or $\epsilon_2\epsilon_3$. Then

$$\begin{aligned} b^2 &= 64 + 16\epsilon_1 + \epsilon_1^2 \approx 64 + 16\epsilon_1, \\ c^2 &= 9 + 6\epsilon_2 + \epsilon_2^2 \approx 9 + 6\epsilon_2, \\ bc &= 24 + 3\epsilon_1 + 8\epsilon_2 + \epsilon_1\epsilon_2 \approx 24 + 3\epsilon_1 + 8\epsilon_2. \end{aligned}$$

For the angle, the compound-angle formula and the small-angle approximations $\cos\epsilon_3 \approx 1$, $\sin\epsilon_3 \approx \epsilon_3$ give

$$\cos A = \cos\tfrac{1}{3}\pi\cos\epsilon_3 - \sin\tfrac{1}{3}\pi\sin\epsilon_3 \approx \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\epsilon_3.$$

(The error in $\cos\epsilon_3 \approx 1$ is of order ϵ_3^2, so it really is negligible at this order.) Multiplying out and again dropping products of small quantities,

$$2bc\cos A \approx 2\big(24 + 3\epsilon_1 + 8\epsilon_2\big)\Big(\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\epsilon_3\Big) \approx 24 + 3\epsilon_1 + 8\epsilon_2 - 24\sqrt{3}\,\epsilon_3.$$

Putting the pieces into the cosine rule,

$$\begin{aligned} a^2 &\approx (64 + 16\epsilon_1) + (9 + 6\epsilon_2) - (24 + 3\epsilon_1 + 8\epsilon_2 - 24\sqrt{3}\,\epsilon_3) \\ &= 49 + 13\epsilon_1 - 2\epsilon_2 + 24\sqrt{3}\,\epsilon_3. \end{aligned}$$

Write $\delta = 13\epsilon_1 - 2\epsilon_2 + 24\sqrt{3}\,\epsilon_3$, which is small. Since $a > 0$,

$$a = \sqrt{49 + \delta} = 7\Big(1 + \frac{\delta}{49}\Big)^{1/2} \approx 7\Big(1 + \frac{\delta}{98}\Big) = 7 + \frac{\delta}{14},$$

using the binomial expansion $(1+x)^{1/2} \approx 1 + \frac{1}{2}x$ for small x. Hence $a \approx 7 + \eta$ with

$$\eta = \frac{13\epsilon_1 - 2\epsilon_2 + 24\sqrt{3}\,\epsilon_3}{14},$$

which is the required result.

Ask yourself Could we have avoided expanding products by hand?

Yes: differentiate $a^2 = b^2 + c^2 - 2bc\cos A$ to get $2a\,\delta a = (2b - 2c\cos A)\,\delta b + (2c - 2b\cos A)\,\delta c + 2bc\sin A\,\delta A$. At $a = 7$, $b = 8$, $c = 3$, $A = \frac{1}{3}\pi$ this reads $14\,\delta a = 13\,\delta b - 2\,\delta c + 24\sqrt{3}\,\delta A$, the same answer in one line. Both methods are just first-order Taylor expansion.

The range of η

The bound on ϵ_2 is printed in the paper as $4\cdot 9 \times 10^{-2}$, with the old-style raised decimal point, so it means $|\epsilon_2| \le 4.9 \times 10^{-2}$.

The three errors vary independently, so by the triangle inequality

$$|\eta| \le \frac{13|\epsilon_1| + 2|\epsilon_2| + 24\sqrt{3}\,|\epsilon_3|}{14} \le \frac{13 \times 2 \times 10^{-3} + 2 \times 4.9 \times 10^{-2} + 24\sqrt{3} \times \sqrt{3} \times 10^{-3}}{14}.$$

The numerator is $0.026 + 0.098 + 0.072 = 0.196$, so

$$|\eta| \le \frac{0.196}{14} = 0.014.$$

The bound is attained: taking $\epsilon_1 = 2 \times 10^{-3}$, $\epsilon_2 = -4.9 \times 10^{-2}$, $\epsilon_3 = \sqrt{3} \times 10^{-3}$ makes every term of η positive and as large as possible, giving $\eta = 0.014$; reversing all three signs gives $\eta = -0.014$. Since η is a continuous (indeed linear) function of the errors, and the set of allowed errors is a box, every value in between is also taken. So the range is

$$-0.014 \le \eta \le 0.014,$$

that is, a lies in roughly 7 ± 0.014.

Checks

- With all errors zero the triangle is 8, 3, 7 with a 60° angle, and indeed $64 + 9 - 24 = 49$.
- Taking the extreme errors above, the exact cosine rule gives $a \approx 7.0141$ and $a \approx 6.9861$, very close to 7 ± 0.014; the small discrepancy is the discarded second-order terms.
- Numerically, the partial derivative of a with respect to A at the base values is $2.9692\ldots = 24\sqrt{3}/14$, as predicted.

Common slips

- Forgetting the square root at the end: $a^2 \approx 49 + \delta$ gives $a \approx 7 + \delta/14$, not $7 + \delta$.
- Taking all three errors positive for the maximum: ϵ_2 enters with a minus sign, so it must be negative to push η up.
- Reading $4 \cdot 9 \times 10^{-2}$ as 36×10^{-2}; that is not a small error at all for a side of length 3.

Beyond STEP

This is error propagation through the total differential: for a function f of several variables, $\delta f \approx \sum (\partial f/\partial x_i)\, \delta x_i$, and the worst case is $\sum |\partial f/\partial x_i|\, |\delta x_i|$. In a first-year course this becomes the multivariable Taylor theorem, with an explicit bound on the second-order remainder that tells you exactly how good "$\approx$" is. Experimental scientists often use instead the root-sum-of-squares rule for independent random errors. Bartle and Sherbert, *Introduction to Real Analysis*, covers Taylor's theorem with remainder carefully.

WORKED SOLUTION

STEP II 2000, Question 4

Pure • Arguments and Arctangent Identities

STEP II 2000, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q4

The key idea

The first two results say that when you multiply complex numbers, their arguments add. So an identity of the form "sum of arctangents $= \pi/4$" is really a statement about a product of Gaussian integers like $5 - \mathrm{i}$ and $1 + \mathrm{i}$: each factor $a + b\mathrm{i}$ with $a > 0$ contributes an angle $\arctan(b/a)$, and $\pi/4$ is the argument of anything on the line $y = x$. The work is to multiply out carefully, then to check that the angles really are equal and not just equal up to a multiple of 2π.

Solution

The product formula and de Moivre's theorem

Expanding and using $\mathrm{i}^2 = -1$,

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)(\cos\phi + \mathrm{i}\sin\phi) &= (\cos\theta\cos\phi - \sin\theta\sin\phi) + \mathrm{i}(\sin\theta\cos\phi + \cos\theta\sin\phi)\\ &= \cos(\theta+\phi) + \mathrm{i}\sin(\theta+\phi),\end{aligned}$$

by the compound-angle formulae for $\cos(A+B)$ and $\sin(A+B)$.

For the second result we use induction on n. When $n = 1$ both sides are $\cos\theta + \mathrm{i}\sin\theta$. Suppose the result holds for $n = k$. Then

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)^{k+1} &= (\cos k\theta + \mathrm{i}\sin k\theta)(\cos\theta + \mathrm{i}\sin\theta)\\ &= \cos(k\theta+\theta) + \mathrm{i}\sin(k\theta+\theta),\end{aligned}$$

using the induction hypothesis and then the product formula with $\phi = k\theta$. So the result holds for $n = k+1$, and hence for every positive integer n.

It is convenient to write $\operatorname{cis}\theta$ for $\cos\theta + \mathrm{i}\sin\theta$. Then we have shown $\operatorname{cis}\theta\operatorname{cis}\phi = \operatorname{cis}(\theta+\phi)$ and $(\operatorname{cis}\theta)^n = \operatorname{cis} n\theta$.

A fact about angles

If $a > 0$ and b is real, then

$$a + b\mathrm{i} = \sqrt{a^2 + b^2}\,\operatorname{cis}\bigl(\arctan(b/a)\bigr),$$

because the angle $\alpha = \arctan(b/a)$ lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\cos\alpha > 0$, and $\tan\alpha = b/a$; these force $\cos\alpha = a/\sqrt{a^2+b^2}$ and $\sin\alpha = b/\sqrt{a^2+b^2}$.

Also, if $R\operatorname{cis} S = R'\operatorname{cis} T$ with $R, R' > 0$, then taking moduli gives $R = R'$, so $\cos S = \cos T$ and $\sin S = \sin T$, and therefore $S - T$ is a multiple of 2π.

The first identity

Let $\alpha = \arctan\frac{1}{5}$. Then $5 - \mathrm{i} = \sqrt{26}\operatorname{cis}(-\alpha)$ and $1 + \mathrm{i} = \sqrt{2}\operatorname{cis}\frac{\pi}{4}$, so by the results above

$$(5-\mathrm{i})^2(1+\mathrm{i}) = 26\sqrt{2}\,\operatorname{cis}\bigl(\tfrac{\pi}{4} - 2\alpha\bigr).$$

On the other hand, multiplying out directly,

$$(5-\mathrm{i})^2 = 24 - 10\mathrm{i}, \qquad (24-10\mathrm{i})(1+\mathrm{i}) = 24 + 24\mathrm{i} - 10\mathrm{i} + 10 = 34 + 14\mathrm{i},$$

and $34 + 14\mathrm{i} = \sqrt{1352}\operatorname{cis}\beta$ with $\beta = \arctan\frac{14}{34} = \arctan\frac{7}{17}$. Comparing, $\frac{\pi}{4} - 2\alpha$ and β differ by a multiple of 2π.

Ask yourself How do we know the multiple of 2π is zero?

We pin both angles down. Since $0 < \frac{1}{5} < 1$, we have $0 < \alpha < \frac{\pi}{4}$, so $-\frac{\pi}{4} < \frac{\pi}{4} - 2\alpha < \frac{\pi}{4}$. Also $0 < \beta < \frac{\pi}{2}$. Two angles in $(-\frac{\pi}{4}, \frac{\pi}{2})$ differ by less than 2π, so the multiple is zero and $\beta = \frac{\pi}{4} - 2\alpha$. That is,

$$\arctan\tfrac{7}{17} + 2\arctan\tfrac{1}{5} = \frac{\pi}{4},$$

as required.

The second identity

The same idea suggests looking at $(4+\mathrm{i})^3(20+\mathrm{i})(1985+\mathrm{i})$, whose argument should be the left-hand side. Multiplying step by step:

$$\begin{aligned}(4+\mathrm{i})^2 &= 15 + 8\mathrm{i},\\ (4+\mathrm{i})^3 &= (15+8\mathrm{i})(4+\mathrm{i}) = 52 + 47\mathrm{i},\\ (4+\mathrm{i})^3(20+\mathrm{i}) &= (52+47\mathrm{i})(20+\mathrm{i}) = 993 + 992\mathrm{i}.\end{aligned}$$

This is already very close to the line $y = x$, and the last factor finishes the job:

$$\begin{aligned}(993+992\mathrm{i})(1985+\mathrm{i}) &= (993\times 1985 - 992) + (993 + 992\times 1985)\mathrm{i}\\ &= 1970113 + 1970113\,\mathrm{i} = 1970113\sqrt{2}\operatorname{cis}\tfrac{\pi}{4}.\end{aligned}$$

(If you wondered where 1985 came from: asking for $(993+992\mathrm{i})(a+\mathrm{i})$ to have equal real and imaginary parts gives $993a-992=992a+993$, so $a=1985$.)

Now let $p=\arctan\frac{1}{4}$, $q=\arctan\frac{1}{20}$ and $r=\arctan\frac{1}{1985}$. By the fact about angles, each of $4+\mathrm{i}$, $20+\mathrm{i}$, $1985+\mathrm{i}$ is a positive real multiple of $\operatorname{cis}p$, $\operatorname{cis}q$, $\operatorname{cis}r$ respectively, so the product is a positive real multiple of $\operatorname{cis}(3p+q+r)$. Comparing with the line above, $3p+q+r-\frac{\pi}{4}$ is a multiple of 2π.

Each of p,q,r lies in $(0,\frac{\pi}{4})$ because its tangent lies in $(0,1)$, so $0<3p+q+r<\frac{5\pi}{4}$. Hence $3p+q+r$ and $\frac{\pi}{4}$ differ by less than 2π, the multiple is zero, and

$$3\arctan\tfrac{1}{4}+\arctan\tfrac{1}{20}+\arctan\tfrac{1}{1985}=\frac{\pi}{4}.$$

Checks

Moduli must match: $|34+14\mathrm{i}|^2=1156+196=1352=26^2\times 2$, as it should. For the second product, $17^3\times 401\times(1985^2+1)$ should equal 2×1970113^2, and it does. Numerically, both left-hand sides evaluate to $0.785398163\ldots=\pi/4$ on a calculator; for instance $\arctan\frac{7}{17}\approx 0.390607$ and $\frac{\pi}{4}-2\arctan\frac{1}{5}\approx 0.390607$.

Common slips

- Concluding equality of angles from equality of complex numbers without ruling out a difference of $2\pi k$. The range argument is a genuine part of the proof.
- Sign errors in $(5-\mathrm{i})^2$: it is $24-10\mathrm{i}$, so $5-\mathrm{i}$ contributes $-\arctan\frac{1}{5}$, which is why the arctangent appears with a plus sign on the other side.
- Arithmetic slips in the large product; check by the moduli, as above.

Beyond STEP

Identities like these are Machin-type formulae; John Machin's $\frac{\pi}{4}=4\arctan\frac{1}{5}-\arctan\frac{1}{239}$, found the same way from $(5+\mathrm{i})^4(239-\mathrm{i})$, was used to compute π to a hundred places in 1706. The underlying structure is the arithmetic of the Gaussian integers $\mathbb{Z}[\mathrm{i}]$, where unique factorisation into Gaussian primes explains which such identities can exist. For the geometry of multiplication as rotation, read Needham, *Visual Complex Analysis*; for Gaussian integers, Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP II 2000, Question 5

Pure • Least Squares Approximation

STEP II 2000, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q5

The key idea

Once you expand the square, the quantity to be minimised is just a quadratic in λ, with coefficients that are fixed numbers (integrals involving f). Completing the square gives the best λ and the smallest error in one go. For the sine example, the point is that $\sin(\pi x/n)$ is almost a straight line through the origin when n is large, and the small-angle approximations make this precise.

Solution

The best value of λ

Write

$$J = \int_0^1 xf(x)\,\mathrm{d}x, \qquad K = \int_0^1 \big(f(x)\big)^2\,\mathrm{d}x.$$

These are constants, not depending on λ. Expanding the square and using $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$,

$$\begin{aligned} I(\lambda) = \int_0^1 \big(f(x) - \lambda x\big)^2\,\mathrm{d}x &= K - 2\lambda J + \tfrac{1}{3}\lambda^2 \\ &= \tfrac{1}{3}\big(\lambda - 3J\big)^2 + K - 3J^2. \end{aligned}$$

The squared term is never negative and is zero only when $\lambda = 3J$, so $I(\lambda)$ is smallest exactly when

$$\lambda = 3\int_0^1 xf(x)\,\mathrm{d}x,$$

as required. (Equivalently, $I'(\lambda) = -2J + \frac{2}{3}\lambda = 0$ gives $\lambda = 3J$, and $I''(\lambda) = \frac{2}{3} > 0$ confirms a minimum.)

The residual error

R^2 is the minimum value of I, which the completed square shows is $K - 3J^2$. Since $J = \lambda/3$, we have $3J^2 = \frac{1}{3}\lambda^2$, so

$$R^2 = \int_0^1 \big(f(x)\big)^2 \,\mathrm{d}x - \tfrac{1}{3}\lambda^2,$$

as required. (If you found λ by differentiating instead, substitute $J = \lambda/3$ into $K - 2\lambda J + \frac{1}{3}\lambda^2$ to get $K - \frac{2}{3}\lambda^2 + \frac{1}{3}\lambda^2$, the same thing.)

The case $f(x) = \sin(\pi x/n)$

To keep the algebra tidy, put $a = \pi/n$, so $f(x) = \sin ax$ and $a \to 0$ as $n \to \infty$. Integrating by parts with $u = x$ and $\mathrm{d}v = \sin ax\,\mathrm{d}x$,

$$\begin{aligned}\int_0^1 x \sin ax \,\mathrm{d}x &= \Big[-\frac{x\cos ax}{a}\Big]_0^1 + \frac{1}{a}\int_0^1 \cos ax \,\mathrm{d}x \\ &= -\frac{\cos a}{a} + \frac{\sin a}{a^2},\end{aligned}$$

so

$$\lambda = \frac{3(\sin a - a\cos a)}{a^2}.$$

Ask yourself Why do we need the θ^3 term in the approximation to $\sin\theta$, rather than just $\sin\theta \approx \theta$?

With only $\sin a \approx a$ and $\cos a \approx 1$, the numerator becomes $a - a = 0$, which tells us nothing: the leading terms cancel, and the answer lives in the next order. Using both approximations from the hint,

$$\sin a - a\cos a \approx \Big(a - \tfrac{1}{6}a^3\Big) - a\Big(1 - \tfrac{1}{2}a^2\Big) = \tfrac{1}{3}a^3,$$

and therefore

$$\lambda \approx \frac{3\cdot\frac{1}{3}a^3}{a^2} = a = \frac{\pi}{n}$$

for large n. This is result (i).

For (ii) we also need K. Using $\sin^2 ax = \frac{1}{2}(1 - \cos 2ax)$,

$$K = \int_0^1 \sin^2 ax \,\mathrm{d}x = \frac{1}{2} - \frac{\sin 2a}{4a}.$$

As $n \to \infty$, $a \to 0$ and $\dfrac{\sin 2a}{2a} \to 1$, so $K \to \frac{1}{2} - \frac{1}{2} = 0$. Also $\lambda \to 0$, since $\lambda \approx \pi/n$. Hence

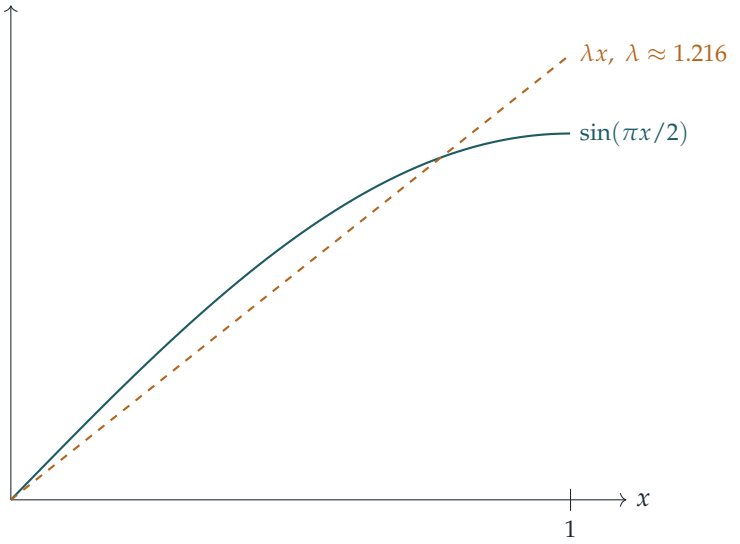
$$R^2 = K - \tfrac{1}{3}\lambda^2 \to 0 - 0 = 0,$$

and because $R \geq 0$ we get $\lim_{n\to\infty} R = 0$. This is result (ii).

In fact the cancellation is much stronger than this. With $\sin 2a \approx 2a - \frac{4}{3}a^3$ we get $K \approx \frac{1}{2} - \big(\frac{1}{2} - \frac{1}{3}a^2\big) = \frac{1}{3}a^2$, and $\frac{1}{3}\lambda^2 \approx \frac{1}{3}a^2$ too. So R^2 is not just small: it is small compared with K, the size of f itself. The straight line captures almost all of f.

Why these results are expected

On $0 \le x \le 1$ the angle $\pi x/n$ lies between 0 and π/n, which is small when n is large. So $\sin(\pi x/n) \approx \pi x/n$ throughout the interval: the function is itself very nearly the linear function λx with $\lambda = \pi/n$. The best-fitting line should therefore have gradient close to π/n, and the fit should become essentially perfect, so the residual error should tend to zero. The picture shows that even for $n = 2$ the curve is not far from a straight line.



Checks

- If $f(x) = cx$ already, then $\lambda = 3c\int_0^1 x^2\,\mathrm{d}x = c$ and $R^2 = \frac{1}{3}c^2 - \frac{1}{3}c^2 = 0$, as it should be.
- Numerically, for $n = 10$ the exact formula gives $\lambda = 0.31107$, against $\pi/10 = 0.31416$; for $n = 100$ the ratio $\lambda/(\pi/n)$ is 0.99990.
- For $n = 10$, quadrature gives $R^2 \approx 6.0 \times 10^{-7}$ while $K \approx 0.032$, so the residual really is tiny compared with f.

Common slips

- Stopping at $\sin a \approx a$, $\cos a \approx 1$ and getting $\lambda \approx 0$: true, but useless. Keep the cubic terms.
- Forgetting that $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$ is where the 3 in $\lambda = 3\int_0^1 xf(x)\,\mathrm{d}x$ comes from.

Beyond STEP

This is least-squares approximation, and it is really geometry. Treat functions on $[0,1]$ as vectors with "dot product" $\langle f, g\rangle = \int_0^1 fg\,\mathrm{d}x$. Then λx is the orthogonal projection of f onto the line spanned by x, and $R^2 = K - \frac{1}{3}\lambda^2$ is Pythagoras' theorem. Pushing the expansion one step further shows $R^2 \approx a^6/1575$, which comes from projecting the x^3 term away from x. Strang's *Introduction to Linear Algebra* develops projections and least squares, and the same ideas lead on to Fourier series.

WORKED SOLUTION

STEP II 2000, Question 6

Pure • Half-Angle Substitution

STEP II 2000, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q6

The key idea

The substitution $t = \tan\frac{1}{2}\theta$ turns any rational function of $\sin\theta$ and $\cos\theta$ into a rational function of t, and here the result is 2 over a quadratic. Completing the square gives an arctangent, and the third identity in the first part is exactly what you need to simplify the arctangent at the upper limit. The second integral then follows from the first by symmetry, without any new integration.

Solution

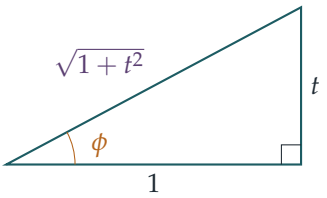
The three identities

Write $\phi = \frac{1}{2}\theta$, so $t = \tan\phi$ and $1 + t^2 = \sec^2\phi$. By the double-angle formulae,

$$\sin\theta = 2\sin\phi\cos\phi = 2\tan\phi\cos^2\phi = \frac{2\tan\phi}{\sec^2\phi} = \frac{2t}{1+t^2},$$

$$\cos\theta = \cos^2\phi - \sin^2\phi = \cos^2\phi\,(1 - \tan^2\phi) = \frac{1-t^2}{1+t^2}.$$

(These hold whenever $\tan\frac{1}{2}\theta$ is defined, that is, $\cos\frac{1}{2}\theta \neq 0$.) The right-angled triangle below, with legs 1 and t and angle ϕ, is a good way to remember them.



For the third, when $\sin\theta \neq 0$, use $1 + \cos\theta = 2\cos^2\phi$ and $\sin\theta = 2\sin\phi\cos\phi$:

$$\frac{1+\cos\theta}{\sin\theta} = \frac{2\cos^2\phi}{2\sin\phi\cos\phi} = \cot\phi = \tan\left(\tfrac{1}{2}\pi - \phi\right) = \tan\left(\tfrac{1}{2}\pi - \tfrac{1}{2}\theta\right),$$

since $\cot\phi = \tan(\frac{1}{2}\pi - \phi)$ for complementary angles. All three identities are shown.

The first integral

With $t = \tan\frac{1}{2}\theta$ we have $\theta = 2\arctan t$, so

$$\mathrm{d}\theta = \frac{2}{1+t^2}\,\mathrm{d}t.$$

When $\theta = 0$, $t = 0$; when $\theta = \frac{1}{2}\pi$, $t = \tan\frac{1}{4}\pi = 1$. Write $c = \cos\alpha$ and $s = \sin\alpha$; both are positive since $0 < \alpha < \frac{1}{2}\pi$. Then

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1 + c\sin\theta} = \int_0^1 \frac{1}{1 + \dfrac{2ct}{1+t^2}} \cdot \frac{2}{1+t^2}\,\mathrm{d}t = \int_0^1 \frac{2}{t^2 + 2ct + 1}\,\mathrm{d}t.$$

Completing the square, and using $1 - c^2 = s^2$,

$$t^2 + 2ct + 1 = (t+c)^2 + s^2.$$

Since $s > 0$, the standard integral $\int \frac{\mathrm{d}u}{u^2+s^2} = \frac{1}{s}\arctan\frac{u}{s}$ gives

$$\int_0^1 \frac{2}{(t+c)^2+s^2}\,\mathrm{d}t = \frac{2}{s}\Big[\arctan\frac{t+c}{s}\Big]_0^1 = \frac{2}{s}\left(\arctan\frac{1+\cos\alpha}{\sin\alpha} - \arctan\frac{\cos\alpha}{\sin\alpha}\right).$$

Ask yourself How do we simplify $\arctan\dfrac{1+\cos\alpha}{\sin\alpha}$?

This is where the first part earns its keep. By the third identity (with α in place of θ), $\frac{1+\cos\alpha}{\sin\alpha} = \tan(\frac{1}{2}\pi - \frac{1}{2}\alpha)$. But we must check the angle lies in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, the range of arctan, before we can cancel arctan against tan. Since $0 < \alpha < \frac{1}{2}\pi$, we have $\frac{1}{4}\pi < \frac{1}{2}\pi - \frac{1}{2}\alpha < \frac{1}{2}\pi$, so indeed

$$\arctan\frac{1+\cos\alpha}{\sin\alpha} = \tfrac{1}{2}\pi - \tfrac{1}{2}\alpha.$$

Similarly $\frac{\cos\alpha}{\sin\alpha} = \cot\alpha = \tan(\frac{1}{2}\pi - \alpha)$ with $0 < \frac{1}{2}\pi - \alpha < \frac{1}{2}\pi$, so $\arctan\frac{\cos\alpha}{\sin\alpha} = \frac{1}{2}\pi - \alpha$. Therefore

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1 + \cos\alpha\sin\theta} = \frac{2}{\sin\alpha}\Big(\big(\tfrac{1}{2}\pi - \tfrac{1}{2}\alpha\big) - \big(\tfrac{1}{2}\pi - \alpha\big)\Big) = \frac{2}{\sin\alpha}\cdot\frac{\alpha}{2} = \frac{\alpha}{\sin\alpha},$$

as required.

The second integral

Substitute $\theta = \frac{1}{2}\pi - u$. Then $\cos\theta = \sin u$, $\mathrm{d}\theta = -\mathrm{d}u$, and the limits 0, $\frac{1}{2}\pi$ swap, which cancels the minus sign:

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1 + \sin\alpha\cos\theta} = \int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}u}{1 + \sin\alpha\sin u}.$$

Now let $\beta = \frac{1}{2}\pi - \alpha$. Then $\sin\alpha = \cos\beta$, and $0 < \beta < \frac{1}{2}\pi$, so the result just proved applies with β in place of α:

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}u}{1 + \cos\beta \sin u} = \frac{\beta}{\sin\beta}.$$

Since $\sin\beta = \cos\alpha$, we conclude that, for $0 < \alpha < \frac{1}{2}\pi$,

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}\theta}{1 + \sin\alpha\cos\theta} = \frac{\frac{1}{2}\pi - \alpha}{\cos\alpha}.$$

Checks

- As $\alpha \to 0$, $\cos\alpha \to 1$ and the first integral should tend to $\int_0^{\pi/2} \frac{\mathrm{d}\theta}{1+\sin\theta} = 1$ (substitute and you get $\int_0^1 \frac{2}{(1+t)^2}\,\mathrm{d}t = 1$); and indeed $\alpha / \sin\alpha \to 1$.
- As $\alpha \to \frac{1}{2}\pi$ the integrand tends to 1, so the first integral should tend to $\frac{1}{2}\pi$, which matches $\frac{\pi/2}{1}$. The second integral likewise tends to 1 as $\alpha \to \frac{1}{2}\pi$, consistent with $\frac{\frac{1}{2}\pi - \alpha}{\cos\alpha} \to 1$.
- Numerically, at $\alpha = \frac{1}{4}\pi$ Simpson's rule gives 1.110721 for the first integral, matching $\frac{\pi\sqrt{2}}{4} \approx 1.110721$; at $\alpha = \frac{1}{3}\pi$ the second integral comes out as $1.047198 \approx \frac{1}{3}\pi$, as the formula predicts.

Common slips

- Forgetting the factor $\frac{2}{1+t^2}$ from $\mathrm{d}\theta$, or the new limits 0 and 1.
- Writing $\arctan(\tan x) = x$ without checking that x lies in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$; here the restriction $0 < \alpha < \frac{1}{2}\pi$ is exactly what makes it valid.
- Redoing the whole integration for the second integral, rather than using $\theta \mapsto \frac{1}{2}\pi - \theta$ and $\alpha \mapsto \frac{1}{2}\pi - \alpha$.

Beyond STEP

The $t = \tan\frac{1}{2}\theta$ substitution (sometimes called the Weierstrass substitution) works because the point $(\cos\theta, \sin\theta) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)$ is a rational parametrisation of the unit circle; taking rational t gives every Pythagorean triple. At university, integrals of the form $\int_0^{2\pi} R(\cos\theta, \sin\theta)\,\mathrm{d}\theta$ are usually done instead by setting $z = e^{i\theta}$ and using the residue theorem, which turns the calculation into finding poles inside the unit circle. Needham's *Visual Complex Analysis* is a beautiful introduction to that idea.

WORKED SOLUTION

STEP II 2000, Question 7

Pure • Lines and Cones

STEP II 2000, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q7

The key idea

Before doing anything else, work out the length of $\mathbf{s}$. The $\cos\theta \pm \sqrt{3}$ terms are set up so that the cross terms cancel and $\cos^2\theta + \sin^2\theta$ appears, so $|\mathbf{s}|$ doesn't depend on θ at all. After that, everything comes from the scalar product. The combination $s_x - s_z$ is also free of θ, which points straight at the direction of m and explains why the plane $x - z =$ constant appears in the final part.

Solution

The angle between l and a general line

Write $c_\theta = \cos\theta$ for short. The length of $\mathbf{s}$ is

$$\begin{aligned} |\mathbf{s}|^2 &= (c_\theta + \sqrt{3})^2 + 2\sin^2\theta + (c_\theta - \sqrt{3})^2 \\ &= 2c_\theta^2 + 6 + 2\sin^2\theta = 8, \end{aligned}$$

since the $\pm 2\sqrt{3}\,c_\theta$ cross terms cancel. So $|\mathbf{s}| = 2\sqrt{2}$ for every θ.

Let $\mathbf{d} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$. Then

$$\mathbf{s}\cdot\mathbf{d} = a(c_\theta + \sqrt{3}) + \sqrt{2}\,b\sin\theta + c(c_\theta - \sqrt{3}) = (a+c)\cos\theta + \sqrt{2}\,b\sin\theta + \sqrt{3}\,(a-c).$$

So the angle ϕ between the two lines satisfies

$$\cos\phi = \frac{(a+c)\cos\theta + \sqrt{2}\,b\sin\theta + \sqrt{3}\,(a-c)}{2\sqrt{2}\,\sqrt{a^2+b^2+c^2}}.$$

A line has no preferred direction, so if you want the acute angle, take the modulus of the right-hand side.

The line m

We want a direction $\mathbf{d}$ that makes $\cos\phi$ independent of θ. Looking at the numerator, the θ-dependence goes away exactly when $a + c = 0$ and $b = 0$. So try $\mathbf{d} = \mathbf{i} - \mathbf{k}$ ($a = 1$,

$b = 0$, $c = -1$):

$$\cos\phi = \frac{\sqrt{3}\cdot 2}{2\sqrt{2}\cdot\sqrt{2}} = \frac{\sqrt{3}}{2}.$$

This is positive, so ϕ is already acute, and $\phi = \pi/6$ for every θ. Hence the line

$$m: \quad \mathbf{r} = \mu(\mathbf{i} - \mathbf{k})$$

through the origin does what is required. As θ varies, l sweeps out a cone with axis m and semi-vertical angle $\pi/6$.

The circle traced by P

On l, $x - z = \lambda(s_x - s_z) = \lambda\big((c_\theta + \sqrt{3}) - (c_\theta - \sqrt{3})\big) = 2\sqrt{3}\,\lambda$. Setting this equal to $4\sqrt{3}$ gives $\lambda = 2$ whatever θ is, so

$$\overrightarrow{OP} = 2\mathbf{s} = (2\cos\theta + 2\sqrt{3})\,\mathbf{i} + 2\sqrt{2}\sin\theta\,\mathbf{j} + (2\cos\theta - 2\sqrt{3})\,\mathbf{k}.$$

The line m meets the plane where $\mu - (-\mu) = 4\sqrt{3}$, that is $\mu = 2\sqrt{3}$, at the point C with

$$\overrightarrow{OC} = 2\sqrt{3}\,\mathbf{i} - 2\sqrt{3}\,\mathbf{k}.$$

Ask yourself Why is C the natural candidate for the centre?

The normal to the plane $x - z = 4\sqrt{3}$ is $\mathbf{i} - \mathbf{k}$, which is the direction of m. So m is the axis of the cone, it is perpendicular to the plane, and any cone cut by a plane perpendicular to its axis gives a circle centred where the axis crosses that plane. Now we check this directly:

$$\overrightarrow{CP} = 2\cos\theta\,\mathbf{i} + 2\sqrt{2}\sin\theta\,\mathbf{j} + 2\cos\theta\,\mathbf{k}, \qquad |\overrightarrow{CP}|^2 = 4\cos^2\theta + 8\sin^2\theta + 4\cos^2\theta = 8.$$

So every position of P lies in the plane and at distance $2\sqrt{2}$ from the fixed point C of that plane. In other words, P lies on the circle in the plane with centre C and radius $2\sqrt{2}$, and C lies on m.

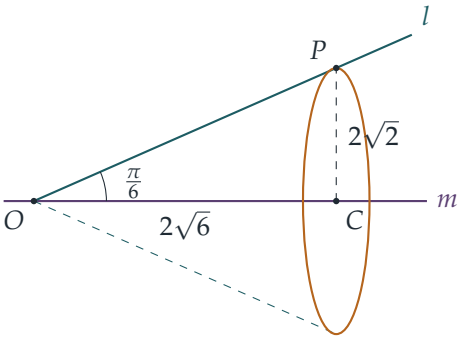
To see that P really traces the whole circle, write

$$\overrightarrow{CP} = 2\sqrt{2}\,(\cos\theta\,\mathbf{u} + \sin\theta\,\mathbf{v}), \qquad \mathbf{u} = \tfrac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{k}), \quad \mathbf{v} = \mathbf{j}.$$

Here $\mathbf{u}$ and $\mathbf{v}$ are perpendicular unit vectors, both perpendicular to $\mathbf{i} - \mathbf{k}$, so they lie along the plane. As θ runs from 0 to 2π, P goes once round the circle. The radius is

$$\boxed{2\sqrt{2}}.$$

A quicker way to get the radius: $|OP| = 2|\mathbf{s}| = 4\sqrt{2}$ and $|OC| = 2\sqrt{6}$, and OC is perpendicular to the plane, so Pythagoras gives $CP = \sqrt{32 - 24} = 2\sqrt{2}$. Equivalently, $CP = OC\tan\frac{\pi}{6} = 2\sqrt{6}/\sqrt{3}$.



Checks

- At $\theta = 0$, $\mathbf{s} = (1+\sqrt{3}, 0, 1-\sqrt{3})$, with $|\mathbf{s}|^2 = (4+2\sqrt{3}) + (4-2\sqrt{3}) = 8$, and $\mathbf{s}\cdot(\mathbf{i}-\mathbf{k}) = 2\sqrt{3}$, giving $\cos\phi = \sqrt{3}/2$.
- The two routes to the radius agree: $\sqrt{(4\sqrt{2})^2 - (2\sqrt{6})^2} = \sqrt{8}$ and $2\sqrt{6}\tan\frac{\pi}{6} = \sqrt{8} \approx 2.828$.
- Numerically, for 20 000 random θ (and random a, b, c for the general formula), the angle with m equals $\pi/6$, P lies in the plane, and $|CP| = 2.8284\ldots$, all to within 10^{-15}.

Common slips

- Forgetting to divide by $|\mathbf{s}|$, or assuming $\mathbf{s}$ is a unit vector. It has length $2\sqrt{2}$.
- Showing only that P lies on a sphere about O (that is, $|OP| = 4\sqrt{2}$). A sphere cut by a plane gives a circle, but you still have to say why its centre is C, on m: it is because OC is perpendicular to the plane.

Beyond STEP

The line l sweeps out a circular cone, and cutting a cone with a plane is how the conic sections got their name. A plane perpendicular to the axis gives a circle; tilt it and you get an ellipse, a parabola or a hyperbola. In a first-year course you would describe the cone as the quadric surface $(\mathbf{r}\cdot\hat{\mathbf{n}})^2 = |\mathbf{r}|^2\cos^2\alpha$ and classify such surfaces using eigenvalues of a symmetric matrix. Strang's *Introduction to Linear Algebra* covers this.

WORKED SOLUTION

STEP II 2000, Question 8

Pure • Separable Differential Equations

STEP II 2000, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q8

The key idea

Both equations are separable: the x-dependence and the y-dependence come as a product, so we can put everything involving y on one side and everything involving x on the other. Each x-side has the form $x\,\mathrm{e}^{ax^2}$, which integrates at once because x is (a multiple of) the derivative of x^2. In part (ii) the power of $(y+3)$ is ours to choose, and we choose it so that the growth of y exactly matches e^{3x^2}.

Solution

Part (i)

Since $y = 6$ at $x = 0$, we have $y + 3 > 0$ near the start, and we may divide by $(y+3)^{1/2}$:

$$(y+3)^{-1/2}\,\frac{\mathrm{d}y}{\mathrm{d}x} = -4x\,\mathrm{e}^{-x^2}.$$

Integrating both sides with respect to x, and using $\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{-x^2} = -2x\,\mathrm{e}^{-x^2}$,

$$2(y+3)^{1/2} = 2\mathrm{e}^{-x^2} + C.$$

Putting $x = 0$, $y = 6$ gives $2 \cdot 3 = 2 + C$, so $C = 4$ and

$$(y+3)^{1/2} = \mathrm{e}^{-x^2} + 2.$$

The right-hand side is always at least 2, so it is a legitimate (positive) square root and $y + 3$ never reaches 0; the division above is justified for all $x \geq 0$. Squaring,

$$\boxed{y = \left(2 + \mathrm{e}^{-x^2}\right)^2 - 3 = 1 + 4\mathrm{e}^{-x^2} + \mathrm{e}^{-2x^2}}.$$

As $x \to \infty$, $\mathrm{e}^{-x^2} \to 0$ and $\mathrm{e}^{-2x^2} \to 0$, so $y \to 1$, as required.

Part (ii)

Here the equation is

$$(y+3)^{k-1}\frac{\mathrm{d}y}{\mathrm{d}x} = x\,\mathrm{e}^{6x^2}.$$

For a general (possibly non-integer) power to make sense we need $y+3>0$, and since $\frac{\mathrm{d}y}{\mathrm{d}x}>0$ for $x>0$, y increases, so $y+3$ stays positive once it starts positive. For $k\neq 0$, integrating gives

$$\frac{(y+3)^k}{k} = \frac{\mathrm{e}^{6x^2}}{12} + C, \qquad \text{so} \qquad (y+3)^k = \frac{k}{12}\,\mathrm{e}^{6x^2} + kC.$$

Ask yourself Which k should we try?

If $k>0$, then for large x the constant is negligible and $y+3 \approx (k/12)^{1/k}\,\mathrm{e}^{6x^2/k}$. We want y to grow like e^{3x^2}, so we need $6/k=3$, that is, $k=2$. (Any other $k>0$ makes $\mathrm{e}^{-3x^2}y$ tend to 0 or to ∞.)

So take $k=2$. Then the equation is $\frac{\mathrm{d}y}{\mathrm{d}x} = x\,\mathrm{e}^{6x^2}/(y+3)$, and the solution is

$$(y+3)^2 = \frac{\mathrm{e}^{6x^2}}{6} + D,$$

where $D=(y_0+3)^2-\frac{1}{6}$ and y_0 is the value of y at $x=0$. The right-hand side increases from $(y_0+3)^2>0$, so it stays positive and the solution exists for every $x\geq 0$:

$$y = \sqrt{\tfrac{1}{6}\mathrm{e}^{6x^2}+D}\;-3.$$

Multiplying by e^{-3x^2} and taking $\mathrm{e}^{-3x^2}=\sqrt{\mathrm{e}^{-6x^2}}$ inside the root,

$$\mathrm{e}^{-3x^2}y = \sqrt{\tfrac{1}{6}+D\,\mathrm{e}^{-6x^2}}\;-\;3\mathrm{e}^{-3x^2} \longrightarrow \sqrt{\tfrac{1}{6}} = \frac{1}{\sqrt{6}}$$

as $x\to\infty$, because both exponentials tend to 0. So with $k=2$ the limit is

$$\boxed{\lim_{x\to\infty}\mathrm{e}^{-3x^2}y = \frac{1}{\sqrt{6}}},$$

whatever the initial value: the constant D has disappeared from the answer, which is why the question can say "any solution".

When $k=2$ the power $(y+3)^{-1}$ also makes sense for $y<-3$. On that branch $y+3=-\sqrt{\frac{1}{6}\mathrm{e}^{6x^2}+D}$, and the same argument gives the limit $-1/\sqrt{6}$. The intended answer is $1/\sqrt{6}$, for solutions with $y>-3$.

Checks

- In (i), $x = 0$ gives $y = (2+1)^2 - 3 = 6$, and differentiating $y = (2 + \mathrm{e}^{-x^2})^2 - 3$ gives $-4x\mathrm{e}^{-x^2}(2 + \mathrm{e}^{-x^2}) = -4x\mathrm{e}^{-x^2}(y+3)^{1/2}$.
- Stepping both equations numerically (Runge–Kutta) agrees with the formulae; for example in (i) $y(2) \approx 1.07360$, and in (ii), starting from $y_0 = 6$, $\mathrm{e}^{-12}y(2) \approx 0.408230$, already close to $1/\sqrt{6} \approx 0.408248$.

Common slips

- Forgetting the constant of integration until after squaring, or fitting $y = 6$ to the wrong expression.
- Integrating $x\,\mathrm{e}^{6x^2}$ as $\mathrm{e}^{6x^2}/6$ rather than $\mathrm{e}^{6x^2}/12$; this does not change k, but it does change the limit.

Beyond STEP

Part (ii) is a small piece of asymptotic analysis: we found how a solution behaves for large x without caring about its initial value, because the constant of integration is swamped by the growing term. Asking which features of a solution are independent of initial conditions is central to the qualitative theory of differential equations. Boyce and DiPrima, *Elementary Differential Equations*, covers separable equations and this style of long-term behaviour.

WORKED SOLUTION

STEP II 2000, Question 9

Mechanics • Motion with Air Resistance

STEP II 2000, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

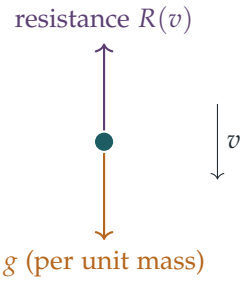
step.maths.org/questions/00-s2-q9

The key idea

Each jumper obeys Newton's second law with a resistance that depends only on speed, so we get a first-order differential equation in v and t that separates. The "speed can never reach" results come from the terminal speed: the speed at which the resistance exactly balances gravity. The solution creeps up towards it but never gets there in finite time. For Karen the quadratic factorises neatly once we write it in terms of the dimensionless speed $u = kv/g$.

Solution

We model each jumper as a particle falling vertically from rest, with constant gravitational acceleration g downwards. We take $v \geq 0$ as the downward speed and measure t from the moment of jumping. Per unit mass, the forces are gravity g downwards and the air resistance upwards, so Newton's second law gives $\dfrac{\mathrm{d}v}{\mathrm{d}t} = g - (\text{resistance per unit mass})$.



Jane's speed bound

Jane's equation of motion is

$$\frac{\mathrm{d}v}{\mathrm{d}t} = g - kv = k\Big(\frac{g}{k} - v\Big).$$

At first $v = 0 < g/k$. While $v < g/k$ we may divide by $g/k - v > 0$ and separate the variables:

$$\int_0^v \frac{\mathrm{d}w}{g/k - w} = \int_0^t k\,\mathrm{d}s \quad\Longrightarrow\quad -\ln\Big(1 - \frac{kv}{g}\Big) = kt,$$

so

$$v = \frac{g}{k}\big(1 - \mathrm{e}^{-kt}\big).$$

This is valid for every $t \geq 0$, because the right-hand side is less than g/k for every finite t, which is the condition we used to divide. Since $\mathrm{e}^{-kt} > 0$, we have $v < g/k$ for all t. So Jane's speed tends to g/k as $t \to \infty$ but never reaches it.

Karen's speed bound

Karen's equation of motion is

$$\frac{\mathrm{d}v}{\mathrm{d}t} = g - kv - \frac{2k^2}{g}v^2.$$

Put $u = kv/g$, a dimensionless speed. Then the right-hand side becomes

$$g\big(1 - u - 2u^2\big) = g(1 - 2u)(1 + u), \qquad \text{and} \qquad \frac{\mathrm{d}u}{\mathrm{d}t} = \frac{k}{g}\frac{\mathrm{d}v}{\mathrm{d}t} = k(1 - 2u)(1 + u).$$

The acceleration is zero when $u = \frac{1}{2}$ (the root $u = -1$ is not a physical speed), that is, when $v = g/(2k)$. This is Karen's terminal speed.

At first $u = 0 < \frac{1}{2}$. While $0 \leq u < \frac{1}{2}$ both factors are positive, so we can separate. In partial fractions,

$$\frac{1}{(1 - 2u)(1 + u)} = \frac{2/3}{1 - 2u} + \frac{1/3}{1 + u},$$

which we can check by putting $u = -1$ and $u = \frac{1}{2}$ in $1 = A(1 + u) + B(1 - 2u)$. Hence

$$kt = \int_0^u \Big(\frac{2/3}{1 - 2w} + \frac{1/3}{1 + w}\Big)\mathrm{d}w = \frac{1}{3}\Big[-\ln(1 - 2w) + \ln(1 + w)\Big]_0^u = \frac{1}{3}\ln\frac{1 + u}{1 - 2u}.$$

Rearranging $\dfrac{1 + u}{1 - 2u} = \mathrm{e}^{3kt}$ gives

$$u = \frac{\mathrm{e}^{3kt} - 1}{2\mathrm{e}^{3kt} + 1} = \frac{1}{2} - \frac{3/2}{2\mathrm{e}^{3kt} + 1} < \frac{1}{2}.$$

This is valid for all $t \geq 0$ for the same reason as before, so Karen's speed satisfies

$$v < \frac{g}{2k} \quad \text{for all } t.$$

That is the corresponding result: her speed approaches $g/(2k)$ but never reaches it.

Jane's time to $g/(3k)$

Setting $v = g/(3k)$ in Jane's solution gives $1 - \mathrm{e}^{-kt} = \frac{1}{3}$, so $\mathrm{e}^{-kt} = \frac{2}{3}$ and

$$t = \frac{1}{k} \ln \frac{3}{2},$$

as required.

Karen's time to $g/(3k)$

> **Ask yourself** Do we need the formula for u in terms of t, or is there a quicker route?

The quicker route is the formula for t in terms of u, which we already have. Speed $g/(3k)$ means $u = \frac{1}{3}$, which is below $\frac{1}{2}$, so she does reach it. Then

$$\frac{1+u}{1-2u} = \frac{4/3}{1/3} = 4, \qquad \text{so} \qquad t = \frac{1}{3k} \ln 4 = \frac{2}{3k} \ln 2.$$

Checks

- Karen feels more resistance than Jane at every positive speed, so she should take longer to reach the same speed. Indeed $\frac{2}{3} \ln 2 \approx 0.462$ is bigger than $\ln \frac{3}{2} \approx 0.405$.
- For small t, both formulas give $v \approx gt$, which is free fall with no resistance, as it should be. Expanding $u = (\mathrm{e}^{3kt} - 1)/(2\mathrm{e}^{3kt} + 1) \approx 3kt/3 = kt$ gives $v \approx gt$.
- We stepped both equations numerically from rest (Runge–Kutta, with $g = 9.8$, $k = 0.7$). The speeds stayed below 14 and 7, which are g/k and $g/(2k)$, and the speed $g/(3k)$ was reached at $t \approx 0.5792$ and $t \approx 0.6601$. These agree with $k^{-1} \ln \frac{3}{2}$ and $\frac{2}{3k} \ln 2$ to about 10^{-9}.

Common slips

- Arguing "the speed can never reach g/k" just from the acceleration being positive. That shows the speed increases, but not that it stays below g/k. You need the explicit solution, or an argument that the time to reach the terminal speed is infinite.
- Getting the sign of the $\ln(1-2u)$ term wrong. Integrating $\frac{1}{1-2u}$ gives $-\frac{1}{2} \ln(1-2u)$.

Beyond STEP

Both equations are autonomous: $\mathrm{d}v/\mathrm{d}t = f(v)$. The terminal speed is a stable equilibrium of such an equation. A uniqueness theorem for ODEs shows that a solution can never reach an equilibrium in finite time when f is smooth, which is the general reason behind both bounds. Karen's equation is a Riccati equation, and the same

logistic-type behaviour appears in population models. Boyce and DiPrima, *Elementary Differential Equations*, covers autonomous equations and their equilibria, and Kleppner and Kolenkow, *An Introduction to Mechanics*, discusses motion with resistance.

WORKED SOLUTION

STEP II 2000, Question 10

Mechanics • Connected Pulley Systems

STEP II 2000, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

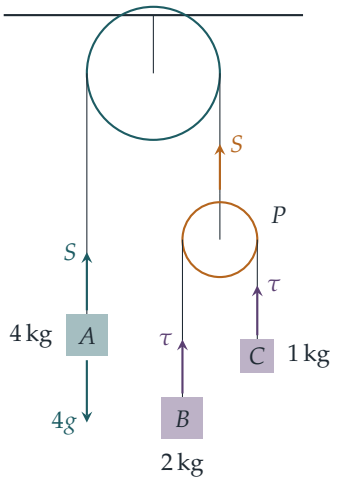
step.maths.org/questions/00-s2-q10

The key idea

This is a double Atwood machine. While A is held, the second pulley is fixed and B, C form an ordinary Atwood machine. Once A is let go, the second pulley moves too, so the clean way in is to measure the motion of B and C *relative to that pulley*, and to use the fact that a light pulley carries no net force: the string above it pulls with exactly twice the tension of the string below.

Solution

We model the particles as point masses, the strings as light and inextensible, and both pulleys as smooth and light. Call the second pulley P. We take *downwards* as positive for every acceleration, write S for the tension in the string through A and τ for the tension in BC, and take g in $\mathrm{m\,s^{-2}}$.



Because P is light, the net force on it must be zero (otherwise it would have infinite

acceleration). It is pulled up by S and down by τ on each side, so

$$S = 2\tau$$

at all times.

First phase: A held

With A held still, the string over the fixed pulley doesn't move, so P is fixed. Then B and C are an ordinary Atwood machine: if B accelerates down at a, then C accelerates up at a. Newton's second law gives

$$2a = 2g - \tau, \qquad 1 \cdot a = \tau - g.$$

Adding, $3a = g$, so

$$a = \tfrac{1}{3}g, \qquad \tau = \tfrac{4}{3}g.$$

So B accelerates downwards and C upwards, each at $\frac{1}{3}g\ \mathrm{m\,s^{-2}}$.

At the end of this phase ($t = T$), B is moving down and C up, each with speed $\frac{1}{3}gT$, and B has fallen $\frac{1}{2} \cdot \frac{1}{3}g\,T^2 = \frac{1}{6}gT^2$. Throughout, A and P are at rest.

Second phase: everything free

Let A have acceleration a_A (downwards). The string over the fixed pulley has constant length, so P has acceleration $-a_A$: it goes up as fast as A goes down.

> **Ask yourself** How do we describe the accelerations of B and C when the pulley they hang from is itself accelerating?

Relative to P, the string BC behaves just as in a fixed Atwood machine: if B moves down relative to P with acceleration r, then C moves up relative to P with acceleration r. Adding P's own acceleration,

$$a_B = -a_A + r, \qquad a_C = -a_A - r.$$

Now Newton's second law for each particle (downwards positive), using $S = 2\tau$:

$$\begin{aligned} A: &\quad 4a_A = 4g - 2\tau, \\ B: &\quad 2(-a_A + r) = 2g - \tau, \\ C: &\quad 1(-a_A - r) = g - \tau. \end{aligned}$$

From A, $\tau = 2g - 2a_A$. Putting this into B gives $-2a_A + 2r = 2a_A$, so $r = 2a_A$. Putting both into C:

$$-3a_A = g - (2g - 2a_A) = -g + 2a_A \quad \Longrightarrow \quad a_A = \tfrac{1}{5}g.$$

Hence $r = \frac{2}{5}g$, $\tau = \frac{8}{5}g$, $S = \frac{16}{5}g$, and

$$a_A = \tfrac{1}{5}g \text{ (down)}, \qquad a_B = -\tfrac{1}{5}g + \tfrac{2}{5}g = \tfrac{1}{5}g \text{ (down)}, \qquad a_C = -\tfrac{1}{5}g - \tfrac{2}{5}g = -\tfrac{3}{5}g,$$

so C accelerates *upwards* at $\frac{3}{5}g\ \mathrm{m\,s^{-2}}$.

These are constant, and at the moment A is released nothing changes abruptly: A and P start from rest, and B, C keep their speeds $\frac{1}{3}gT$.

Speed of A

In the first phase B moved $\frac{1}{6}gT^2$, so it still has

$$0.6\,gT^2 - \tfrac{1}{6}gT^2 = \tfrac{3}{5}gT^2 - \tfrac{1}{6}gT^2 = \tfrac{13}{30}gT^2$$

to go. In the second phase B starts with downward velocity $\frac{1}{3}gT$ and has downward acceleration $\frac{1}{5}g$, so its velocity stays positive and the distance it travels equals its displacement. If this takes time t, then $s = ut + \frac{1}{2}at^2$ gives

$$\tfrac{1}{3}gT\,t + \tfrac{1}{10}gt^2 = \tfrac{13}{30}gT^2.$$

Write $t = kT$ and multiply by $30/(gT^2)$:

$$3k^2 + 10k - 13 = 0 \quad\Longrightarrow\quad (k-1)(3k+13) = 0.$$

Since $t > 0$, $k = 1$: the second phase lasts exactly T seconds. A started this phase from rest with acceleration $\frac{1}{5}g$, so its speed is then

$$v_A = \tfrac{1}{5}g \cdot T = \tfrac{gT}{5}\ \mathrm{m\,s^{-1}}.$$

Checks

- Each equation of motion holds with the values found: for A, $4 \cdot \frac{1}{5}g = 4g - \frac{16}{5}g$; for C, $-\frac{3}{5}g = g - \frac{8}{5}g$.
- An independent check: the strings force $y_C = \text{const} - 2y_A - y_B$ (downward positions), and energy methods with the two free coordinates y_A, y_B give the same $a_A = a_B = \frac{1}{5}g$, $a_C = -\frac{3}{5}g$.
- Stepping the motion numerically with $g = 9.8$, $T = 1.7$, B has travelled $0.6gT^2$ at $t = 3.400$ (that is, $2T$), when A's speed is $3.332 = gT/5$.
- Total weight is $7g$ but the fixed pulley only takes $2S = \frac{32}{5}g$, less than $7g$, so the system's centre of mass accelerates downwards in phase two, as it should.

Common slips

- Treating P as fixed in the second phase and reusing $\pm\frac{1}{3}g$ for B and C.
- Forgetting that $S = 2\tau$ for a light pulley, or giving the pulley a mass term.
- Taking the full $0.6gT^2$ as the second-phase distance, instead of subtracting the $\frac{1}{6}gT^2$ already covered.

Beyond STEP

Choosing positions that automatically satisfy the string constraints, as in the second check, is the starting point of Lagrangian mechanics: write the kinetic and potential energy in terms of the free coordinates and the equations of motion follow, with no tensions in sight. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats constrained systems such as pulleys carefully and is a good next step.

WORKED SOLUTION

STEP II 2000, Question 11

Mechanics • Elastic Strings and SHM

STEP II 2000, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q11

The key idea

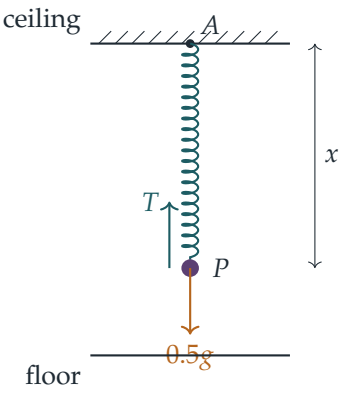
Energy settles both qualitative claims quickly: compare the elastic energy stored at release with the gravitational energy needed to reach the natural length and the ceiling. For the timing, split the motion into two phases: while the string is taut the particle moves in simple harmonic motion about its equilibrium position, and once the string is slack it moves freely under gravity. The trip back is the trip up run in reverse, so we double the time taken to go up.

Solution

We model P as a particle on a light elastic string obeying Hooke's law, with no air resistance. Let x be the distance of P below A, measured downwards, in metres. The string is taut when $x > 1.5$, and then its extension is $x - 1.5$ and its tension is

$$T = \frac{\lambda}{l}(x - 1.5) = \frac{5g}{1.5}(x - 1.5) = \frac{10g}{3}(x - 1.5).$$

The room is 2.5 m high, so P starts at $x = 2$, with extension 0.5 m.



The string becomes slack

At release, the elastic energy stored is

$$\frac{\lambda e^2}{2l} = \frac{5g\,(0.5)^2}{2(1.5)} = \frac{5g}{12}\ \text{J}.$$

For the string to reach its natural length, P must rise 0.5 m, gaining potential energy $0.5g \times 0.5 = \frac{g}{4}$ J. Since $\frac{5g}{12} > \frac{g}{4}$, P arrives at $x = 1.5$ still moving upwards, with kinetic energy

$$\tfrac{1}{2}(0.5)v^2 = \frac{5g}{12} - \frac{g}{4} = \frac{g}{6}, \qquad \text{so} \qquad v^2 = \frac{2g}{3}.$$

It therefore passes above the point where the string has its natural length, and the string becomes slack.

P does not reach the ceiling

To reach the ceiling, P would have to rise 2 m from its starting point, gaining $0.5g \times 2 = g$ J of potential energy. All the energy available is the $\frac{5g}{12}$ J stored in the string at release (once the string is slack it stores nothing), and $\frac{5g}{12} < g$. So P cannot reach the ceiling.

More precisely, after the string goes slack P is a projectile moving vertically with speed v, so it rises a further $\dfrac{v^2}{2g} = \dfrac{1}{3}$ m. Its highest point is at $x = 1.5 - \frac{1}{3} = \frac{7}{6}$, still $\frac{7}{6}$ m below the ceiling.

Simple harmonic motion while taut

For $x > 1.5$, Newton's second law (downwards positive) gives

$$0.5\,\ddot{x} = 0.5g - \frac{10g}{3}(x - 1.5).$$

Equilibrium is where the right-hand side vanishes: $x - 1.5 = 0.15$, so $x = 1.65$. Put $y = x - 1.65$, the displacement below equilibrium. Then

$$\ddot{y} = 2g - \frac{20g}{3}(y + 0.15) = -\frac{20g}{3}\,y,$$

which is simple harmonic motion about $y = 0$ with $\omega^2 = \dfrac{20g}{3}$. This holds whenever the string is in tension, as required.

The time to return

P starts at rest at $y = 2 - 1.65 = 0.35$, so while the string is taut

$$y = 0.35\cos\omega t.$$

The string goes slack when $x = 1.5$, that is $y = -0.15$. So the first time this happens, t_1, satisfies

$$\cos\omega t_1 = -\frac{0.15}{0.35} = -\frac{3}{7}.$$

Ask yourself Which solution of $\cos\omega t_1 = -\frac{3}{7}$ do we want, and how do we write it using arccos?

As t increases from 0, ωt increases from 0, and the first time $\cos\omega t$ reaches $-\frac{3}{7}$ is when ωt is in $(\frac{\pi}{2}, \pi)$. Since $\arccos\frac{3}{7} \in (0, \frac{\pi}{2})$ and $\cos(\pi - \theta) = -\cos\theta$, that value is $\omega t_1 = \pi - \arccos\frac{3}{7}$. Hence

$$t_1 = \frac{1}{\omega}\left(\pi - \arccos\tfrac{3}{7}\right) = \left(\frac{3}{20g}\right)^{1/2}\left(\pi - \arccos\tfrac{3}{7}\right).$$

(As a check, the speed then is $\omega\sqrt{0.35^2 - 0.15^2} = \sqrt{\frac{20g}{3} \times 0.1} = \sqrt{\frac{2g}{3}}$, matching the energy calculation.)

While the string is slack, P moves under gravity alone. It leaves $x = 1.5$ moving up with speed $v = \sqrt{2g/3}$, and it comes back to $x = 1.5$ moving down with the same speed, after time

$$t_2 = \frac{2v}{g} = \frac{2}{g}\sqrt{\frac{2g}{3}} = \left(\frac{8}{3g}\right)^{1/2}.$$

The motion under a force that depends only on position is reversible. Having re-entered the taut region at $x = 1.5$ with the same speed as it left, P retraces the SHM phase in reverse, taking another t_1 to come to rest at $y = 0.35$, its original position. The total time is

$$\begin{aligned} 2t_1 + t_2 &= 2\left(\frac{3}{20g}\right)^{1/2}\left(\pi - \arccos\tfrac{3}{7}\right) + \left(\frac{8}{3g}\right)^{1/2} \\ &= \left(\frac{8}{3g}\right)^{1/2} + \left(\frac{3}{5g}\right)^{1/2}\left(\pi - \arccos\tfrac{3}{7}\right), \end{aligned}$$

using $2\sqrt{3/(20g)} = \sqrt{12/(20g)} = \sqrt{3/(5g)}$. This is the required result.

Checks

- The two routes to the speed at the natural length (energy, and $v^2 = \omega^2(a^2 - y^2)$ for SHM) both give $v^2 = \frac{2g}{3}$.
- With $g = 9.8$, the formula gives about 1.0199 s. Stepping the equation of motion numerically (with the tension switched off when $x < 1.5$) gives the same time to return to rest at the start, and a highest point $\frac{7}{6}$ m below the ceiling.
- Without the slack phase, the SHM period would be $2\pi/\omega \approx 0.77$ s. Our $2t_1 \approx 0.50$ s is less than this, as it should be, since only part of the oscillation is completed.

Common slips

- Writing the SHM about the natural length instead of the equilibrium position. Gravity shifts the centre to an extension of 0.15 m.

- Taking $\omega t_1 = \arccos(-\frac{3}{7})$ and then not recognising that this is $\pi - \arccos\frac{3}{7}$.
- Forgetting the return journey: the time asked for includes going up *and* coming back, so both phases are doubled.

Beyond STEP

This is a piecewise-defined, or "non-smooth", dynamical system: different equations of motion hold in different regions, and solutions are pieced together at the boundaries by matching position and velocity. The same idea governs bouncing balls, impacts and switching circuits, and the reversibility argument we used is a first glimpse of time-reversal symmetry in conservative systems. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 2000, Question 12

Probability and Statistics • Sums of Normal Variables

STEP II 2000, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q12

The key idea

Henry is late exactly when his *total* journey time exceeds the time he has before 0900, so for each route we add the two independent normal journey times to get a single normal variable. Standardising turns "late" into "a standard normal exceeds some number", and the two routes can then be compared without any tables, because Φ is increasing. The rest is the law of total probability and Bayes' theorem.

Solution

Comparing the two routes

We assume that, on each route, the train journey and the onward journey are independent.

King's Cross. Let $X_1 \sim N(55, 25)$ be the train time and $Y_1 \sim N(30, 144)$ the onward time. Since they are independent normal variables, their sum is normal, with means and variances adding:

$$T_1 = X_1 + Y_1 \sim N(85,\ 169), \qquad \text{standard deviation } 13.$$

The train leaves at 0715, so Henry has 105 minutes, and he is late when $T_1 > 105$. With $Z \sim N(0, 1)$,

$$\mathrm{P}(\text{late} \mid \text{King's Cross}) = \mathrm{P}\left(Z > \frac{105 - 85}{13}\right) = 1 - \Phi\left(\tfrac{20}{13}\right) = A.$$

Liverpool Street. Similarly $T_2 = X_2 + Y_2$ with $X_2 \sim N(65, 16)$ and $Y_2 \sim N(25, 9)$ independent, so

$$T_2 \sim N(90,\ 25), \qquad \text{standard deviation } 5.$$

This train leaves at 0720, leaving 100 minutes, so

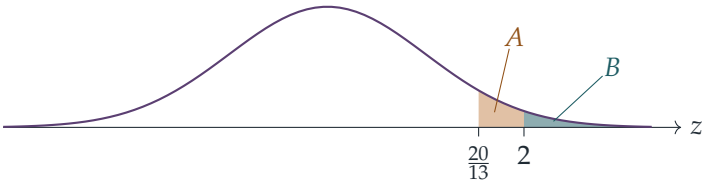
$$\mathrm{P}(\text{late} \mid \text{Liverpool Street}) = \mathrm{P}\left(Z > \frac{100 - 90}{5}\right) = 1 - \Phi(2) = B.$$

Ask yourself How can we compare A and B without tables?

Φ is a cumulative distribution function with a positive density, so it is strictly increasing. Since $\frac{20}{13} < \frac{26}{13} = 2$, we have $\Phi(\frac{20}{13}) < \Phi(2)$, and therefore

$$A = 1 - \Phi\left(\tfrac{20}{13}\right) > 1 - \Phi(2) = B.$$

So Henry is more likely to be late if he catches the King's Cross train, as required. (Interestingly, that route is *faster* on average; it loses because the Underground leg from King's Cross is so variable.)



The shaded tails of the standard normal curve: A is the whole shaded area, B the darker part beyond 2.

The expected number of late arrivals

On a single journey let K be the event that Henry takes the King's Cross train, so $\mathrm{P}(K) = p$, and let D be the event that he is late. By the law of total probability,

$$\mathrm{P}(D) = \mathrm{P}(D \mid K)\,\mathrm{P}(K) + \mathrm{P}(D \mid K')\,\mathrm{P}(K') = pA + (1-p)B.$$

Assuming the M journeys are independent, the number of late arrivals is binomial with parameters M and $\mathrm{P}(D)$, so

$$L = M(pA + (1-p)B) = M(B + p(A-B)).$$

Since $A - B > 0$ and $0 \le p \le 1$, the expression $B + p(A-B)$ increases with p, from B at $p = 0$ to A at $p = 1$. Hence for every possible value of p,

$$BM \le L \le AM,$$

with equality on the left only when $p = 0$ and on the right only when $p = 1$. In words: L/M is a weighted average of A and B, so it must lie between them.

Estimating p

The proportion of late occasions on which he had caught the King's Cross train estimates $\mathrm{P}(K \mid D)$, and because M is large we take the observed $\frac{3}{5}$ as equal to this probability. By Bayes' theorem,

$$\mathrm{P}(K \mid D) = \frac{\mathrm{P}(D \mid K)\,\mathrm{P}(K)}{\mathrm{P}(D)} = \frac{pA}{pA + (1-p)B}.$$

Setting this equal to $\frac{3}{5}$:

$$5pA = 3pA + 3(1-p)B$$
$$2pA + 3pB = 3B,$$

so the estimate is

$$\hat{p} = \frac{3B}{2A + 3B}.$$

Checks

- Numerically, $A \approx 0.0620$ and $B \approx 0.0228$, so King's Cross is late about 6% of the time and Liverpool Street about 2%. A simulation of 4×10^5 journeys on each route agreed to three decimal places.
- With these values $\hat{p} \approx 0.355$. Substituting back, $pA/(pA + (1-p)B) = 0.6$ exactly, and a simulation of the whole process with this p gave about 60% of late arrivals coming from King's Cross.
- Sanity check on the last answer: if the two routes were equally reliable ($A = B$), the late arrivals would reflect only the choice of train, so $\frac{3}{5}$ of them via King's Cross would mean $p = \frac{3}{5}$. The formula gives $3B/5B = \frac{3}{5}$. Because King's Cross is the less reliable route, it is over-represented among late days, so p must be smaller than $\frac{3}{5}$, as it is.

Common slips

- Adding standard deviations instead of variances: the total for King's Cross has variance $25 + 144 = 169$, so its standard deviation is 13, not $5 + 12 = 17$.
- Using the same 105-minute allowance for both trains: the Liverpool Street train leaves five minutes later.
- Setting $p = \frac{3}{5}$ directly. The $\frac{3}{5}$ is $\mathrm{P}(K \mid D)$, not $\mathrm{P}(K)$, and the two differ because the routes are not equally reliable.

Beyond STEP

Two ideas here recur throughout a first probability course. The fact that a sum of independent normal variables is normal is most cleanly proved with moment generating functions, and it is a special case of the stability of the normal family that underlies the central limit theorem. The final part is statistical inference in miniature: we observed an effect (late arrivals) and used Bayes' theorem to infer the hidden cause (the choice of train). Grimmett and Welsh, *Probability: An Introduction*, covers both, and Blitzstein and Hwang, *Introduction to Probability*, has many more Bayes problems of exactly this flavour.

WORKED SOLUTION

STEP II 2000, Question 13

Probability and Statistics • Capture-Recapture Estimation

STEP II 2000, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 13.2, *Counting Voles*: https://beyondstep.keithmcnulty.org/ch13/voles.html

The key idea

This is capture–recapture: the second sample is a random choice of 200 voles from N, of which 200 are marked, so the number of marked voles in it has a hypergeometric distribution. To find where p_N is largest, don't differentiate (factorials of N aren't differentiable in any easy sense); instead compare consecutive terms through the ratio p_N/p_{N-1}, where almost everything cancels. The final identity then comes free: the third-sample probabilities must add up to 1.

Solution

The probability p_N

Model the second sample as a choice of 200 of the N voles, every subset of size 200 being equally likely. There are $\binom{N}{200}$ such subsets. Of the N voles, 200 are marked and $N-200$ are not. A sample with exactly 11 marked voles is a choice of 11 marked ones and 189 unmarked ones, so

$$p_N = \frac{\binom{200}{11}\binom{N-200}{189}}{\binom{N}{200}}.$$

Writing the two binomial coefficients that involve N as factorials,

$$p_N = \binom{200}{11} \cdot \frac{(N-200)!}{189!\,(N-389)!} \cdot \frac{200!\,(N-200)!}{N!}$$
$$= k\,\frac{((N-200)!)^2}{N!\,(N-389)!}, \qquad \text{where } k = \binom{200}{11}\frac{200!}{189!},$$

which does not depend on N. This is the required form. (It needs $N \geq 389$, since the two samples together contain $200 + 189 = 389$ different voles.)

Maximising p_N

For $N \geq 390$, divide p_N by p_{N-1}. Using $(N-200)! = (N-200)\,(N-201)!$, $N! = N\,(N-1)!$ and $(N-389)! = (N-389)\,(N-390)!$, everything cancels except

$$\frac{p_N}{p_{N-1}} = \frac{(N-200)^2}{N(N-389)}.$$

Both the numerator and the denominator are positive, so

$$\begin{aligned} p_N > p_{N-1} &\iff (N-200)^2 > N(N-389) \\ &\iff N^2 - 400N + 40000 > N^2 - 389N \\ &\iff 40000 > 11N \\ &\iff N < 3636.\overline{36}. \end{aligned}$$

Ask yourself What does this tell us about the shape of the sequence $p_{389}, p_{390}, \ldots$?

For $390 \leq N \leq 3636$ the ratio is greater than 1, so the sequence strictly increases up to p_{3636}. For $N \geq 3637$ the ratio is less than 1 (equality would need $11N = 40000$, which has no integer solution), so the sequence strictly decreases from then on. Hence p_N has a unique maximum, at

$$N = 3636.$$

This is the integer part of $\frac{200\times200}{11}$. It is what you'd guess by saying that the proportion marked in the sample, $\frac{11}{200}$, should match the proportion marked in the population, $\frac{200}{N}$.

The third sample

Take the population to be 3636, the estimate just found. After the second release, the marked voles are the original 200 plus the 189 newly marked ones, so 389 are marked and $3636 - 389 = 3247$ are unmarked. Reasoning exactly as before, the probability that a random sample of 200 contains exactly j marked voles is

$$\frac{\binom{389}{j}\binom{3247}{200-j}}{\binom{3636}{200}}, \qquad j = 0, 1, \ldots, 200.$$

Since $389 \geq 200$ and $3247 \geq 200$, every j from 0 to 200 is possible, and these are all the possible outcomes. They are mutually exclusive, so their probabilities add up to 1:

$$\sum_{j=0}^{200} \frac{\binom{389}{j}\binom{3247}{200-j}}{\binom{3636}{200}} = 1.$$

Multiplying by $\binom{3636}{200}$ gives

$$\sum_{j=0}^{200} \binom{389}{j}\binom{3247}{200-j} = \binom{3636}{200},$$

as required.

Checks

- Compute p_N exactly with integer binomial coefficients: it agrees with the factorial formula for several values of N (for example 389, 1000 and 5000), and a search over $389 \leq N < 20000$ puts the maximum at $N = 3636$.
- At the turning point, $p_{3636}/p_{3635} = 3436^2/(3636 \times 3246) \approx 1.0003 > 1$, but $p_{3637}/p_{3636} < 1$.
- The two sides of the final identity, computed exactly as integers, are equal. Note that $389 + 3247 = 3636$, as it must be.

Common slips

- Treating N as continuous and differentiating. Use the ratio of consecutive terms instead, and decide the inequality exactly.
- Forgetting that after the second release 389 voles, not 200, are marked.
- Saying "maximum at $N = 3636.36$": the population size must be a whole number, and you should check which neighbour wins.

Beyond STEP

Choosing the value of the parameter that makes the observed data most probable is the method of *maximum likelihood*, the central idea of first-year statistics. Here it gives the Lincoln–Petersen estimator for capture–recapture. The final identity is Vandermonde's identity, and the argument used, counting the same thing in two ways, is a powerful technique in combinatorics. Grimmett and Welsh, *Probability: An Introduction*, covers the hypergeometric distribution and this style of argument.

WORKED SOLUTION

STEP II 2000, Question 14

Probability and Statistics • Distribution of the Median

STEP II 2000, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q14

This question has a full guided treatment in *Beyond STEP*, Section 13.4, *Mean or Median?*: https://beyondstep.keithmcnulty.org/ch13/mean-or-median.html

The key idea

The density of the median is a polynomial of the form $y^r(1-y)^s$, so the given integral does all the work: once for the normalising constant, then again with the power of y raised by one and by two for the first two moments. For the final part, both Y and $\bar{X}$ are centred at $\frac{1}{2}$, and with normal distributions the probability of landing within a fixed distance of the centre depends only on the standard deviation. The variable with the smaller spread wins.

Solution

The constant k

A density must integrate to 1. Using the given result with $r = s = n$,

$$1 = \int_0^1 k y^n (1-y)^n \, \mathrm{d}y = k \, \frac{n!\, n!}{(2n+1)!}, \qquad \text{so} \qquad k = \frac{(2n+1)!}{(n!)^2},$$

as required.

The mean

The density is symmetric about $y = \frac{1}{2}$ (replacing y by $1-y$ leaves it unchanged), so $\mathrm{E}(Y) = \frac{1}{2}$. To confirm this with the integral, take $r = n+1$, $s = n$:

$$\mathrm{E}(Y) = k \int_0^1 y^{n+1}(1-y)^n \, \mathrm{d}y = \frac{(2n+1)!}{(n!)^2} \cdot \frac{(n+1)!\, n!}{(2n+2)!} = \frac{n+1}{2n+2} = \frac{1}{2}.$$

The variance

Now take $r = n+2$, $s = n$:

$$\mathrm{E}(Y^2) = \frac{(2n+1)!}{(n!)^2} \cdot \frac{(n+2)!\,n!}{(2n+3)!} = \frac{(n+2)(n+1)}{(2n+3)(2n+2)} = \frac{n+2}{2(2n+3)}.$$

Hence

$$\mathrm{Var}(Y) = \frac{n+2}{2(2n+3)} - \frac{1}{4} = \frac{2(n+2)-(2n+3)}{4(2n+3)} = \frac{1}{4(2n+3)}.$$

Comparing the median with the mean

Each X_i is uniform on $[0,1]$, so $\mathrm{E}(X_i) = \frac{1}{2}$ and $\mathrm{Var}(X_i) = \frac{1}{12}$. Since the X_i are independent,

$$\mathrm{E}(\bar{X}) = \tfrac{1}{2}, \qquad \mathrm{Var}(\bar{X}) = \frac{1}{(2n+1)^2} \cdot (2n+1) \cdot \frac{1}{12} = \frac{1}{12(2n+1)}.$$

> **Ask yourself** Both variables are centred at $\frac{1}{2}$. What single number decides which is more likely to fall within $d/\sqrt{n}$ of the centre?

The standard deviation. For large n we may take $Y \sim \mathrm{N}(\frac{1}{2}, \sigma_Y^2)$ and $\bar{X} \sim \mathrm{N}(\frac{1}{2}, \sigma_X^2)$. If $W \sim \mathrm{N}(\frac{1}{2}, \sigma^2)$ and $c > 0$, then, writing Φ for the standard normal distribution function,

$$\mathrm{P}\big(|W - \tfrac{1}{2}| < c\big) = \mathrm{P}\Big(|Z| < \frac{c}{\sigma}\Big) = 2\Phi\Big(\frac{c}{\sigma}\Big) - 1,$$

where $Z \sim \mathrm{N}(0,1)$. This is a strictly *decreasing* function of σ, because Φ is strictly increasing. So with $c = d/\sqrt{n}$ the required inequality holds exactly when $\sigma_Y > \sigma_X$, that is, when $\mathrm{Var}(Y) > \mathrm{Var}(\bar{X})$:

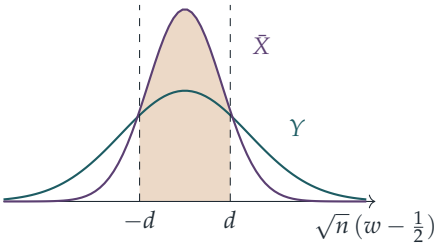
$$\begin{aligned} \frac{1}{4(2n+3)} > \frac{1}{12(2n+1)} &\iff 12(2n+1) > 4(2n+3) \\ &\iff 24n + 12 > 8n + 12 \\ &\iff n > 0. \end{aligned}$$

This is true for every $n \geq 1$. So once n is large enough for the normal approximations to be valid, the probability for Y is strictly less than the probability for $\bar{X}$, whatever the positive number d. This completes the proof.

To see how big the gap is, standardise:

$$\frac{d/\sqrt{n}}{\sigma_Y} = 2d\sqrt{\frac{2n+3}{n}} \to 2\sqrt{2}\,d, \qquad \frac{d/\sqrt{n}}{\sigma_X} = d\sqrt{\frac{12(2n+1)}{n}} \to 2\sqrt{6}\,d.$$

Asymptotically, the mean's standard deviation is smaller by a factor of $\sqrt{3}$.



The figure shows the two limiting densities on the scale $\sqrt{n}\,(w-\frac{1}{2})$. The narrower curve (the mean) puts more probability in any central band.

Checks

- $n=0$: the "median" of one uniform variable is just X_1, and our formulas give $k=1$ and $\mathrm{Var}(Y)=\frac{1}{12}$, the uniform variance, as they should.
- Both variances equal $\frac{1}{12}$ when $n=0$ and split apart for $n\geq 1$, which matches the inequality $n>0$ above.
- A simulation with $n=50$ (101 uniforms) gave $\mathrm{Var}(Y)\approx 0.00243$ against $1/412\approx 0.00243$. With $d=0.3$ it gave $\mathrm{P}(|Y-\frac{1}{2}|<d/\sqrt{n})\approx 0.607$ and $\mathrm{P}(|\bar{X}-\frac{1}{2}|<d/\sqrt{n})\approx 0.859$, close to the normal values 0.611 and 0.860.

Common slips

- Writing $\mathrm{Var}(\bar{X})=\frac{1}{12}$, or dividing by $2n+1$ only once instead of getting $(2n+1)^2$ in the denominator and then multiplying by $2n+1$.
- Getting the direction wrong: a *larger* variance gives a *smaller* probability of being near the centre.
- Forgetting that $\mathrm{E}(Y^2)$ uses $r=n+2$, not $r=2n$.

Beyond STEP

The density of the median is a Beta distribution, $\mathrm{Beta}(n+1,n+1)$, and the given integral is the Beta function. The comparison you have just made is the idea of *asymptotic relative efficiency*: for uniform data the sample mean estimates the centre with a third of the median's variance, while for heavy-tailed data the median can win. Order statistics and the Beta function are covered well in Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP III 2000, Question 1

Pure • Tangents to Hyperbolas

STEP III 2000, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q1

The key idea

Both curves are rectangular hyperbolas centred at the origin, and both are unchanged by the half-turn $(x, y) \mapsto (-x, -y)$. That symmetry hands us Q for free. For the tangents, the neat trick is not to solve the simultaneous equations by brute force but to *check* that the proposed point lies on both lines, using the two facts $ab = 1$ and $a^2 - b^2 = 2$. Once we have the four points, a quarter-turn about the origin explains the square.

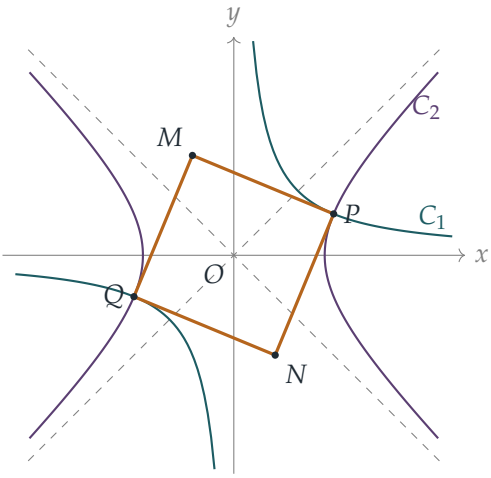
Solution

The sketch

C_1, $xy = 1$, is the familiar rectangular hyperbola in the first and third quadrants, with the coordinate axes as asymptotes.

C_2, $x^2 - y^2 = 2$, crosses the x-axis at $(\pm\sqrt{2}, 0)$ and never meets the y-axis (that would need $-y^2 = 2$). Writing $y^2 = x^2 - 2$, we see that $|y| < |x|$ and that $y \approx \pm x$ for large $|x|$, so its asymptotes are $y = \pm x$. It is C_1 rotated through $45°$ about the origin (and scaled by $\sqrt{2}$), with branches opening left and right.

Where do they meet? Since $xy = 1 > 0$, any common point lies in the first or third quadrant. Substituting $y = 1/x$ into C_2 gives $x^4 - 2x^2 - 1 = 0$, so $x^2 = 1 + \sqrt{2}$ (the other root, $1 - \sqrt{2}$, is negative). So there are exactly two intersections, one in each of those quadrants.



The figure shows C_1 in teal and C_2 in plum, with dashed asymptotes $y = \pm x$ for C_2. The amber square $PMQN$ is what we are about to find.

The point Q

Replacing (x, y) by $(-x, -y)$ leaves both xy and $x^2 - y^2$ unchanged. So if (a, b) lies on both curves, so does $(-a, -b)$. As there are only two intersection points,

$$Q = (-a, -b).$$

Throughout, we use the two facts that P lies on both curves:

$$ab = 1, \qquad a^2 - b^2 = 2.$$

The tangents

For C_1, differentiating $xy = 1$ implicitly gives $y + x\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so the gradient at a point (p, q) of C_1 is $-q/p$. The tangent there is $p(y - q) = -q(x - p)$, that is,

$$qx + py = 2pq = 2.$$

For C_2, differentiating $x^2 - y^2 = 2$ gives $2x - 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so the gradient at (p, q) is p/q. The tangent is $q(y - q) = p(x - p)$, that is,

$$px - qy = p^2 - q^2 = 2.$$

(Both forms also work at points where a gradient is vertical; they are just the gradient vector of the curve dotted with (x, y).) Putting in $P = (a, b)$ and $Q = (-a, -b)$:

$$\begin{aligned}
\text{tangent to } C_1 \text{ at } P: &\quad bx + ay = 2,\\
\text{tangent to } C_2 \text{ at } Q: &\quad -ax + by = 2,\\
\text{tangent to } C_2 \text{ at } P: &\quad ax - by = 2,\\
\text{tangent to } C_1 \text{ at } Q: &\quad -bx - ay = 2.
\end{aligned}$$

The point M

M is where the first two lines meet. Try $(x, y) = (-b, a)$:

$$bx + ay = -b^2 + a^2 = 2,$$
$$-ax + by = ab + ab = 2ab = 2.$$

So $(-b, a)$ lies on both lines. The lines are not parallel, because the determinant of the coefficients is $b \cdot b - a \cdot (-a) = a^2 + b^2 \neq 0$, so they meet in exactly one point. Hence

$$M = (-b, a),$$

as required.

> **Ask yourself** Isn't checking a given answer weaker than solving for it?

Not here. Two non-parallel lines meet in exactly one point, so any point lying on both of them *is* the intersection. Solving directly would also work, but we would then have to use $ab = 1$ and $a^2 - b^2 = 2$ to tidy up the answer anyway.

The point N

In the same way, N is where $ax - by = 2$ meets $-bx - ay = 2$. The point $(b, -a)$ gives

$$ab + ab = 2, \qquad -b^2 + a^2 = 2,$$

and the determinant is $a(-a) - (-b)(-b) = -(a^2 + b^2) \neq 0$. So

$$N = (b, -a).$$

$PMQN$ is a square

Look at the four position vectors:

$$P = (a, b), \quad M = (-b, a), \quad Q = (-a, -b), \quad N = (b, -a).$$

A rotation by $90°$ anticlockwise about O sends (x, y) to $(-y, x)$. So it takes $P \to M \to Q \to N \to P$. That already suggests the answer; here is a direct argument.

- The diagonals PQ and MN both have midpoint O, so they bisect each other and $PMQN$ is a parallelogram.
- Both diagonals have length $2\sqrt{a^2 + b^2}$, so they are equal and the parallelogram is a rectangle.
- $\overrightarrow{OP} \cdot \overrightarrow{OM} = a(-b) + b(a) = 0$, so the diagonals are perpendicular and the rectangle is a rhombus.

A quadrilateral that is both a rectangle and a rhombus is a square. The vertices also come in the right order: P, M, Q, N are at successive quarter-turns round O, so $PMQN$ goes round the square rather than crossing over itself. Hence $PMQN$ is a square. ■

Its side has length squared $|PM|^2 = (a + b)^2 + (b - a)^2 = 2(a^2 + b^2)$. With $a^2 = 1 + \sqrt{2}$ and $b^2 = 1/a^2 = \sqrt{2} - 1$, this is $4\sqrt{2}$.

Checks

- Numerically, $a = \sqrt{1+\sqrt{2}} \approx 1.5538$ and $b \approx 0.6436$. Intersecting the tangent lines numerically gives $M \approx (-0.6436, 1.5538)$ and $N \approx (0.6436, -1.5538)$, as claimed. All four sides come out as $2.3784 \approx \sqrt{4\sqrt{2}}$, and consecutive sides are perpendicular.
- The angle of P is $\arctan(b/a) = \arctan(\sqrt{2}-1) = 22.5°$, halfway between the asymptote $y = 0$ of C_1 and the asymptote $y = x$ of C_2, so the square is tilted by $22.5°$, just as the sketch shows.

Common slips

- Writing the tangent to C_2 as $px + qy = 2$. The minus sign matters, and it is what makes the arithmetic work.
- Taking $Q = (b, a)$ or $(-b, -a)$ by guessing a reflection symmetry. Reflection in $y = x$ preserves C_1 but not C_2; only the half-turn about O preserves both.
- Showing four equal sides only. That gives a rhombus; you also need a right angle (or equal diagonals).

Beyond STEP

The tangent formulas we used, $qx + py = 2$ and $px - qy = 2$, are examples of the general rule that the tangent to a conic $\mathbf{x}^{\mathsf{T}} A\mathbf{x} = c$ at $\mathbf{p}$ is $\mathbf{p}^{\mathsf{T}} A\mathbf{x} = c$. In a first-year course on quadratic forms you learn to diagonalise A by a rotation, which is exactly why $xy = 1$ and $x^2 - y^2 = 2$ are the same curve turned through $45°$. Strang's *Introduction to Linear Algebra* covers quadratic forms and rotations clearly, and Siklos's *Advanced Problems in Mathematics* has more STEP-style coordinate geometry.

WORKED SOLUTION

STEP III 2000, Question 2

Pure • Integration by Substitution

STEP III 2000, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q2

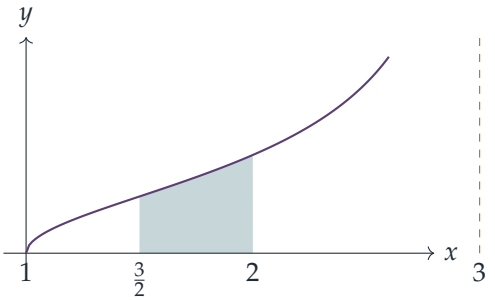
The key idea

A square root of a quotient like $(x-1)/(3-x)$ is begging for the half-angle formulae: with $x = 2 - \cos\theta$ the top becomes $1 - \cos\theta$ and the bottom $1 + \cos\theta$, and their ratio is a perfect square, $\tan^2 \frac{\theta}{2}$. For the general result we don't need to redo the work. A linear change of variable carries the interval from a to b onto the interval from 1 to 3, and turns the general integral into a multiple of the first one.

Solution

The first integral

The integrand is defined for $1 \leq x < 3$, and our range $\frac{3}{2} \leq x \leq 2$ lies inside it. The shaded area below is the integral we want.



Put $x = 2 - \cos\theta$, so $\mathrm{d}x = \sin\theta\,\mathrm{d}\theta$. When $x = \frac{3}{2}$, $\cos\theta = \frac{1}{2}$, and when $x = 2$, $\cos\theta = 0$, so we take θ from $\frac{\pi}{3}$ to $\frac{\pi}{2}$. Then

$$x - 1 = 1 - \cos\theta = 2\sin^2\tfrac{\theta}{2}, \qquad 3 - x = 1 + \cos\theta = 2\cos^2\tfrac{\theta}{2},$$

so

$$\left(\frac{x-1}{3-x}\right)^{\frac{1}{2}} = \left|\tan\tfrac{\theta}{2}\right| = \tan\tfrac{\theta}{2},$$

because $0 < \frac{\theta}{2} < \frac{\pi}{2}$ on our range, so the tangent is positive.

Ask yourself Once the integrand is $\tan\frac{\theta}{2}\sin\theta$, how do we avoid integrating a tangent?

Write $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. The cosine cancels:

$$\tan\tfrac{\theta}{2}\,\sin\theta = \frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\cdot 2\sin\tfrac{\theta}{2}\cos\tfrac{\theta}{2} = 2\sin^2\tfrac{\theta}{2} = 1-\cos\theta.$$

Therefore

$$\begin{aligned}\int_{3/2}^{2}\left(\frac{x-1}{3-x}\right)^{\frac{1}{2}}\mathrm{d}x &= \int_{\pi/3}^{\pi/2}(1-\cos\theta)\,\mathrm{d}\theta = \Big[\theta-\sin\theta\Big]_{\pi/3}^{\pi/2}\\ &= \left(\frac{\pi}{2}-1\right)-\left(\frac{\pi}{3}-\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}-1+\frac{\sqrt{3}}{2}\\ &= \frac{\pi+3\sqrt{3}-6}{6}.\end{aligned}$$

The general integral

The first integral is the case $a=1$, $b=3$, with $p = \frac{3+3}{4} = \frac{3}{2}$ and $q = \frac{1+3}{2} = 2$. So we look for a linear substitution that sends $u=1$ to $x=a$ and $u=3$ to $x=b$:

$$x = a + \tfrac{1}{2}(b-a)(u-1), \qquad \mathrm{d}x = \tfrac{1}{2}(b-a)\,\mathrm{d}u.$$

Then

$$x-a = \tfrac{1}{2}(b-a)(u-1), \qquad b-x = (b-a)-\tfrac{1}{2}(b-a)(u-1) = \tfrac{1}{2}(b-a)(3-u),$$

and since $b-a \neq 0$ the common factor cancels:

$$\frac{x-a}{b-x} = \frac{u-1}{3-u}.$$

For the limits: $u=\frac{3}{2}$ gives $x = a+\frac{1}{4}(b-a) = \frac{3a+b}{4} = p$, and $u=2$ gives $x = a+\frac{1}{2}(b-a) = \frac{a+b}{2} = q$. Since $b-a>0$, the substitution is increasing, so the limits keep their order. Hence

$$\begin{aligned}\int_p^q\left(\frac{x-a}{b-x}\right)^{\frac{1}{2}}\mathrm{d}x &= \frac{b-a}{2}\int_{3/2}^{2}\left(\frac{u-1}{3-u}\right)^{\frac{1}{2}}\mathrm{d}u\\ &= \frac{b-a}{2}\cdot\frac{\pi+3\sqrt{3}-6}{6} = \frac{(b-a)(\pi+3\sqrt{3}-6)}{12},\end{aligned}$$

using the first result. This is what we had to show.

(Equivalently, you could substitute $x = \frac{a+b}{2} - \frac{b-a}{2}\cos\theta$ directly. The limits are again $\frac{\pi}{3}$ and $\frac{\pi}{2}$, and the working is the same with an extra factor $\frac{b-a}{2}$.)

Checks

- Numerically, $\frac{1}{6}(\pi + 3\sqrt{3} - 6) \approx 0.38962$. Simpson's rule applied to the original integrand on $[\frac{3}{2}, 2]$ gives the same value to ten decimal places.
- Plausibility: the integrand rises from $\sqrt{1/3} \approx 0.577$ to 1 over an interval of length $\frac{1}{2}$, so the area must lie between about 0.29 and 0.5. It does.
- The general formula agreed with Simpson's rule for 20 random choices of $a < b$, including negative a.
- Scaling: stretching the interval $[a, b]$ by a factor k leaves the integrand's values unchanged but stretches $[p, q]$ by k. So the answer must be proportional to $b - a$, as it is.

Common slips

- Forgetting $\mathrm{d}x = \sin\theta\,\mathrm{d}\theta$ and trying to integrate $\tan\frac{\theta}{2}$ alone with respect to θ.
- Taking the limits as $\theta = -\frac{\pi}{3}$ or $\frac{5\pi}{3}$. Any choice must make θ increase from $\frac{\pi}{3}$ to $\frac{\pi}{2}$ through a range where $\sin\theta > 0$ and $\tan\frac{\theta}{2} > 0$.
- Missing the factor $\frac{1}{2}(b - a)$ from $\mathrm{d}x$ in the general case, which gives 6 in the denominator instead of 12.

Beyond STEP

The second part shows that a definite integral transforms predictably under an affine change of variable. This is why quantities like this one depend on a and b only through the length $b - a$. The integrand itself is a beta-function-type integrand: over the whole interval, $\int_a^b \left(\frac{x-a}{b-x}\right)^{1/2}\mathrm{d}x = \frac{\pi}{2}(b - a)$, and the functions $\int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\,\mathrm{d}t$ recur throughout probability and analysis. Blitzstein and Hwang, *Introduction to Probability*, meets them as the normalising constants of the Beta distribution. Bartle and Sherbert, *Introduction to Real Analysis*, proves the substitution rule carefully.

WORKED SOLUTION

STEP III 2000, Question 3

Pure • Complex Numbers and Geometry

STEP III 2000, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 5.6, *The Fermat Point*: https://beyondstep.keithmcnulty.org/ch05/fermat-point.html

The key idea

In the Argand plane, rotating a vector through 60° is just multiplication by $\alpha = e^{i\pi/3}$, so the third vertex of an equilateral triangle on a given side can be written down in one line. After that, everything comes from the identity $1 + \alpha^2 = \alpha$ and its relatives, such as $\alpha^3 = -1$. They collapse the vertex expressions into very simple forms, and once you see those, the final part is almost immediate.

Solution

The identity

Since $\alpha^2 = e^{2i\pi/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$,

$$1 + \alpha^2 = \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i = \cos\tfrac{\pi}{3} + i\sin\tfrac{\pi}{3} = \alpha.$$

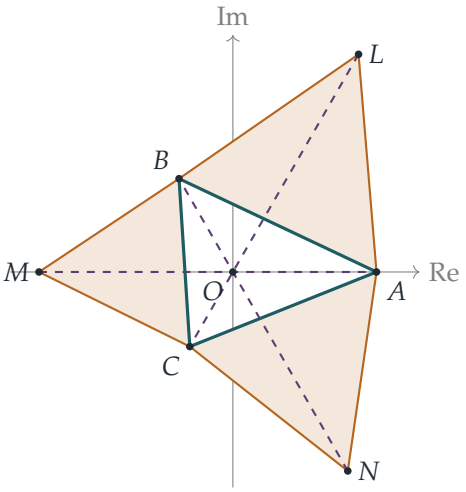
We shall also use these consequences:

- $\alpha^3 = e^{i\pi} = -1$;
- $1 - \alpha = -\alpha^2$ (this is the identity rearranged);
- $\alpha^2 - \alpha = -1$ (also a rearrangement);
- $1 - \alpha = \bar{\alpha} = e^{-i\pi/3}$, since $\alpha + \bar{\alpha} = 2\cos\frac{\pi}{3} = 1$.

The last one says that multiplying by $1 - \alpha$ rotates through 60° *clockwise*.

The triangle ABC

$A = p$ lies on the positive real axis, $B = q\alpha^2 = q\,\mathrm{e}^{2\mathrm{i}\pi/3}$ lies on the ray at argument 120°, and $C = -r\alpha = r\,\mathrm{e}^{4\mathrm{i}\pi/3}$ lies on the ray at argument 240°. So the three vertices sit on three rays from O that are 120° apart, at distances p, q, r from O. The origin is inside the triangle, and $A \to B \to C$ goes *anticlockwise*. The sketch below uses $p = 2$, $q = 1.5$, $r = 1.2$, and shows the erected triangles too.



The vertices N, L and M

If XYZ is an equilateral triangle lettered clockwise, then going round from X, the side XZ is the side XY turned clockwise through 60° about X. So

$$Z - X = (Y - X)\,\mathrm{e}^{-\mathrm{i}\pi/3} = (Y - X)(1 - \alpha), \qquad \text{that is,} \qquad Z = \alpha X + (1 - \alpha)Y.$$

Ask yourself Why does "clockwise" put the new triangles *outside* ABC?

Since $A \to B \to C$ is anticlockwise, C lies to the left of the directed side AB. Turning AB clockwise about A swings it to the right, so L lies on the opposite side of AB from C. The same holds for the other two sides, which matches the picture.

N (triangle CAN, so $X = C$, $Y = A$):

$$N = \alpha(-r\alpha) + (1 - \alpha)p = (1 - \alpha)p - \alpha^2 r,$$

as required. Using $1 - \alpha = -\alpha^2$, this tidies to $N = -\alpha^2(p + r)$.

L (triangle ABL, so $X = A$, $Y = B$):

$$L = \alpha p + (1 - \alpha)q\alpha^2 = \alpha p + q(\alpha^2 - \alpha^3) = \alpha p + q(1 + \alpha^2) = (p + q)\alpha,$$

using $\alpha^3 = -1$ and then $1 + \alpha^2 = \alpha$.

M (triangle BCM, so $X = B$, $Y = C$):

$$M = \alpha \cdot q\alpha^2 + (1-\alpha)(-r\alpha) = q\alpha^3 + r(\alpha^2 - \alpha) = -q - r = -(q+r),$$

using $\alpha^3 = -1$ and $\alpha^2 - \alpha = -1$.

So, in the same style as the given form of N,

$$L = \alpha p + \alpha q, \qquad M = -q - r, \qquad N = (1-\alpha)p - \alpha^2 r.$$

The three lines

Now look at each pair of points as a positive multiple of a unit complex number:

$$\begin{aligned} L &= (p+q)\,\mathrm{e}^{\mathrm{i}\pi/3}, & C &= -r\alpha = r\,\mathrm{e}^{\mathrm{i}\pi/3}\cdot(-1), \\ M &= (q+r)\cdot(-1), & A &= p\cdot 1, \\ N &= (p+r)\cdot(-\alpha^2), & B &= q\,\alpha^2. \end{aligned}$$

In each pair the two points are positive real multiples of u and $-u$ for a unit complex number u (namely $u = \alpha$, $u = 1$ and $u = \alpha^2$). So the two points lie on a common line through O, on *opposite* sides of O. Hence each of the lines LC, MA, NB passes through the origin, and in fact the origin lies on each segment. Then, because the points are on opposite sides of O, the length of each segment is the sum of the distances from O:

$$LC = (p+q) + r, \qquad MA = (q+r) + p, \qquad NB = (p+r) + q,$$

all equal to $p + q + r$, as required.

Checks

With $p = 1$, $q = 2$, $r = 3$ the formulas give $L = 3\alpha \approx 1.5 + 2.598\mathrm{i}$, $M = -5$ and $N = -4\alpha^2 \approx 2 - 3.464\mathrm{i}$. Measuring $|L - A| = |L - B| = |A - B| = \sqrt{7}$ confirms that ABL is equilateral. A computer built each triangle directly from the two rotations by $\pm 60^\circ$, kept the clockwise one, and confirmed the three formulas, the concurrence at O and the common length $p + q + r$ for 10 000 random choices of p, q, r. A nice special case: when $p = q = r$, triangle ABC is equilateral with centre O, and the figure has the expected threefold symmetry.

Common slips

- Rotating the wrong way, by α instead of $\bar{\alpha} = 1 - \alpha$. That puts L, M, N inside the triangle, and the given form of N won't come out.
- Showing only that O is on each line and forgetting that the length then needs O to lie *between* the endpoints. The signs of the real multiples settle this.

Beyond STEP

This is a special case of Napoleon's configuration: for *any* triangle with equilateral triangles erected outwards, the lines AM, BN, CL meet at a single point (the Fermat point) and have equal lengths. The Fermat point minimises the total distance to the three vertices; here it is O, where each side subtends $120°$. Complex numbers turn rotations into multiplications, and that is why proofs like this become short algebra. Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

WORKED SOLUTION

STEP III 2000, Question 4

Pure • Rational Function Sketching

STEP III 2000, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q4

The key idea

Setting $f(x) = x + c$ and clearing the denominator, the cubic terms cancel and we are left with a *quadratic* in x. Whether the line meets the curve then comes down to the sign of a discriminant, and that discriminant is itself a quadratic in c, so we take a discriminant of a discriminant. For the sketches, splitting f into a linear part plus partial fractions tells us everything: the asymptotes, which side of them the curve lies, and where it crosses the oblique one.

Solution

Where the line meets the curve

For $x \neq \pm 1$, the equation $f(x) = x + c$ is equivalent to

$$x(x-2)(x-a) = (x+c)(x^2-1).$$

Expanding, $x^3 - (a+2)x^2 + 2ax = x^3 + cx^2 - x - c$. The x^3 terms cancel, leaving

$$q(x) \equiv (a+2+c)x^2 - (2a+1)x - c = 0. \qquad (*)$$

So the line meets the curve exactly when $(*)$ has a real root other than $x = \pm 1$.

? **Ask yourself** Could a root of $(*)$ be one of the excluded values $x = \pm 1$?

Substituting, $q(1) = (a+2+c) - (2a+1) - c = 1 - a$ and $q(-1) = (a+2+c) + (2a+1) - c = 3(a+1)$. Neither depends on c, so $x = 1$ is a root only when $a = 1$, and $x = -1$ is a root only when $a = -1$. For $|a| \neq 1$ every real root of $(*)$ gives a genuine intersection.

The case $a + 2 + c = 0$. Then $(*)$ is linear: $-(2a+1)x + (a+2) = 0$. If $|a| > 1$ then $2a + 1 \neq 0$, so there is the root $x = \frac{a+2}{2a+1}$.

The case $a+2+c\neq 0$. Then $(*)$ is a genuine quadratic with discriminant

$$D(c)=(2a+1)^2+4c(a+2+c)=4c^2+4(a+2)c+(2a+1)^2.$$

Think of D as a quadratic in c with positive leading coefficient. Its own discriminant is

$$\begin{aligned}16(a+2)^2-16(2a+1)^2&=16\big[(a+2)-(2a+1)\big]\big[(a+2)+(2a+1)\big]\\&=16(1-a)\cdot 3(1+a)=48(1-a^2).\end{aligned}$$

If $|a|>1$, this is negative, so $D(c)>0$ for every c; then $(*)$ has two distinct real roots, neither equal to ± 1. Together with the linear case, the line $y=x+c$ meets the curve for every c.

If $|a|=1$, $D(c)\geq 0$ for every c, so $(*)$ still always has real roots, which is the algebraic content of the claim. But here the curve itself degenerates: a factor cancels and the curve is a hyperbola with a single point missing. For instance, with $a=1$ we get $f(x)=x-3+\frac{3}{x+1}$ $(x\neq 1)$, and the line $y=x-3$ (its asymptote) does not meet it; in $(*)$ this is $c=-3$, where the only root is the excluded $x=1$. So for $|a|=1$ the statement holds for all but these exceptional values of c, and it holds without exception for $|a|>1$.

If $|a|<1$, the quadratic $D(c)$ has two real roots and is negative strictly between them. An explicit choice is $c=-\frac{1}{2}(a+2)$, the midpoint of the roots:

$$D\big(-\tfrac{1}{2}(a+2)\big)=(a+2)^2-2(a+2)^2+(2a+1)^2=3(a^2-1)<0.$$

For this c, $a+2+c=\frac{1}{2}(a+2)\neq 0$ (since $a>-1$), so $(*)$ is a genuine quadratic with negative discriminant and has no real roots. Hence the line $y=x-\frac{1}{2}(a+2)$ misses the curve, as required.

The oblique asymptote

Dividing $x^3-(a+2)x^2+2ax$ by x^2-1,

$$f(x)=x-(a+2)+\frac{(2a+1)x-(a+2)}{x^2-1}.$$

The fraction tends to 0 as $x\to\pm\infty$, so the oblique asymptote is

$$y=x-a-2.$$

For the sketches it helps to split the fraction into partial fractions:

$$f(x)=x-(a+2)+\frac{a-1}{2(x-1)}+\frac{3(a+1)}{2(x+1)}.$$

This gives us everything:

- zeros at $x=0,\ 2,\ a$, and vertical asymptotes at $x=\pm 1$;
- the curve meets its oblique asymptote only where $(2a+1)x=a+2$, that is at $x_0=\frac{a+2}{2a+1}$ (this is the linear case of $(*)$ again);

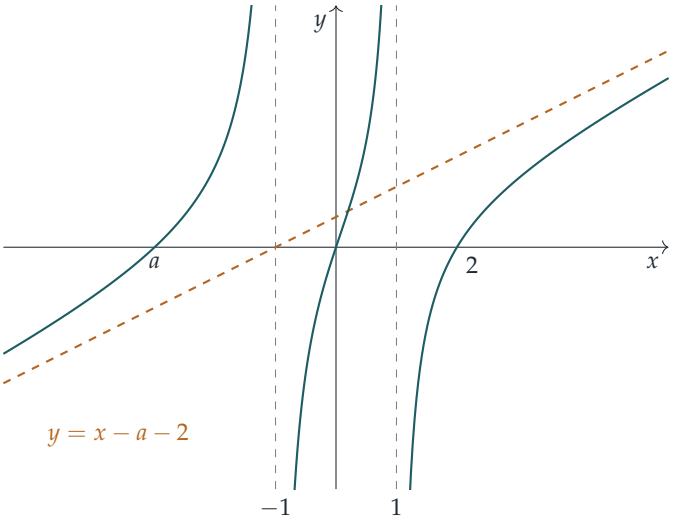
- for large $|x|$ the gap $f(x) - (x - a - 2)$ behaves like $\frac{2a+1}{x}$. In both cases $2a + 1 < 0$, so the curve lies *below* the asymptote as $x \to +\infty$ and *above* it as $x \to -\infty$;
- the coefficients $\frac{a-1}{2}$ and $\frac{3(a+1)}{2}$ fix the behaviour on each side of $x = 1$ and $x = -1$.

Case (i): $a < -1$

Both coefficients are negative: $\frac{a-1}{2} < 0$ and $\frac{3(a+1)}{2} < 0$. So $f(x) \to +\infty$ as $x \to 1^-$ or $x \to -1^-$, and $f(x) \to -\infty$ as $x \to 1^+$ or $x \to -1^+$. Moreover

$$f'(x) = 1 - \frac{a-1}{2(x-1)^2} - \frac{3(a+1)}{2(x+1)^2} > 0,$$

since both subtracted terms are negative, so f is increasing on each of its three branches and there are no turning points at all. The crossing point satisfies $x_0 + 1 = \frac{3(a+1)}{2a+1} > 0$ and $x_0 - 1 = \frac{1-a}{2a+1} < 0$, so $-1 < x_0 < 1$: the curve crosses its asymptote on the middle branch. The zeros are at $a < -1$ (left branch), 0 (middle) and 2 (right).



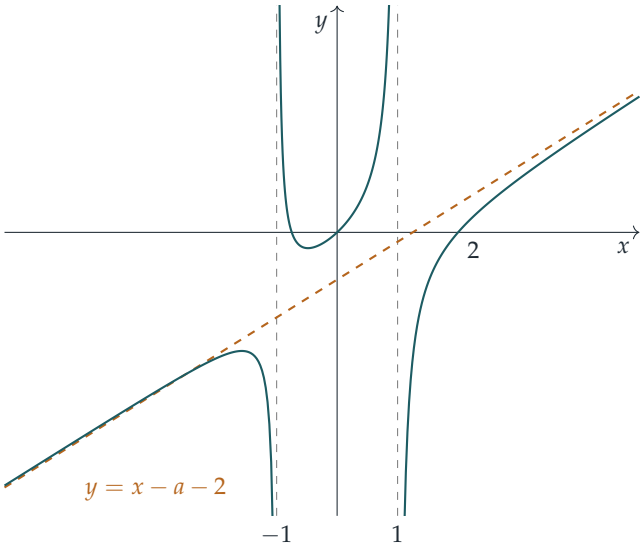
(Drawn for $a = -3$, where the asymptote is $y = x + 1$ and $x_0 = \frac{1}{5}$.)

Case (ii): $-1 < a < -\frac{1}{2}$

Now $\frac{a-1}{2} < 0$ but $\frac{3(a+1)}{2} > 0$. Near $x = 1$ the behaviour is as before ($+\infty$ from the left, $-\infty$ from the right), but near $x = -1$ it flips: $f(x) \to -\infty$ as $x \to -1^-$ and $f(x) \to +\infty$ as $x \to -1^+$. Now $x_0 + 1 = \frac{3(a+1)}{2a+1} < 0$, so $x_0 < -1$: the crossing is on the left branch. The zeros a and 0 both lie in $(-1, 0)$.

Ask yourself Where must the turning points be, even though we are not asked to find them?

On the left branch, the curve comes up from $-\infty$ above the asymptote, crosses it at x_0, then falls to $-\infty$ as $x \to -1^-$; so it has a maximum. It has no zeros there (all three zeros are > -1), so the whole branch lies below the x-axis. On the middle branch $f \to +\infty$ at both ends, it dips below the axis between a and 0 and is positive elsewhere, so there is a minimum between a and 0. On the right branch, $\frac{3(a+1)}{2(x+1)^2} < \frac{3}{4} \cdot \frac{1}{4} < 1$ for $x > 1$, so $f'(x) > 0$: the curve rises from $-\infty$ through $x = 2$ and approaches the asymptote from below.



(Drawn for $a = -\frac{3}{4}$: asymptote $y = x - \frac{5}{4}$, crossing at $x_0 = -\frac{5}{2}$, and a shallow dip below the axis between $-\frac{3}{4}$ and 0.)

Checks

- With $a = 0$ and $c = -1$ (our choice $c = -\frac{1}{2}(a+2)$), $(*)$ becomes $x^2 - x + 1 = 0$, whose discriminant is -3: no intersection, as predicted.
- With $a = -3$: $f(-1.1) \approx 30.9$ and $f(-0.9) \approx -28.8$, which matches the signs found for $x \to -1^{\mp}$ in case (i). With $a = -\frac{3}{4}$, $f(-1.1) \approx -5.7$ and $f(-0.9) \approx 2.1$, the flipped behaviour of case (ii).
- Counting sign changes of f' numerically for several a in each range confirms no turning points in case (i), and exactly one on each of the left and middle branches in case (ii).

Common slips

- Forgetting that roots $x = \pm 1$ of $(*)$ are not intersections; this is exactly what goes wrong when $|a| = 1$.

- Ignoring the case $a + 2 + c = 0$, where $(*)$ is linear and the discriminant argument does not apply.
- Getting the side of the vertical asymptote $x = -1$ wrong in case (ii): the sign of $a + 1$ changes the picture there.

Beyond STEP

When $|a| < 1$, the two roots of $D(c)$ give lines $y = x + c$ that are tangent to the curve, and the lines between them miss it. As a crosses ± 1 these tangents appear or disappear and the curve changes shape: a zero passes through a vertical asymptote and the behaviour beside it flips. Studying how the qualitative shape of a family of curves changes with a parameter is the start of bifurcation theory, met in first-year courses on differential equations; see Boyce and DiPrima, *Elementary Differential Equations*. For more practice at this kind of sketching, Siklos, *Advanced Problems in Mathematics*, is excellent.

WORKED SOLUTION

STEP III 2000, Question 5

Pure • Linear Transformations

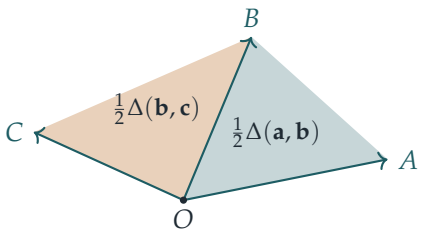
STEP III 2000, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q5

The key idea

$\Delta(\mathbf{a}, \mathbf{b})$ is the determinant of the matrix whose columns are $\mathbf{a}$ and $\mathbf{b}$, so it is twice the signed area of the triangle OAB, and it is non-zero exactly when $\mathbf{a}$ and $\mathbf{b}$ are not parallel. A linear map multiplies every such area by the same factor, its determinant. That one fact gives necessity at once. For sufficiency, the identity in (ii) writes $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ using only Δ's, so once $\mathbf{M}$ is pinned down by A and B, it has no choice about where C goes.



Solution

Throughout, write $[\mathbf{u}\ \mathbf{v}]$ for the 2×2 matrix with columns $\mathbf{u}$ and $\mathbf{v}$. Then

$$\Delta(\mathbf{u}, \mathbf{v}) = u_1 v_2 - u_2 v_1 = \det[\mathbf{u}\ \mathbf{v}].$$

For non-zero $\mathbf{u}$, $\mathbf{v}$, we have $\Delta(\mathbf{u}, \mathbf{v}) = 0$ exactly when $\mathbf{u}$ and $\mathbf{v}$ are parallel: the determinant vanishes precisely when one column is a multiple of the other. So all six of $\Delta(\mathbf{a}, \mathbf{b})$, $\Delta(\mathbf{b}, \mathbf{c})$, $\Delta(\mathbf{c}, \mathbf{a})$, $\Delta(\mathbf{p}, \mathbf{q})$, $\Delta(\mathbf{q}, \mathbf{r})$, $\Delta(\mathbf{r}, \mathbf{p})$ are non-zero.

Two more facts get used repeatedly. Since $\mathbf{M}[\mathbf{u}\ \mathbf{v}] = [\mathbf{M}\mathbf{u}\ \ \mathbf{M}\mathbf{v}]$ (column by column) and determinants multiply,

$$\Delta(\mathbf{M}\mathbf{u}, \mathbf{M}\mathbf{v}) = \det \mathbf{M}\ \Delta(\mathbf{u}, \mathbf{v}). \qquad (*)$$

And $\Delta(\mathbf{v}, \mathbf{u}) = -\Delta(\mathbf{u}, \mathbf{v})$.

Part (i)

We need $\mathbf{Ma} = \mathbf{p}$ and $\mathbf{Mb} = \mathbf{q}$, which together say

$$\mathbf{M}\,[\mathbf{a}\ \mathbf{b}] = [\mathbf{p}\ \mathbf{q}].$$

Since $\mathbf{a}$ and $\mathbf{b}$ are not parallel, $\det[\mathbf{a}\ \mathbf{b}] = \Delta(\mathbf{a},\mathbf{b}) \neq 0$, so $[\mathbf{a}\ \mathbf{b}]$ is invertible, and

$$\mathbf{M} = [\mathbf{p}\ \mathbf{q}]\,[\mathbf{a}\ \mathbf{b}]^{-1} = \frac{1}{\Delta(\mathbf{a},\mathbf{b})}\begin{pmatrix} p_1 & q_1 \\ p_2 & q_2 \end{pmatrix}\begin{pmatrix} b_2 & -b_1 \\ -a_2 & a_1 \end{pmatrix}$$

does the job. In fact this $\mathbf{M}$ is the *only* such matrix, because we can multiply on the right by the inverse. Taking determinants, $\det\mathbf{M} = \Delta(\mathbf{p},\mathbf{q})/\Delta(\mathbf{a},\mathbf{b})$, which is non-zero.

Part (ii): the identity

Take the first component of the left-hand side:

$$\begin{aligned}&(a_1b_2 - a_2b_1)c_1 + (c_1a_2 - c_2a_1)b_1 + (b_1c_2 - b_2c_1)a_1\\ &\quad = a_1b_2c_1 - a_2b_1c_1 + a_2b_1c_1 - a_1b_1c_2 + a_1b_1c_2 - a_1b_2c_1 = 0,\end{aligned}$$

since the six terms cancel in pairs. The second component is the same calculation with every subscript on $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ outside the Δ's changed from 1 to 2:

$$\begin{aligned}&(a_1b_2 - a_2b_1)c_2 + (c_1a_2 - c_2a_1)b_2 + (b_1c_2 - b_2c_1)a_2\\ &\quad = a_1b_2c_2 - a_2b_1c_2 + a_2b_2c_1 - a_1b_2c_2 + a_2b_1c_2 - a_2b_2c_1 = 0.\end{aligned}$$

So

$$\Delta(\mathbf{a},\mathbf{b})\,\mathbf{c} + \Delta(\mathbf{c},\mathbf{a})\,\mathbf{b} + \Delta(\mathbf{b},\mathbf{c})\,\mathbf{a} = \mathbf{0}, \tag{1}$$

as required. The identity holds for *any* three vectors, so it holds for $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ too:

$$\Delta(\mathbf{p},\mathbf{q})\,\mathbf{r} + \Delta(\mathbf{r},\mathbf{p})\,\mathbf{q} + \Delta(\mathbf{q},\mathbf{r})\,\mathbf{p} = \mathbf{0}. \tag{2}$$

Ask yourself What exactly does the ratio condition say, as an equation?

All the Δ's are non-zero, so the condition $\Delta(\mathbf{a},\mathbf{b}) : \Delta(\mathbf{b},\mathbf{c}) : \Delta(\mathbf{c},\mathbf{a}) = \Delta(\mathbf{p},\mathbf{q}) : \Delta(\mathbf{q},\mathbf{r}) : \Delta(\mathbf{r},\mathbf{p})$ means there is a single number $k \neq 0$ with

$$\Delta(\mathbf{p},\mathbf{q}) = k\,\Delta(\mathbf{a},\mathbf{b}), \qquad \Delta(\mathbf{q},\mathbf{r}) = k\,\Delta(\mathbf{b},\mathbf{c}), \qquad \Delta(\mathbf{r},\mathbf{p}) = k\,\Delta(\mathbf{c},\mathbf{a}). \tag{3}$$

In words: the three signed triangle areas OPQ, OQR, ORP are the areas OAB, OBC, OCA all scaled by the same factor.

Part (ii): necessity

Suppose $\mathbf{Ma} = \mathbf{p}$, $\mathbf{Mb} = \mathbf{q}$ and $\mathbf{Mc} = \mathbf{r}$. By $(*)$,

$$\Delta(\mathbf{p},\mathbf{q}) = \det\mathbf{M}\,\Delta(\mathbf{a},\mathbf{b}), \quad \Delta(\mathbf{q},\mathbf{r}) = \det\mathbf{M}\,\Delta(\mathbf{b},\mathbf{c}), \quad \Delta(\mathbf{r},\mathbf{p}) = \det\mathbf{M}\,\Delta(\mathbf{c},\mathbf{a}).$$

This is (3) with $k = \det\mathbf{M}$, and $k \neq 0$ because $\Delta(\mathbf{p},\mathbf{q}) \neq 0$. So the ratios are equal.

Part (ii): sufficiency

Now suppose (3) holds for some $k \neq 0$. By part (i) there is a matrix $\mathbf{M}$ with $\mathbf{Ma} = \mathbf{p}$ and $\mathbf{Mb} = \mathbf{q}$; we must show that this same $\mathbf{M}$ also sends $\mathbf{c}$ to $\mathbf{r}$.

Apply $\mathbf{M}$ to (1). Matrix multiplication is linear and the Δ's are just numbers, so

$$\Delta(\mathbf{a},\mathbf{b})\,\mathbf{Mc} = -\Delta(\mathbf{c},\mathbf{a})\,\mathbf{q} - \Delta(\mathbf{b},\mathbf{c})\,\mathbf{p}.$$

Multiply by k and use (3) on each term:

$$\Delta(\mathbf{p},\mathbf{q})\,\mathbf{Mc} = -\Delta(\mathbf{r},\mathbf{p})\,\mathbf{q} - \Delta(\mathbf{q},\mathbf{r})\,\mathbf{p}.$$

By (2), the right-hand side equals $\Delta(\mathbf{p},\mathbf{q})\,\mathbf{r}$. So

$$\Delta(\mathbf{p},\mathbf{q})\,\mathbf{Mc} = \Delta(\mathbf{p},\mathbf{q})\,\mathbf{r},$$

and since $\Delta(\mathbf{p},\mathbf{q}) \neq 0$ (as $\mathbf{p}$, $\mathbf{q}$ are not parallel), $\mathbf{Mc} = \mathbf{r}$. Hence P, Q, R are the images of A, B, C under $\mathbf{M}$.

Necessity and sufficiency together prove that the ratio condition is necessary and sufficient.

Checks

- Try $\mathbf{a} = \binom{1}{0}$, $\mathbf{b} = \binom{0}{1}$, $\mathbf{c} = \binom{1}{1}$. Then $\Delta(\mathbf{a},\mathbf{b}) = 1$, $\Delta(\mathbf{b},\mathbf{c}) = -1$, $\Delta(\mathbf{c},\mathbf{a}) = -1$, and (1) reads $\binom{1}{1} - \binom{0}{1} - \binom{1}{0} = \mathbf{0}$. With $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ we get $\mathbf{p} = \binom{2}{0}$, $\mathbf{q} = \binom{1}{3}$, $\mathbf{r} = \binom{3}{3}$, and the three Δ's are 6, -6, -6: the ratio $1 : -1 : -1$ again, with $k = 6 = \det \mathbf{M}$.
- Numerically, we tested the identity on 20,000 random integer triples, using exact fractions. For about 17,700 triples with no two vectors parallel, we built $\mathbf{M}$ from part (i) and confirmed necessity, and that $\mathbf{Mc} = \mathbf{r}$ exactly when (3) holds.

Common slips

- Writing the ratio condition as one equation of fractions without saying the Δ's are non-zero. The hypotheses that no two vectors are parallel are there precisely to make every division legal.
- Proving only necessity. The real content is sufficiency: a matrix is fixed by its action on two independent vectors, so the condition must force the third image.
- Sign errors from the order inside Δ: the cyclic order $\mathbf{b},\mathbf{c}$ then $\mathbf{c},\mathbf{a}$ matters, since $\Delta(\mathbf{a},\mathbf{c}) = -\Delta(\mathbf{c},\mathbf{a})$.

Beyond STEP

Identity (1) is Cramer's rule in disguise: it gives the coordinates of $\mathbf{c}$ in the basis $\mathbf{a}$, $\mathbf{b}$ as ratios of determinants. The fact $(*)$, that a linear map scales every signed area by $\det \mathbf{M}$,

is the geometric definition of the determinant, and it is behind the Jacobian factor in a change of variables for double integrals. Strang's *Introduction to Linear Algebra* develops both ideas.

WORKED SOLUTION

STEP III 2000, Question 6

Pure • Factorising Quartic Polynomials

STEP III 2000, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q6

The key idea

Comparing coefficients turns the factorisation into three equations linking p, q, r to a, b, c. To get an equation for a alone we must eliminate b and c, and the identity $4bc = (b+c)^2 - (b-c)^2$ does exactly that, because $b+c$ and $b-c$ are the combinations that appear naturally. For the final quartic the cubic needs the y^3 term removed first, which a shift $y = x + k$ achieves.

Solution

Finding p, q and r

Expanding the right-hand side,

$$(x^2 - ax + b)(x^2 + ax + c) = x^4 + (b + c - a^2)x^2 + a(b-c)x + bc,$$

where the x^3 terms $ax^3 - ax^3$ cancel. Comparing coefficients,

$$p = b + c - a^2, \qquad q = a(b-c), \qquad r = bc.$$

The cubic for a^2

Write $u = a^2$. From the first equation, $b + c = p + u$.

If $a \neq 0$, the second equation gives $b - c = q/a$, so $(b-c)^2 = q^2/u$. Now

$$4bc = (b+c)^2 - (b-c)^2 \quad \Longrightarrow \quad 4r = (p+u)^2 - \frac{q^2}{u}.$$

Multiplying by $u \neq 0$,

$$4ru = u(u^2 + 2pu + p^2) - q^2 \quad \Longrightarrow \quad u^3 + 2pu^2 + (p^2 - 4r)u - q^2 = 0,$$

so $u = a^2$ is a root.

If $a = 0$, then $q = a(b - c) = 0$, and $u = a^2 = 0$ is a root because every term of the cubic then vanishes at $u = 0$. So in every case a^2 is a root, as required.

Ask yourself Why must there be a root with $u \geq 0$, whatever p, q and r are?

Call the cubic $f(u)$. At $u = 0$ we have $f(0) = -q^2 \leq 0$, and since the leading term is u^3, $f(u) \to +\infty$ as $u \to +\infty$. If $q = 0$ then $u = 0$ is itself a root. Otherwise $f(0) < 0$ and $f(U) > 0$ for some large U, so because f is continuous, the intermediate value theorem gives a root strictly between 0 and U. Either way there is a non-negative root.

This matters because we need $a = \sqrt{u}$ to be real. A non-negative root always gives a real a, so every real quartic with no x^3 term can be split into two real quadratics of the given form. (If $u > 0$ we can recover b and c from $b + c = p + u$ and $b - c = q/\sqrt{u}$.)

The case $p = -1$, $q = -6$, $r = 15$

Here $p^2 - 4r = 1 - 60 = -59$, so the cubic is

$$u^3 - 2u^2 - 59u - 36 = 0.$$

At $u = 9$: $729 - 162 - 531 - 36 = 0$, so $u = 9$ is a root. (Dividing out gives $(u - 9)(u^2 + 7u + 4)$, whose quadratic factor has two negative roots, so $u = 9$ is the only non-negative root.)

Factorising the quartic

The given quartic has a y^3 term, so we shift to remove it. The roots sum to 8, so their mean is 2, and we put $y = x + 2$. Expanding each term:

$$\begin{aligned}
(x+2)^4 &= x^4 + 8x^3 + 24x^2 + 32x + 16,\\
-8(x+2)^3 &= -8x^3 - 48x^2 - 96x - 64,\\
23(x+2)^2 &= 23x^2 + 92x + 92,\\
-34(x+2) &= -34x - 68,
\end{aligned}$$

and adding these and 39,

$$y^4 - 8y^3 + 23y^2 - 34y + 39 = x^4 - x^2 - 6x + 15.$$

This is precisely the case $p = -1$, $q = -6$, $r = 15$, which is why that case appeared.

Take $u = a^2 = 9$ with $a = 3$. Then

$$b + c = p + a^2 = 8, \qquad b - c = \frac{q}{a} = -2,$$

so $b = 3$ and $c = 5$. As a check, $bc = 15 = r$. Hence

$$x^4 - x^2 - 6x + 15 = (x^2 - 3x + 3)(x^2 + 3x + 5).$$

Finally substitute $x = y - 2$:

$$x^2 - 3x + 3 = (y^2 - 4y + 4) - (3y - 6) + 3 = y^2 - 7y + 13,$$
$$x^2 + 3x + 5 = (y^2 - 4y + 4) + (3y - 6) + 5 = y^2 - y + 3.$$

So

$$y^4 - 8y^3 + 23y^2 - 34y + 39 = (y^2 - 7y + 13)(y^2 - y + 3).$$

Taking $a = -3$ instead swaps b and c and gives the same two factors in the other order.

Checks

- Expand the answer: $(y^2 - 7y + 13)(y^2 - y + 3) = y^4 - 8y^3 + (3 + 7 + 13)y^2 - (21 + 13)y + 39$, which matches.
- Put $y = 1$: the quartic is $1 - 8 + 23 - 34 + 39 = 21$, and the factors give $7 \times 3 = 21$.
- Both quadratic factors have negative discriminant (-3 and -11), so neither splits further over the reals, and the quartic has no real roots. That fits $y = 1$ giving a positive value.

We also tested the formulae for p, q, r and the cubic on 2000 random rational choices of a, b, c, and found a non-negative root by bisection for 2000 random (p, q, r).

Common slips

- Dividing by a to get $b - c = q/a$ without dealing with the case $a = 0$.
- Arguing that "a cubic always has a real root", which is true but doesn't show the root is non-negative. The sign of $f(0) = -q^2$ is the point.
- Forgetting to substitute back from x to y at the end.

Beyond STEP

This is essentially Descartes' (and Ferrari's) method for solving the general quartic: shift away the cubic term, then factor into two quadratics by solving an auxiliary cubic, the *resolvent cubic*. Combined with Cardano's formula for cubics, it shows that every quartic can be solved by radicals. Why this stops at degree four, with no such formula for the general quintic, is the subject of Galois theory. Allenby's *Rings, Fields and Groups* is a good place to start.

WORKED SOLUTION

STEP III 2000, Question 7

Pure • Inequalities and Limits

STEP III 2000, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q7

The key idea

Both inequalities come from comparing a product with something we already understand. Expanding $\left(1+\frac{1}{n}\right)^n$ by the binomial theorem gives terms that look just like $\frac{1}{r!}$, only a little smaller. For the product $\mathrm{P}(n)$, the AM–GM inequality turns a product of n factors into an nth power of their average. The factors are $1+\frac{1}{2^k}$, and their average is just under $1+\frac{1}{n}$, which links straight back to the first part.

Solution

The bound $\left(1+\frac{1}{n}\right)^n < \mathrm{e}$

By the binomial theorem,

$$\left(1+\frac{1}{n}\right)^n = \sum_{r=0}^{n}\binom{n}{r}\frac{1}{n^r} = \sum_{r=0}^{n}\frac{1}{r!}\cdot\frac{n(n-1)(n-2)\cdots(n-r+1)}{n^r}.$$

The second fraction is a product of r factors,

$$\frac{n(n-1)\cdots(n-r+1)}{n^r} = 1\cdot\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{r-1}{n}\right),$$

each lying in $(0,1]$ because $0 \le j \le r-1 \le n-1$. So the product is at most 1, and every term of the expansion is at most $\frac{1}{r!}$. Hence

$$\left(1+\frac{1}{n}\right)^n \le \sum_{r=0}^{n}\frac{1}{r!} < \sum_{r=0}^{\infty}\frac{1}{r!} = \mathrm{e},$$

where the last inequality is strict because the omitted terms $\frac{1}{(n+1)!}, \frac{1}{(n+2)!}, \ldots$ are all positive. This holds for every positive integer n.

The bound $\mathrm{P}(n) < \mathrm{e}$

Write $\mathrm{P}(n) = a_1 a_2 \cdots a_n$ with $a_k = \dfrac{2^k + 1}{2^k} = 1 + \dfrac{1}{2^k}$, which are all positive, so AM–GM applies. The arithmetic mean is

$$\frac{a_1 + \cdots + a_n}{n} = \frac{1}{n}\left(n + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n}\right) = 1 + \frac{1 - 2^{-n}}{n},$$

by summing the geometric series $\frac{1}{2} + \cdots + \frac{1}{2^n} = 1 - 2^{-n}$. Since $1 - 2^{-n} < 1$, this mean is strictly less than $1 + \frac{1}{n}$. So AM–GM gives

$$\mathrm{P}(n)^{1/n} \leq 1 + \frac{1 - 2^{-n}}{n} < 1 + \frac{1}{n}.$$

Both sides are positive, so raising to the nth power keeps the inequality:

$$\mathrm{P}(n) < \left(1 + \frac{1}{n}\right)^n < \mathrm{e},$$

using the first part for the second inequality. This holds for every positive integer n.

> **Ask yourself** Where does the strict inequality come from? AM–GM only gives $\leq$.

The strictness comes from $1 - 2^{-n} < 1$, not from AM–GM. In fact, for $n \geq 2$ the factors a_k are all different, so AM–GM is itself strict. But when $n = 1$ there is only one factor, and AM–GM is an equality ($\frac{3}{2} = \frac{3}{2}$). The step $1 + \frac{1-2^{-n}}{n} < 1 + \frac{1}{n}$ is strict in every case, so the argument covers all n.

The limit

Each factor $a_{n+1} = 1 + 2^{-(n+1)}$ is greater than 1, so $\mathrm{P}(n+1) = a_{n+1}\mathrm{P}(n) > \mathrm{P}(n)$. The sequence $\mathrm{P}(n)$ is therefore strictly increasing, and it is bounded above by e. An increasing sequence that is bounded above converges, so $\mathrm{P}(n)$ tends to a limit L.

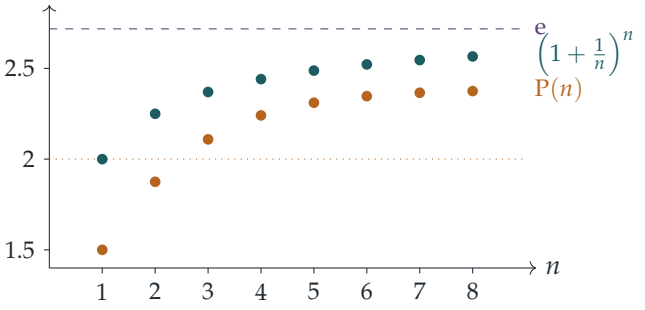
For the upper bound: every $\mathrm{P}(n) < \mathrm{e}$, so the limit satisfies $L \leq \mathrm{e}$. (Taking limits only preserves weak inequalities, which is why we get $\leq$ and not $<$.)

For the lower bound, compute a few terms:

$$\mathrm{P}(1) = \frac{3}{2}, \qquad \mathrm{P}(2) = \frac{3}{2} \cdot \frac{5}{4} = \frac{15}{8}, \qquad \mathrm{P}(3) = \frac{15}{8} \cdot \frac{9}{8} = \frac{135}{64} > \frac{128}{64} = 2.$$

Since the sequence is increasing, $L \geq \mathrm{P}(4) > \mathrm{P}(3) > 2$. Together,

$$2 < L \leq \mathrm{e}.$$



Checks

- $n = 1$: $(1+1)^1 = 2 <$ e and $\mathrm{P}(1) = 1.5 < 2$. For $n = 4$: $\mathrm{P}(4) = \frac{2295}{1024} \approx 2.241 < \left(\frac{5}{4}\right)^4 \approx 2.441$.
- Computing the product numerically, $\mathrm{P}(n)$ settles at $L \approx 2.3842$ (by $n = 60$ it is constant to many decimal places), which is comfortably between 2 and e ≈ 2.7183.

Common slips

- Claiming $L <$ e just because each $\mathrm{P}(n) <$ e. Strict inequalities can become equalities in the limit; only $L \leq$ e follows. (Here $L <$ e is in fact true, but it needs a separate argument.)
- Showing $\mathrm{P}(n) > 2$ for some n but forgetting to say that the sequence is increasing, which is what passes the bound on to L.
- Applying AM–GM without checking that the a_k are positive.

Beyond STEP

The step "increasing and bounded above, so convergent" is the monotone convergence theorem, which is equivalent to the completeness of the real numbers. Proving it from first principles is one of the first things a university analysis course does. Taking logarithms turns $\mathrm{P}(n)$ into the series $\sum \ln\left(1+2^{-k}\right)$, which converges by comparison with $\sum 2^{-k}$. That is the standard way to study infinite products. Abbott's *Understanding Analysis* treats both ideas beautifully.

WORKED SOLUTION

STEP III 2000, Question 8

Pure • Recurrence Relations

STEP III 2000, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q8

The key idea

The defining rule is non-linear and involves division, so it is awkward to work with directly. The trick is to rewrite it as a product, $a_n a_{n-2} - a_{n-1}^2 = 1$. This says a certain combination of three consecutive terms is always 1. Writing the same fact one step further on and subtracting the two versions turns the non-linear rule into the linear one. Once the recurrence is linear, the closed form is the standard second-order recurrence machinery: roots of the auxiliary quadratic, then matching two starting values.

Solution

A preliminary: the sequence is well defined

Each step divides by a_{n-2}, so we should make sure no term is zero. In fact every term is positive. We have $a_0 = a_1 = 1 > 0$. If a_{n-1} and a_{n-2} are positive, then $1 + a_{n-1}^2 > 0$ and $a_{n-2} > 0$, so $a_n > 0$. By induction, $a_n > 0$ for all n. The first few terms are

$$a_0 = 1, \quad a_1 = 1, \quad a_2 = \frac{1+1}{1} = 2, \quad a_3 = \frac{1+4}{1} = 5, \quad a_4 = \frac{1+25}{2} = 13.$$

The linear recurrence

Multiplying the definition by a_{n-2} gives, for every $n \geq 2$,

$$a_n a_{n-2} = 1 + a_{n-1}^2. \qquad (*)$$

Let $P(k)$ be the statement $a_k = 3a_{k-1} - a_{k-2}$. We prove $P(k)$ for all $k \geq 2$ by induction on k.

Base case. $a_2 = 2$ and $3a_1 - a_0 = 3 - 1 = 2$, so $P(2)$ is true.

Inductive step. Suppose $P(k)$ is true for some $k \geq 2$, so that

$$a_k + a_{k-2} = 3a_{k-1}.$$

We need to show $a_{k+1} = 3a_k - a_{k-1}$. By definition, $a_{k+1} = (1 + a_k^2)/a_{k-1}$. The idea is to replace the awkward 1 in the numerator using $(*)$ with $n = k$, which says $1 = a_k a_{k-2} - a_{k-1}^2$:

$$\begin{aligned} a_{k+1} &= \frac{a_k a_{k-2} - a_{k-1}^2 + a_k^2}{a_{k-1}} \\ &= \frac{a_k\,(a_k + a_{k-2}) - a_{k-1}^2}{a_{k-1}} \\ &= \frac{a_k \cdot 3a_{k-1} - a_{k-1}^2}{a_{k-1}} && \text{by } P(k) \\ &= 3a_k - a_{k-1} && \text{since } a_{k-1} > 0. \end{aligned}$$

So $P(k+1)$ is true. By induction, $a_n = 3a_{n-1} - a_{n-2}$ for all $n \geq 2$.

> **Ask yourself** Where did the inductive hypothesis come in, and why is $(*)$ at $n = k$ fair to use?

The hypothesis was used once, to replace $a_k + a_{k-2}$ by $3a_{k-1}$. That bracket only appears because we swapped the 1 for $a_k a_{k-2} - a_{k-1}^2$. Using $(*)$ needs no hypothesis at all: it is just the definition of a_k multiplied out, valid for every $k \geq 2$. In other words, the quantity $a_n a_{n-2} - a_{n-1}^2$ is the same at every step, and that is what carries the linear relation forward.

The closed form

Write $\beta = \alpha^{-1}$. Since $\alpha = \frac{1+\sqrt{5}}{2}$,

$$\beta = \frac{2}{1+\sqrt{5}} = \frac{2(\sqrt{5}-1)}{5-1} = \frac{\sqrt{5}-1}{2}, \qquad \text{so} \qquad \alpha + \beta = \sqrt{5}, \quad \alpha\beta = 1.$$

Hence

$$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 5 - 2 = 3, \qquad \alpha^2\beta^2 = 1,$$

so α^2 and β^2 are the two roots of the auxiliary equation $x^2 - 3x + 1 = 0$ of our recurrence. (Explicitly, they are $\frac{3\pm\sqrt{5}}{2}$.)

Now define, for $n \geq 1$,

$$b_n = \frac{\alpha^{2n-1} + \beta^{2n-1}}{\sqrt{5}}.$$

We show $b_n = a_n$ by checking that b_n has the same first two terms as a_n and obeys the same recurrence.

It obeys the recurrence. If x is a root of $x^2 = 3x - 1$, then multiplying by x^{n-2} gives $x^n = 3x^{n-1} - x^{n-2}$. Since $\alpha^{2n-1} = \beta\,(\alpha^2)^n$ (using $\alpha\beta = 1$), taking $x = \alpha^2$ and multiplying by β gives

$$\alpha^{2n-1} = 3\alpha^{2n-3} - \alpha^{2n-5}.$$

In the same way, with $x = \beta^2$ and multiplying by α, we get $\beta^{2n-1} = 3\beta^{2n-3} - \beta^{2n-5}$. Adding and dividing by $\sqrt{5}$:

$$b_n = 3b_{n-1} - b_{n-2} \qquad (n \geq 3).$$

The starting values agree.

$$b_1 = \frac{\alpha + \beta}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{5}} = 1 = a_1,$$
$$b_2 = \frac{\alpha^3 + \beta^3}{\sqrt{5}} = \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)}{\sqrt{5}} = \frac{5\sqrt{5} - 3\sqrt{5}}{\sqrt{5}} = 2 = a_2.$$

Conclusion. Both (a_n) and (b_n) satisfy $u_n = 3u_{n-1} - u_{n-2}$ for $n \geq 3$, and they agree at $n = 1$ and $n = 2$. A recurrence like this determines each term from the two before it, so by (strong) induction $a_n = b_n$ for every $n \geq 1$. That is,

$$a_n = \frac{\alpha^{2n-1} + \alpha^{-(2n-1)}}{\sqrt{5}} \qquad (n \geq 1),$$

as required.

Ask yourself Why start the comparison at $n = 1$ rather than $n = 0$?

The linear recurrence for a_n at $n = 3$ involves a_2 and a_1, so matching b_1 and b_2 is all we need for $n \geq 1$. In fact the formula works at $n = 0$ too: $b_0 = (\alpha^{-1} + \alpha)/\sqrt{5} = 1 = a_0$. So nothing is lost either way.

Checks

- The terms are $1, 1, 2, 5, 13, 34, 89, 233, \ldots$, the Fibonacci numbers $F_1, F_3, F_5, \ldots$ (with $a_0 = 1 = F_{-1}$). Each is three times the previous one minus the one before: $3 \cdot 13 - 5 = 34$.
- Check $(*)$ directly: $34 \cdot 5 - 13^2 = 170 - 169 = 1$.
- Numerically, $a_{10} = 4181$, and $(\alpha^{19} + \alpha^{-19})/\sqrt{5} \approx 4181.000$. The term $\alpha^{-19}/\sqrt{5}$ is tiny, so a_n is the nearest integer to $\alpha^{2n-1}/\sqrt{5}$.
- Computing the sequence exactly from the original rule up to $n = 39$ gives integers satisfying the linear recurrence every time, matching the closed form.

Common slips

- Trying to prove $a_{k+1} = 3a_k - a_{k-1}$ by substituting $a_k = 3a_{k-1} - a_{k-2}$ into $(1 + a_k^2)/a_{k-1}$ and expanding. This just makes a mess. The 1 has to be replaced using the definition one step back.

- Forgetting that the definition divides by a_{n-2}. A complete answer notes that the terms are positive, so the division and the final cancellation of a_{k-1} are legitimate.
- Checking only that b_n satisfies the recurrence, without matching *two* starting values. A second-order recurrence needs both.

Beyond STEP

The constant $a_n a_{n-2} - a_{n-1}^2$ is an instance of Cassini's identity for Fibonacci numbers, $F_{m+1}F_{m-1} - F_m^2 = (-1)^m$. Here we only see odd indices, so the sign is always $+1$. More generally, $u_{n+1}u_{n-1} - u_n^2$ changes by a constant factor at each step for any linear second-order recurrence (it is a 2×2 determinant, and determinants multiply). Rational recurrences that, surprisingly, always produce integers, like this one, are linked to the study of "Somos sequences" and the Laurent phenomenon. For the number theory of Fibonacci-type sequences and continued fractions (note that α is the golden ratio), Davenport's *The Higher Arithmetic* is a lovely read.

WORKED SOLUTION

STEP III 2000, Question 9

Mechanics • Oblique Elastic Collisions

STEP III 2000, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

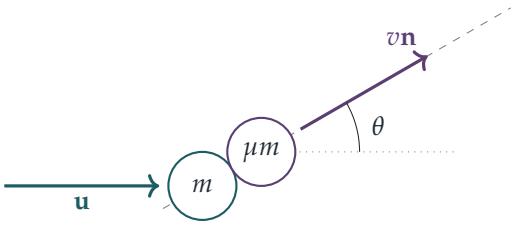
step.maths.org/questions/00-s3-q9

The key idea

The discs are smooth, so during the impact the only force between them acts along the line of centres. The disc that starts at rest therefore moves off along that line, so $\mathbf{n}$ *is* the line of centres. Write its velocity as $v\mathbf{n}$, use conservation of momentum (a vector equation) to get the other disc's velocity, and then let conservation of kinetic energy fix v. Part (ii) is then one more energy equation, and the inequality comes from insisting that a cosine is at most 1.

Solution

We model the discs as smooth particles on a smooth horizontal plane, so there is no friction between them or with the surface. Let θ be the angle between $\mathbf{u}$ and $\mathbf{n}$, and write $u = |\mathbf{u}|$, so that $\mathbf{u} \cdot \mathbf{n} = u\cos\theta$.



At the moment of impact the line of centres (dashed) points in the direction $\mathbf{n}$, making angle θ with the direction of $\mathbf{u}$ (dotted). The figure is only schematic; we do not need it for the algebra.

Part (i)

The disc of mass μm starts at rest and the only horizontal force on it during the impact is the contact force from the other disc, which acts along the line of centres. Since it moves off in direction $\mathbf{n}$, its velocity afterwards is $v\mathbf{n}$ for some speed $v > 0$.

Let the disc of mass m have velocity $\mathbf{w}$ afterwards. Conservation of momentum gives

$$m\mathbf{u} = m\mathbf{w} + \mu m v\,\mathbf{n} \qquad \Longrightarrow \qquad \mathbf{w} = \mathbf{u} - \mu v\,\mathbf{n}.$$

The collision is perfectly elastic, so kinetic energy is conserved:

$$\tfrac{1}{2}m\,\mathbf{u}\cdot\mathbf{u} = \tfrac{1}{2}m\,\mathbf{w}\cdot\mathbf{w} + \tfrac{1}{2}\mu m v^2.$$

Using $\mathbf{n}\cdot\mathbf{n} = 1$,

$$\mathbf{w}\cdot\mathbf{w} = (\mathbf{u} - \mu v\mathbf{n})\cdot(\mathbf{u} - \mu v\mathbf{n}) = u^2 - 2\mu v\,\mathbf{u}\cdot\mathbf{n} + \mu^2 v^2.$$

Substituting and cancelling $\tfrac{1}{2}m$ and u^2,

$$0 = -2\mu v\,\mathbf{u}\cdot\mathbf{n} + \mu^2 v^2 + \mu v^2 = \mu v\big((1+\mu)v - 2\,\mathbf{u}\cdot\mathbf{n}\big).$$

Now $\mu > 0$, and $v \neq 0$ because the struck disc does move. So we may divide by μv to get

$$v = \frac{2\,\mathbf{u}\cdot\mathbf{n}}{1+\mu},$$

which is the required speed. As a by-product, $v > 0$ tells us that $\mathbf{u}\cdot\mathbf{n} > 0$, so $0 \le \theta < \tfrac{1}{2}\pi$.

Ask yourself Why don't we need Newton's experimental law here?

We could use it: with $e = 1$, the components of relative velocity along $\mathbf{n}$ are reversed, and the components perpendicular to $\mathbf{n}$ are unchanged because the discs are smooth. But "perfectly elastic" means kinetic energy is conserved, and that single scalar equation, with momentum, already pins down v. The working above is exactly equivalent and avoids splitting into components.

Part (ii)

The total kinetic energy after the collision equals the initial $\tfrac{1}{2}mu^2$. If the two discs share it equally, each has $\tfrac{1}{4}mu^2$, so for the struck disc

$$\tfrac{1}{2}\mu m v^2 = \tfrac{1}{4}mu^2 \qquad \Longrightarrow \qquad \mu v^2 = \tfrac{1}{2}u^2.$$

Using part (i) with $\mathbf{u}\cdot\mathbf{n} = u\cos\theta$, we have $v = \dfrac{2u\cos\theta}{1+\mu}$, so

$$\frac{4\mu u^2\cos^2\theta}{(1+\mu)^2} = \frac{u^2}{2} \qquad \Longrightarrow \qquad \cos^2\theta = \frac{(1+\mu)^2}{8\mu}.$$

Since $\cos\theta > 0$ (from the end of part (i)), we take the positive root:

$$\cos\theta = \frac{1+\mu}{2\sqrt{2\mu}}.$$

For such an angle to exist we need $\cos\theta \le 1$, that is $\cos^2\theta \le 1$:

$$\frac{(1+\mu)^2}{8\mu} \le 1 \iff (1+\mu)^2 \le 8\mu \qquad \text{(multiplying by } 8\mu > 0\text{)}$$
$$\iff \mu^2 - 6\mu + 1 \le 0.$$

The quadratic $\mu^2 - 6\mu + 1$ has roots $\mu = 3 \pm \sqrt{9-1} = 3 \pm \sqrt{8}$, and it is negative or zero exactly between them. Hence

$$3 - \sqrt{8} \le \mu \le 3 + \sqrt{8},$$

as required. (The lower bound is positive, $3 - \sqrt{8} \approx 0.172$, so it is consistent with $\mu > 0$.) Conversely, for every μ in this range the formula gives a valid cosine in $(0, 1]$, so equal sharing really is possible; at either end $\theta = 0$, a head-on collision.

Checks

- Head-on case: if $\mathbf{n}$ is parallel to $\mathbf{u}$, part (i) gives $v = 2u/(1+\mu)$, the familiar one-dimensional elastic result. With $\mu = 1$ this is $v = u$: the moving disc stops and passes all its motion on, as in snooker.
- Equal masses, $\mu = 1$: part (ii) gives $\cos\theta = 2/(2\sqrt{2}) = 1/\sqrt{2}$, so $\theta = 45°$. This fits the classic fact that equal smooth discs move off at right angles after an elastic collision; equal energies then force the two speeds to be equal, so each makes $45°$ with $\mathbf{u}$.
- Numerically, we simulated 10 000 random smooth elastic collisions (impulse along $\mathbf{n}$), confirmed conservation of momentum and energy and the speed in part (i), and confirmed equal kinetic energies at the angle found in part (ii) for μ between 0.172 and 5.83. For example, $\mu = 0.1$ gives $\cos^2\theta = 1.51 > 1$, which is impossible.

Common slips

- Treating $\mathbf{u}$ and $\mathbf{n}$ as parallel. The collision is oblique; only the component $\mathbf{u}\cdot\mathbf{n}$ along the line of centres is transferred.
- Dividing by v without saying why it is non-zero ($v = 0$ is the "no collision" solution of the energy equation).
- Forgetting to rule out the negative square root for $\cos\theta$.

Beyond STEP

The trick of writing the struck particle's velocity along the impulse and then using momentum and energy is the heart of elastic scattering in physics. It is cleanest in the centre-of-mass frame, where an elastic collision just rotates both velocities without changing their sizes; the angle range in part (ii) then becomes a statement about which

lab-frame angles are reachable. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions in both frames very clearly.

WORKED SOLUTION

STEP III 2000, Question 10

Mechanics • Limiting Friction Equilibrium

STEP III 2000, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q10

The key idea

Start with the geometry: the distances $\sqrt{3}\,a$ and a are chosen so that the beam meets the ground at a special angle, and that fixes every direction in the problem. Then take moments about the two points where unknown forces drop out. About the hinge X, the beam's equation contains only the normal reaction from the sphere, because the friction from the sphere acts along the beam itself. About the centre of the sphere, every force except the two frictions passes through the centre, so the two frictions must be equal.

Solution

Model and geometry

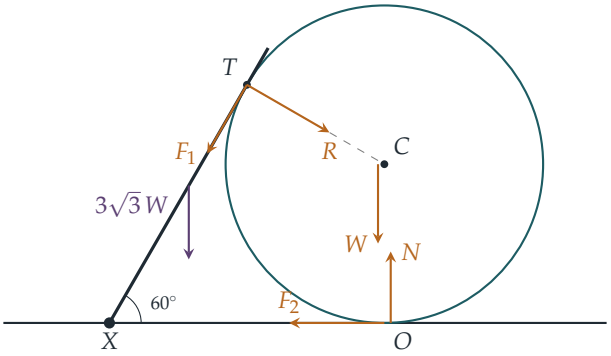
We treat the beam as a uniform rod and the sphere as a uniform rigid body whose weight acts at its centre C. Both contacts are rough. Let O be the point where the sphere touches the ground, so $OX = \sqrt{3}\,a$ and $OC = a$, with OC vertical.

The beam touches the sphere at a point T, and it is tangent to the sphere there. The ground is the other tangent from X. In the right-angled triangle XOC,

$$\tan \angle OXC = \frac{a}{\sqrt{3}\,a} = \frac{1}{\sqrt{3}}, \qquad \text{so} \qquad \angle OXC = 30^\circ.$$

The two tangents from X are symmetric about the line XC, so the beam makes an angle $2 \times 30^\circ = 60^\circ$ with the ground. Tangent lengths from a point are equal, so $XT = XO = \sqrt{3}\,a$. Since $\sqrt{3}\,a < 2a$, the contact point really does lie on the beam, a little beyond its midpoint.

The radius CT is perpendicular to the beam. So the normal reaction at T makes 30° with the horizontal, and the friction at T acts along the beam, at 60° to the horizontal.



The amber arrows are the forces on the sphere. The beam feels R and F_1 in the opposite directions, together with its weight $3\sqrt{3}\,W$ at its midpoint (a distance a from X) and an unknown reaction at the hinge.

We have guessed the directions of the frictions: on the sphere, F_1 acts down the beam towards X, and F_2 acts along the ground towards X. If a guess is wrong, that force will come out negative.

The beam: moments about X

The hinge reaction acts at X, so it has no moment about X. The friction from the sphere acts along the beam, and the beam's line passes through X, so it has no moment either. The normal reaction R is perpendicular to the beam at distance $XT = \sqrt{3}\,a$. The weight acts at horizontal distance $a\cos 60^\circ = \frac{1}{2}a$ from X. Hence

$$R \cdot \sqrt{3}\,a = 3\sqrt{3}\,W \cdot \tfrac{1}{2}a, \qquad \text{so} \qquad R = \tfrac{3}{2}W.$$

The sphere: moments about C

The forces R, W and N all pass through C. (The normal reactions act along radii.) The two frictions are tangential, each at distance a from C. F_1 at T turns the sphere one way about C and F_2 at O turns it the other way, so

$$F_1 a = F_2 a, \qquad \text{so} \qquad F_1 = F_2.$$

Call this common value F.

> **Ask yourself** Why can we be sure the two frictions are equal, before knowing anything else about them?

The sphere is in equilibrium, so the total moment about any point is zero. The centre is the clever choice: the weight and both normal reactions pass through it, so only the frictions are left, and they have equal lever arms a.

The sphere: resolving

Horizontally, taking the direction away from X as positive: R has component $R\cos 30^\circ$, F_1 has component $-F\cos 60^\circ$, and F_2 contributes $-F$. So

$$\frac{\sqrt{3}}{2}R = \frac{1}{2}F + F = \frac{3}{2}F, \qquad \text{so} \qquad F = \frac{R}{\sqrt{3}} = \frac{\sqrt{3}}{2}W.$$

This is positive, so the guessed friction directions were right. Vertically, R pushes down with component $R\sin 30^\circ$, and F_1 pulls down with component $F\sin 60^\circ$. So

$$N = W + \tfrac{1}{2}R + \tfrac{\sqrt{3}}{2}F = W + \tfrac{3}{4}W + \tfrac{3}{4}W = \tfrac{5}{2}W.$$

The coefficients of friction

The sphere is on the point of slipping at both contacts, so friction is limiting at both: $F_1 = \mu_1 R$ and $F_2 = \mu_2 N$. Therefore

$$\mu_1 = \frac{F}{R} = \frac{\frac{\sqrt{3}}{2}W}{\frac{3}{2}W} = \frac{1}{\sqrt{3}}, \qquad \mu_2 = \frac{F}{N} = \frac{\frac{\sqrt{3}}{2}W}{\frac{5}{2}W} = \frac{\sqrt{3}}{5}.$$

Checks

- A numerical check with coordinates ($a = 1.3$, $W = 2$) confirmed that T lies on the sphere with CT perpendicular to the beam. Solving the sphere's three equilibrium equations directly gave $R = 1.5W$, $F_1 = F_2 \approx 0.8660W$ and $N = 2.5W$, so $\mu_1 \approx 0.5774 = 1/\sqrt{3}$ and $\mu_2 \approx 0.3464 = \sqrt{3}/5$.
- An independent equation: take moments for the sphere about O. Now N, F_2 and W have no moment. The line of R passes through C, so its moment is that of its horizontal component at height a, namely $\frac{\sqrt{3}}{2}R\,a$. The moment of F_1 about O works out to $\frac{3}{2}aF$. Equating these gives $F = R/\sqrt{3}$ again, without using the moment equation about C.
- Both coefficients are less than 1, which is physically sensible for ordinary rough contacts.

Common slips

- Taking the beam's angle to be 30° (the angle of XC) instead of 60°.
- Forgetting the vertical component of F_1 when finding N. This gives $N = \frac{7}{4}W$ and the wrong μ_2.
- Including the friction from the sphere in the beam's moment equation about X. It acts along the beam, so its moment about X is zero.

Beyond STEP

A rigid body in a plane has three degrees of freedom, so it gives three independent equilibrium equations. Here the sphere has five unknown contact forces, and the beam supplies R to cut this down. The two "limiting friction" conditions then turn the problem into a solvable system. Counting equations against unknowns like this, and spotting when a system is statically indeterminate, is the starting point of structural mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and torque carefully.

WORKED SOLUTION

STEP III 2000, Question 11

Mechanics • Rotation About Rough Pivot

STEP III 2000, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q11

The key idea

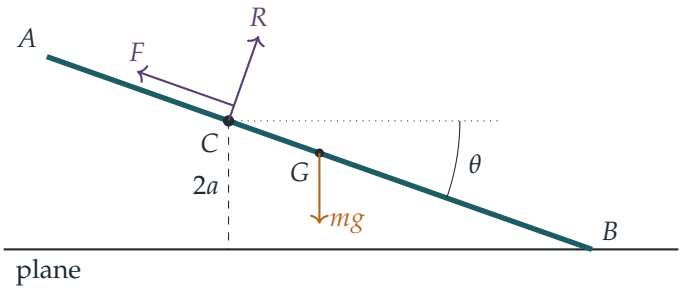
As long as there is no slipping, the rod simply pivots about the point C on the beam, so it is a rigid body rotating about a fixed axis. Energy gives the angular speed and the moment equation gives the angular acceleration. Then Newton's second law for the centre of mass, resolved along and across the rod, tells us the friction and normal reaction the beam must supply. Slipping is avoided while friction stays within μ times the normal reaction, and we need this to hold all the way until B reaches the plane.

Solution

The model and the geometry

We model the rod as a uniform rigid rod and the beam as a fixed thin rough line, so the beam exerts a single contact force at C: a normal reaction R perpendicular to the rod and a friction force F along it.

The centre of mass G is the midpoint, $4.5a$ from A. Since $AC = 3a$, G lies on the B side of C with $CG = \frac{3}{2}a$, and $CB = 6a$. So the weight has a moment about C, and the rod tips with B going down. Let θ be the angle the rod has turned through, so that the rod makes angle θ with the horizontal.



B reaches the plane when it has dropped $2a$, that is when $6a\sin\theta = 2a$, or

$$\sin\theta = \tfrac{1}{3}, \qquad \tan\theta = \frac{1}{2\sqrt{2}}.$$

The rotation about C

The moment of inertia about C is, by the parallel axes theorem,

$$I_C = \tfrac{1}{12}m(9a)^2 + m\left(\tfrac{3}{2}a\right)^2 = \tfrac{27}{4}ma^2 + \tfrac{9}{4}ma^2 = 9ma^2.$$

While there is no slipping, C is fixed and the contact force does no work. G has fallen $\frac{3}{2}a\sin\theta$, so conservation of energy (starting from rest) gives

$$\tfrac{1}{2}(9ma^2)\dot\theta^2 = mg\cdot\tfrac{3}{2}a\sin\theta \quad\Longrightarrow\quad \dot\theta^2 = \frac{g\sin\theta}{3a}.$$

Taking moments about C (only the weight has a moment):

$$9ma^2\,\ddot\theta = mg\cdot\tfrac{3}{2}a\cos\theta \quad\Longrightarrow\quad \ddot\theta = \frac{g\cos\theta}{6a}.$$

(Differentiating the energy equation with respect to t gives the same thing, a useful check.)

The contact forces

G moves on a circle of radius $\frac{3}{2}a$ about C, so its acceleration has a component $\frac{3}{2}a\dot\theta^2$ along the rod towards C and a component $\frac{3}{2}a\ddot\theta$ perpendicular to the rod, in the direction of increasing θ (downwards across the rod).

Ask yourself Which way does the friction act, and why is it not simply $mg\sin\theta$?

The component of the weight down the rod, $mg\sin\theta$, would pull the rod off the beam towards B; in addition G needs a centripetal force towards C. Both must be supplied by friction, which therefore acts up the rod, towards A. Resolving along the rod, taking the direction towards C as positive for the acceleration of G:

$$F - mg\sin\theta = m\cdot\tfrac{3}{2}a\dot\theta^2 = m\cdot\tfrac{3}{2}a\cdot\frac{g\sin\theta}{3a} = \tfrac{1}{2}mg\sin\theta,$$

so

$$F = \tfrac{3}{2}mg\sin\theta.$$

Resolving perpendicular to the rod, in the direction in which G is accelerating:

$$mg\cos\theta - R = m\cdot\tfrac{3}{2}a\ddot\theta = m\cdot\tfrac{3}{2}a\cdot\frac{g\cos\theta}{6a} = \tfrac{1}{4}mg\cos\theta,$$

so

$$R = \tfrac{3}{4}mg\cos\theta,$$

which is positive throughout, so the rod stays in contact with the beam.

The condition for no slipping

The rod does not slip at C as long as $F \le \mu R$, that is

$$\frac{F}{R} = \frac{\frac{3}{2}mg\sin\theta}{\frac{3}{4}mg\cos\theta} = 2\tan\theta \le \mu.$$

The ratio $2\tan\theta$ increases with θ, starting from 0 at release, so the condition is hardest to satisfy at the last moment, just as B reaches the plane. There $\tan\theta = 1/(2\sqrt{2})$, so we need

$$\mu \ge 2 \cdot \frac{1}{2\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

Hence the minimum value of μ is

$$\boxed{\mu = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.707.}$$

With exactly this value, friction becomes limiting only at the instant B strikes the plane; for any smaller μ slipping starts before then.

Checks

- At release $\dot\theta = 0$, so $F = 0$ and $R = \frac{3}{4}mg$: the beam carries less than the full weight because G starts accelerating downwards with $\frac{3}{2}a\ddot\theta = \frac{1}{4}g$. That is the familiar "a released plank feels lighter" effect.
- Units: F and R are both multiples of mg, and μ is dimensionless, as it must be.
- We stepped the equation $\ddot\theta = g\cos\theta/(6a)$ numerically from rest, computed the acceleration of G by finite differences, and recovered F and R from Newton's second law. They match $\frac{3}{2}mg\sin\theta$ and $\frac{3}{4}mg\cos\theta$, and F/R climbs steadily to $0.7071\ldots = 1/\sqrt{2}$ when $\sin\theta = \frac{1}{3}$.

Common slips

- Taking moments about the centre or forgetting the parallel axes term: $I_C = 9ma^2$, not $\frac{27}{4}ma^2$.
- Treating the problem as statics, $F = mg\sin\theta$, $R = mg\cos\theta$, which gives $\mu \ge \tan\theta$. Both the centripetal and the tangential acceleration of G change the forces.
- Checking the friction condition only at release, where $F = 0$, rather than at the point where it is most demanding.

Beyond STEP

This question is a first taste of rigid-body dynamics with constraint forces: we found the motion from energy and moments, and only then used Newton's law for the centre

of mass to work backwards to the forces the constraint must provide. The same pattern (solve the motion, then check whether the constraint can actually supply the force needed) decides when a ladder starts to slip, when a toppling pencil leaves the table, or when a bead leaves a wire. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops fixed-axis rotation and the link between torque and the motion of the centre of mass carefully.

WORKED SOLUTION

STEP III 2000, Question 12

Probability and Statistics • Poisson Approximation

STEP III 2000, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q12

The key idea

Once you know your own ticket wins, the only randomness left is in the other players' tickets. Each of them independently matches the winning number with probability $\frac{1}{N}$, so the number of *other* winners is binomial. With many players and a small success probability, that binomial is close to a Poisson distribution, and the given limit $\left(1-\frac{a}{n}\right)^n \approx \mathrm{e}^{-a}$ is exactly what makes this work.

Solution

The exact probability

Suppose you hold a winning ticket. Let X be the number of the other $2N-1$ players who also hold the winning number, so the total number of winners is $W = 1 + X$.

The machine chooses each player's number uniformly and independently of everyone else's, and independently of the draw. So, given the winning number (whatever it is) and given your ticket, each of the other $2N-1$ players matches it with probability $\frac{1}{N}$, independently of the others. Hence

$$X \sim \mathrm{B}\left(2N-1,\ \tfrac{1}{N}\right).$$

Three or fewer winners means $X \leq 2$, so the exact probability is

$$\mathrm{P}(W \leq 3) = \sum_{k=0}^{2} \binom{2N-1}{k} \left(\frac{1}{N}\right)^k \left(1-\frac{1}{N}\right)^{2N-1-k}.$$

Written out in full, this is

$$\left(1-\frac{1}{N}\right)^{2N-1} + \frac{2N-1}{N}\left(1-\frac{1}{N}\right)^{2N-2} + \frac{(2N-1)(2N-2)}{2N^2}\left(1-\frac{1}{N}\right)^{2N-3}.$$

The approximation

For each fixed $k \in \{0, 1, 2\}$, split the power:

$$\left(1 - \frac{1}{N}\right)^{2N-1-k} = \left[\left(1 - \frac{1}{N}\right)^{N}\right]^{2} \left(1 - \frac{1}{N}\right)^{-(1+k)}.$$

Using the given fact with $a = 1$ and $n = N$, the bracket is about e^{-1}. The last factor is a fixed small power of a number very close to 1, so it is about 1. Hence each power is about e^{-2}.

For the binomial coefficients, k is at most 2 and N is large, so

$$\binom{2N-1}{k} \frac{1}{N^k} = \frac{(2N-1)(2N-2)\cdots(2N-k)}{k!\, N^k} \approx \frac{(2N)^k}{k!\, N^k} = \frac{2^k}{k!}.$$

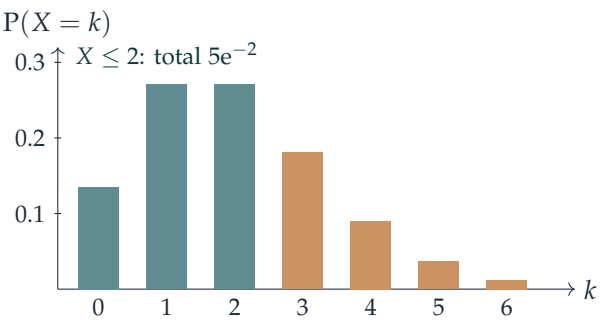
Putting these together, the terms for $k = 0, 1, 2$ are about e^{-2}, $2\mathrm{e}^{-2}$ and $2\mathrm{e}^{-2}$, so

$$\mathrm{P}(W \leq 3) \approx \mathrm{e}^{-2}\left(1 + 2 + \frac{2^2}{2!}\right) = 5\mathrm{e}^{-2} \approx 0.677 \approx \frac{2}{3},$$

using $\mathrm{e}^2 \approx 7.39$. This is the required result.

Ask yourself Why does X behave like a Poisson variable with mean 2?

The calculation above is the standard Poisson limit of a binomial. The number of trials, $2N - 1$, is large, and the success probability $\frac{1}{N}$ is small, while their product stays close to 2. In that regime $\binom{n}{k} p^k (1-p)^{n-k} \to \mathrm{e}^{-\lambda}\lambda^k/k!$ with $\lambda = np$, and here $\lambda \to 2$.



The limiting distribution of X is Poisson with mean 2. The teal bars ($X \leq 2$, so at most three winners in all) make up about two-thirds of the total.

Players choosing their own numbers

If players choose, the numbers are no longer uniform: people favour birthdays (numbers up to 31), lucky numbers and patterns. Suppose a player picks number i with probability p_i, where $\sum p_i = 1$, and the winning number is still drawn uniformly. Then, given

that you win, your number is i with probability proportional to $p_i \cdot \frac{1}{N}$, that is, with probability p_i. So popular numbers are over-represented among winning tickets. If your number is i, the number of other winners is $\mathrm{B}(2N-1, p_i)$, with mean $(2N-1)p_i$. Averaging over i, the expected number of other winners is

$$(2N-1)\sum_i p_i^2 \;\geq\; (2N-1)\cdot\frac{1}{N},$$

by the Cauchy–Schwarz inequality $\left(\sum p_i\right)^2 \leq N\sum p_i^2$, with equality only when every $p_i = \frac{1}{N}$. So non-uniform choices mean more people sharing your win, on average, and the probability of three or fewer winners would typically *decrease*. (A single player who deliberately picks an unpopular number would do better than average, but players as a whole do worse.)

The expected number of winners

Since $X \sim \mathrm{B}\left(2N-1, \frac{1}{N}\right)$, its mean is $np = (2N-1)\cdot\frac{1}{N}$. (Equivalently, write X as a sum of $2N-1$ indicator variables, one for each other player, each with mean $\frac{1}{N}$.) Therefore

$$\mathrm{E}(W) = 1 + \mathrm{E}(X) = 1 + \frac{2N-1}{N} = 3 - \frac{1}{N} = 3 - N^{-1},$$

as required. Note this is exact, not an approximation.

Checks

- For $N = 2$ and $N = 3$, enumerating every assignment of numbers to the $2N$ players and every winning number gives exactly the binomial expression and exactly $3 - \frac{1}{N}$.
- The exact probability is 0.7054 for $N = 10$, 0.6794 for $N = 100$ and 0.6767 for $N = 10\,000$, approaching $5\mathrm{e}^{-2} = 0.67668$.
- A simulation with $N = 50$ (10^5 weeks) gave 0.683 against the exact 0.682, and a mean of 2.981 winners against $3 - \frac{1}{50} = 2.98$.
- With $N = 10$ and half the numbers twice as popular as the others, the probability falls from 0.705 to 0.641, and the expected number of winners rises from 2.9 to 3.11.

Common slips

- Using $2N$ trials instead of $2N-1$: your own ticket is already known to win, so only the other players are random. With $2N$ trials the expectation would come out as $1 + 2 = 3$, not $3 - \frac{1}{N}$.
- Using the unconditional mean: across all weeks the expected number of winners is $2N \cdot \frac{1}{N} = 2$, but knowing that you win adds one certain winner and leaves $2N-1$ others.
- Forgetting the $k = 2$ term, or writing it as $2^2\mathrm{e}^{-2}$ without the $\frac{1}{2!}$.

Beyond STEP

The approximation here is the law of rare events: a binomial with many trials and a small success probability converges to a Poisson distribution. The final part touches on the inspection paradox, or size-biased sampling. Conditioning on *your* ticket winning makes crowded numbers more likely, which is why the conditional mean is 3 rather than 2. Both ideas are treated well in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP III 2000, Question 13

Probability and Statistics • Probability Generating Functions

STEP III 2000, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q13

The key idea

Both random variables are easiest to handle through "tail" or "cumulative" events. The first six comes after roll r exactly when *every* die misses on every one of the first r rolls; the last six has come by roll r exactly when *each* die has already shown a six. Each of these is an intersection of independent events, so its probability is a product, and the distribution follows by subtracting consecutive values. Once the first-six variable is recognised as geometric, everything about it is quick.

Solution

Throughout we assume the dice behave independently of each other and from roll to roll. For a single die, let X_i be the number of the roll on which die i first shows a six, so

$$\mathrm{P}(X_i > r) = q^r \qquad (r = 0, 1, 2, \dots),$$

since $X_i > r$ means r consecutive non-sixes. Then $R = \min_i X_i$ and $S = \max_i X_i$.

The first six

The event $R > r$ says that none of the n dice has shown a six in the first r rolls, that is, all nr individual throws were non-sixes. By independence,

$$\mathrm{P}(R > r) = q^{nr}.$$

Hence, for $r \geq 1$,

$$\mathrm{P}(R = r) = \mathrm{P}(R > r - 1) - \mathrm{P}(R > r) = q^{n(r-1)} - q^{nr} = q^{nr}\left(q^{-n} - 1\right),$$

which is the required result.

Writing $Q = q^n$, this is $\mathrm{P}(R = r) = Q^{r-1}(1 - Q)$: R is geometric, each roll of the whole set being a "success" (at least one six) with probability $1 - Q$. The probability

generating function is therefore a geometric series:

$$G_R(t) = \sum_{r=1}^{\infty} t^r Q^{r-1}(1-Q) = (1-Q)\,t \sum_{r=1}^{\infty} (Qt)^{r-1}$$
$$= \frac{(1-q^n)\,t}{1-q^n t},$$

valid for $|t| < q^{-n}$ (in particular near $t = 1$).

For the mean, differentiate by the quotient rule:

$$G_R'(t) = (1-q^n)\,\frac{(1-q^n t) + q^n t}{(1-q^n t)^2} = \frac{1-q^n}{(1-q^n t)^2},$$

so

$$\mathrm{E}(R) = G_R'(1) = \frac{1}{1-q^n}.$$

With $n = 2$ and $p = \frac{1}{6}$, $q^2 = \frac{25}{36}$, so $\mathrm{E}(R) = 1/\frac{11}{36} = \frac{36}{11}$, as required.

The last six

The event $S \le r$ says that every die has shown a six somewhere in the first r rolls. For one die this has probability $1 - q^r$, and the dice are independent, so

$$\mathrm{P}(S \le r) = (1-q^r)^n.$$

Subtracting consecutive values, for $r \ge 1$,

$$\mathrm{P}(S = r) = \mathrm{P}(S \le r) - \mathrm{P}(S \le r-1) = \left(1-q^r\right)^n - \left(1-q^{r-1}\right)^n,$$

as required. (For $r = 1$ the second term is $0^n = 0$, correctly.)

Ask yourself Summing $t^r\left[(1-q^r)^2 - (1-q^{r-1})^2\right]$ looks messy. How can we turn it into geometric series?

Expand the squares when $n = 2$:

$$\mathrm{P}(S = r) = \left(1 - 2q^r + q^{2r}\right) - \left(1 - 2q^{r-1} + q^{2r-2}\right)$$
$$= 2q^{r-1}(1-q) - q^{2(r-1)}(1-q^2)$$
$$= 2p\,q^{r-1} - (1-q^2)\,(q^2)^{r-1}.$$

Each term is (a multiple of) a geometric distribution, so

$$G_S(t) = \sum_{r=1}^{\infty} t^r\,\mathrm{P}(S = r) = \frac{2pt}{1-qt} - \frac{(1-q^2)\,t}{1-q^2 t}.$$

This makes sense: since $S = X_1 + X_2 - R$, its distribution is "twice a geometric with parameter p, minus the distribution of R". To simplify, put everything over $(1-qt)(1-q^2t)$ and use $1 - q^2 = p(1+q)$:

$$2p(1-q^2t) - p(1+q)(1-qt) = p\left[2 - 2q^2t - 1 - q + qt + q^2t\right]$$
$$= p\left[(1-q) + qt(1-q)\right] = p^2(1+qt).$$

Hence

$$G_S(t) = \frac{p^2\, t\,(1+qt)}{(1-qt)(1-q^2t)}.$$

As a sanity check, $G_S(1) = \dfrac{p^2(1+q)}{p\cdot p(1+q)} = 1$.

For the mean, the split form is easiest to differentiate, since each piece is a geometric PGF whose derivative at 1 is its mean scaled by its coefficient. For $\dfrac{(1-a)t}{1-at}$ the derivative at $t=1$ is $\dfrac{1}{1-a}$ (exactly as for G_R above), so

$$\mathrm{E}(S) = G_S'(1) = 2\cdot\frac{1}{p} - \frac{1}{1-q^2} = \frac{2(1+q)-1}{p(1+q)} = \frac{1+2q}{p(1+q)}.$$

With $p=\frac{1}{6}$, $q=\frac{5}{6}$:

$$\mathrm{E}(S) = 12 - \frac{36}{11} = \frac{96}{11} \approx 8.73.$$

Checks

- $\mathrm{E}(R)+\mathrm{E}(S) = \frac{36}{11}+\frac{96}{11} = 12 = \mathrm{E}(X_1)+\mathrm{E}(X_2)$, as it must be since $\min+\max = X_1+X_2$.
- With $n=1$ the first and last six coincide, and G_R reduces to the single-die geometric PGF $pt/(1-qt)$.
- A Monte Carlo simulation of 2×10^5 pairs of dice gave sample means 3.274 and 8.709, against $\frac{36}{11}\approx 3.273$ and $\frac{96}{11}\approx 8.727$; summing the series for G_S numerically agrees with the closed form at several values of t.

Common slips

- Writing $\mathrm{P}(S=r)$ as "some die first shows a six on roll r and the others earlier", and forgetting the case where both dice first show a six on the same roll. Working with $\mathrm{P}(S\le r)$ avoids this entirely.
- Differentiating the simplified G_S instead of the split form: correct, but far more algebra and many chances for error.

Beyond STEP

R and S are the minimum and maximum of independent random variables, the simplest examples of *order statistics*; the trick of using $\mathrm{P}(\min > r)$ and $\mathrm{P}(\max \le r)$ works for any distribution. The memoryless geometric distribution is the discrete cousin of the exponential, and in the continuous limit the expected maximum of n exponentials becomes the harmonic sum $1+\frac{1}{2}+\cdots+\frac{1}{n}$ (times the mean), a key step in the coupon

collector's problem. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions and these ideas in depth.

WORKED SOLUTION

STEP III 2000, Question 14

Probability and Statistics • Covariance and Independence

STEP III 2000, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q14

The key idea

Fill in the missing fourth probability, then write out $\mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)$ term by term. Everything collects into a single factor $a - pq$ times something that cannot vanish, and $a = pq$ is exactly the statement that the top-left cell factorises. Once one cell factorises in a 2×2 table, the margins force the other three to as well. With three values there is more room, and a symmetric table breaks the link.

Solution

The identity

The four probabilities sum to 1, so

$$\mathrm{P}(X = x_2,\, Y = y_2) = 1 - a - (q - a) - (p - a) = 1 - p - q + a.$$

Adding along rows and columns gives the marginal distributions:

$$\mathrm{P}(X = x_1) = a + (q - a) = q, \qquad \mathrm{P}(Y = y_1) = a + (p - a) = p,$$

so $\mathrm{P}(X = x_2) = 1 - q$ and $\mathrm{P}(Y = y_2) = 1 - p$. This is what p and q really mean.

	y_1	y_2	
x_1	a	$q - a$	q
x_2	$p - a$	$1 - p - q + a$	$1 - q$
	p	$1 - p$	

Now

$$\begin{aligned}\mathrm{E}(XY) &= a\,x_1y_1 + (q-a)\,x_1y_2 + (p-a)\,x_2y_1 + (1-p-q+a)\,x_2y_2,\\ \mathrm{E}(X)\mathrm{E}(Y) &= \big(qx_1 + (1-q)x_2\big)\big(py_1 + (1-p)y_2\big)\\ &= pq\,x_1y_1 + q(1-p)\,x_1y_2 + p(1-q)\,x_2y_1 + (1-p)(1-q)\,x_2y_2.\end{aligned}$$

Subtract coefficient by coefficient:

$$\begin{aligned}x_1y_1:&\quad a-pq,\\ x_1y_2:&\quad (q-a)-(q-pq) = -(a-pq),\\ x_2y_1:&\quad (p-a)-(p-pq) = -(a-pq),\\ x_2y_2:&\quad (1-p-q+a)-(1-p-q+pq) = a-pq.\end{aligned}$$

Hence

$$\mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y) = (a-pq)(x_1y_1 - x_1y_2 - x_2y_1 + x_2y_2) = (a-pq)(x_1-x_2)(y_1-y_2).$$

So if $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$, the left-hand side is zero and $(a-pq)(x_1-x_2)(y_1-y_2) = 0$, as required.

Independence

Any pair of random variables each taking two distinct values has a joint table of the form above, for suitable a, p, q (just name the cells). If $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$ then the product above is zero. Since $x_1 \neq x_2$ and $y_1 \neq y_2$, the last two factors are non-zero, so $a = pq$.

> **Ask yourself** We now know one cell factorises. Why does that settle the other three?

Because each other cell is a margin minus a cell we already know:

$$\begin{aligned}\mathrm{P}(X=x_1, Y=y_2) &= q - pq = q(1-p) = \mathrm{P}(X=x_1)\,\mathrm{P}(Y=y_2),\\ \mathrm{P}(X=x_2, Y=y_1) &= p - pq = p(1-q) = \mathrm{P}(X=x_2)\,\mathrm{P}(Y=y_1),\\ \mathrm{P}(X=x_2, Y=y_2) &= 1-p-q+pq = (1-p)(1-q) = \mathrm{P}(X=x_2)\,\mathrm{P}(Y=y_2).\end{aligned}$$

Every cell is the product of its marginals, so X and Y are independent. (The converse, that independence gives $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$, is standard, so for two-valued variables the two conditions are equivalent.)

The three-valued example

Since A and B are each uniform on $\{-1, 0, 1\}$, we have $\mathrm{E}(A) = \mathrm{E}(B) = 0$, so we need a table with every row and column summing to $\frac{1}{3}$, with $\mathrm{E}(AB) = 0$ but not every cell

equal to $\frac{1}{9}$. A symmetric choice works:

	$B=-1$	$B=0$	$B=1$
$A=-1$	$\frac{1}{6}$	0	$\frac{1}{6}$
$A=0$	0	$\frac{1}{3}$	0
$A=1$	$\frac{1}{6}$	0	$\frac{1}{6}$

Each row and each column sums to $\frac{1}{3}$, so A and B each take $-1, 0, 1$ with probability $\frac{1}{3}$. The only non-zero products ab come from the corners:

$$\mathrm{E}(AB) = \tfrac{1}{6}(1) + \tfrac{1}{6}(-1) + \tfrac{1}{6}(-1) + \tfrac{1}{6}(1) = 0 = \mathrm{E}(A)\mathrm{E}(B).$$

But $\mathrm{P}(A=0, B=0) = \frac{1}{3} \neq \frac{1}{9} = \mathrm{P}(A=0)\,\mathrm{P}(B=0)$, so A and B are not independent. In words: $B=0$ exactly when $A=0$, and otherwise B is ± 1 with equal chances whatever A is.

Checks

- Try $X=Y$ with $x_1=y_1=1$, $x_2=y_2=0$: then $a=p=q$, and the identity says $\mathrm{Var}(X) = p - p^2$, which is right for a Bernoulli variable.
- The identity holds as a polynomial identity: we confirmed it exactly with rational arithmetic on 2000 random choices of x_i, y_i, a, p, q. It also checked the three-valued table: margins all $\frac{1}{3}$, $\mathrm{E}(AB)=0$, and $\mathrm{P}(A=0, B=0) = \frac{1}{3}$.

Common slips

- Showing $a = pq$ and stopping. Independence needs *every* cell to factorise, so the other three need checking (quickly).
- Forgetting to say why we may cancel $(x_1 - x_2)(y_1 - y_2)$: it is non-zero because the values are distinct.
- In the last part, offering $B = A^2$ or $B = |A|$. These are dependent with zero covariance, but B is never -1, so they don't have the required marginals.

Beyond STEP

The quantity $\mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)$ is the covariance, and this question shows that "uncorrelated" is weaker than "independent", except for two-valued (indicator-like) variables, where the identity reduces everything to the covariance of two indicators. A related fact you will meet: for jointly normal random variables, zero covariance does imply independence. For more, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

Beyond STEP

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